Department of Mechanical, Materials and Manufacturing Engineering

Modelling a grid-wide integrated hydrogen energy storage system for a net-zero power network in the UK in 2035

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1 Summary

This report details a final year bachelor's project for a Mechanical Engineering degree. The project explores the use and potential of green hydrogen for energy storage in the UK power system through a simulation approach using MATLAB. A model of the UK power system expected in 2035 is constructed based on the current geographical distribution of renewable energy sources and their predicted capacities in 2035. Initial simulations allow for calculations of power and energy surplus over the course of the year 2035, which were employed to estimate a suitable hydrogen energy storage (HES) network with a total charging capacity of 21 GW and a storage capacity of 8.4 TWh. The HES network is based on hydrogen production via water electrolysis in a polymer-exchange membrane (PEM) electrolyser and conversion back to electricity via the use of a fuel cell. The inclusion of the HES network in the simulation revealed huge potential to reduce associated CO_2 emissions. The total energy deficit over the year 2035 was shown to reduce by over 80% with the inclusion of HES. The corresponding reduction in emissions due to a reduced reliance on fossil-fuel based power is quantified and discussed. Three potential HES network topologies are outlined and evaluated for their suitability in the UK power system. It is suggested that a blend of approaches to HES may be required to build operational experience and inform future developments beyond 2035.

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2 Nomenclature

Abbreviations and Acronyms

Instance	Meaning
RES	Renewable Energy Sources
HES	Hydrogen Energy Storage
BEIS	Department for Business, Energy, and Industrial Strategy
CB6	Sixth Carbon Budget
PEM	Polymer Electrolyte Membrane
PEMFC	Polymer Electrolyte Membrane Fuel Cell
PV	Photo-voltaic (e.g., solar photo-voltaic cells)
CCGT	Combined-Cycle Gas Turbine
LCOE	Levelized Cost of Electricity
CfD	Contracts for Difference
CO_2e	Carbon dioxide equivalent
H_2	Hydrogen (gas)
WGS84	World Geodetic System 84 (coordinate system)
OSBNG36	Ordinance Survey British National Grid 36 (coordinate system)
REPD	Renewable Energy Planning Database (beis)
£m	£1,000,000
£bn	£1,000,000,000

Units

Symbol	Quantity	SI Unit
W	Watts, power	$J.s^{-1}$
Wh	Watt-hours, energy	3600 J
M	Mega	10^6
G	Giga	10^9
T	Tera	10^{12}
Mt	Mega-tonne (metric)	10^9 kg
g	Grams	10^{-3} kg

3 Introduction

Climate change is widely accepted to be one of the greatest threats to modern human civilisation across the world. Figure 1 shows the relationship between global carbon dioxide (CO_2) concentrations and average annual temperature deviations; this correlation is the known cause of climate change.

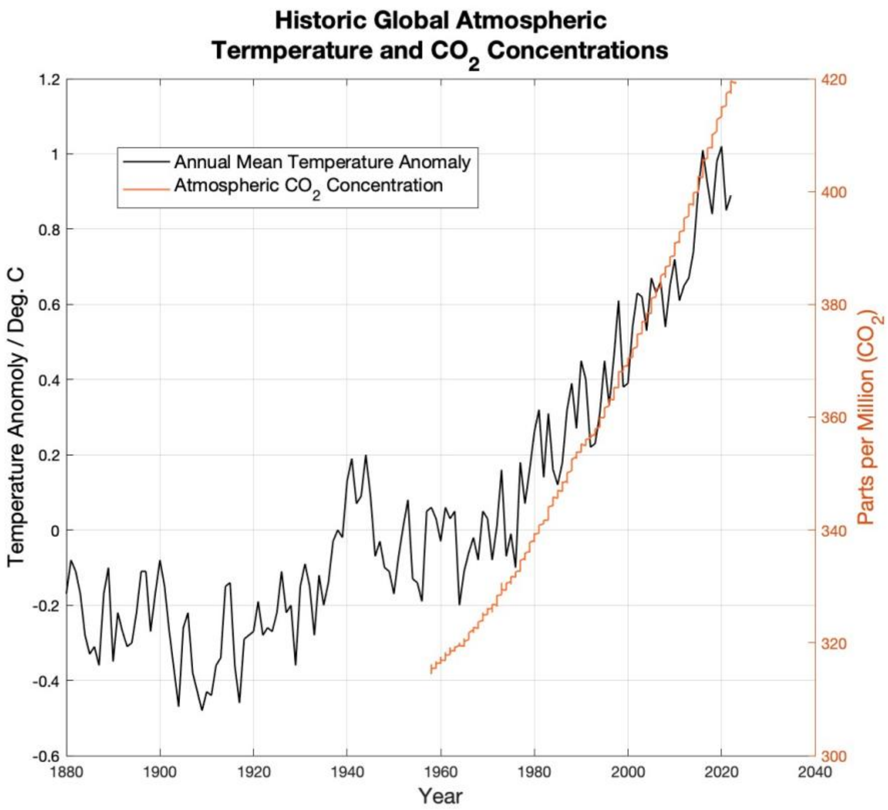

Figure 1 – Global CO_2 concentrations versus Temperature [1]

The effects of uncontrolled climate change are expected to be indiscriminate, unpredictable and to worsen exponentially. Climate scientists believe this phenomenon will occur irreversibly should the annual temperature anomaly exceed 2.3°C [2]. To avoid this, governments, companies, and NGOs are all changing the way they interact with energy.

Existing, global models have shown that the challenges faced in limiting global surface temperature rise to 2 degrees Celsius are likely to be too great unless a state of increased momentum towards net-zero is achieved and the potential of hydrogen is exploited [3].

The UK Government's Sixth Carbon Budget (CB6) is a legally binding commitment to various climate related targets, including for the entire UK to be net-zero by 2050. Another major constituent of CB6 is the commitment to a net-zero power grid, operational by 2035 [4].

The UK Government has indicated a £240m expenditure on a 'Net-Zero Hydrogen Fund' by 2035 in a publication intended to incite investment into the growing UK hydrogen sector [5]. The Department for Business, Energy and Industrial Strategy (BEIS) have stated that 'any option' to remaining on track to achieve their CB6 commitments will require the advancement and construction of hydrogen energy storage infrastructure [6].

The International Energy Agency (IEA) has also stated the importance of modelling dynamic hydrogen storage systems for a number of developed economies, including the UK [7]. One existing model has addressed the economic viability of hydrogen energy solutions, demonstrating that potential savings resultant of power-based hydrogen production could exceed £2bn per year for the UK [8]. These examples are clear evidence that the UK Government and authorities on the global energy system expect hydrogen energy storage (HES) to be of great significance in the changes required to abate climate change.

In 2021, 2.3 TWh of wind generated, clean power was curtailed (disconnected from the grid, wasted) due to insufficient demand. This equates to the total annual consumption of 800,000 UK homes [9]. Had this surplus energy been diverted to storage, pressure on fossil-based energy sources may have been reduced at times of a shortfall in available renewable energy and a residual power demand.

Members of the UK public were warned by Chief Executive of the National Grid to prepare for power outages for up to three hours on particularly cold days in the early months of 2023 [10]. The warning comes after a huge shift in energy security resultant of the political tensions surrounding the Russian invasion of Ukraine in 2022. Not only does this highlight the harmful impacts of a lack of energy sovereignty, but also the potential to mitigate such risks by developing a more robust, efficient, and autonomous energy system.

There are numerous, critical pretexts manifesting the necessity for storage of energy from renewable energy sources (RES). Whilst the majority of these stem from the challenges posed by global climate change, there are many others which will become more prevalent as our reliance on electrical power becomes ever more encompassing.

Hydrogen is one of many energy storage possibilities and will form the focus of this project. A time-based, regional model of the UK power system is to be developed in order to meet the aims and objectives presented in the following section.

3.1 Project aim

This project will endeavour to answer the following question through the tasks and mechanisms described herein:

To what extent could a grid-integrated hydrogen production and storage system powered by renewable electricity aid the UK to achieve HM Government's goal of a net-zero power network in 2035?

3.2 Key objectives

In order to fulfil the aim of this project, the following objectives have been ascribed:

1. Determine an estimate for the amount of power (in Wh) that could be supplied to meet instantaneous demand by storing surplus electricity from renewable sources due to curtailment, insufficient demand etc. To be achieved by:
    a. Forecast the renewable energy generation capacity in 2035.
    b. Determine a required capacity based on a likely efficiency for production and storage of surplus electricity as hydrogen.
    c. Present reasonable control systems for the production and expenditure of stored hydrogen.
    d. Simulate (over a period of a year) the operation of a hydrogen storage network coupled to UK renewable energy sources to reveal the efficacy of the system.
2. Perform a high-level comparison of the carbon dioxide emissions resultant of a scenario with no HES and of the fully commissioned H_2 storage scenario modelled.
3. Present three simple examples of HES network solutions for the UK based on existing and/or expected RES locations.
4. Discuss potential risks, downsides, and challenges in the process of designing, developing and operating an HES network.
    a. Discuss risks associated with construction and operation to function and personnel.
    b. Present an evaluative comparison with other potential energy storage mechanisms and technologies.
    c. Suggest key challenges to be overcome in establishing an effective HES network in the UK.

4 Literature Review

The following section will divulge on some key information, data and studies pertaining to hydrogen energy storage and the use of renewable energy in the UK power grid. The information presented will form the basis of calculations and conclusions hereafter, and thus, this section should be referred to for further reading, verification and repetition unless otherwise stated.

4.1 Hydrogen technologies

Zhang, et al. [11] give a thorough overview of existing, advancing, and potential hydrogen related technologies in a frequently cited report. Attention is drawn upon critical decisions for design of a water-electrolysis based hydrogen production, generation, and storage system:

4.1.1 Type of water electrolysis:

Of discussed electrolysis technologies, the polymer-electrolyte membrane (PEM) method is the logical choice for integration with RES in the UK. This is due to shorter start-up times than alkaline electrolysers [12] - an essential property for limiting curtailment losses. Further, whilst alkaline systems are more mature, the global installation rate of PEM electrolysers is much greater than that of alkaline [13] and thus the technology should be seen to have already achieved a greater manufacturing capacity and higher investment.

4.1.2 Storage mechanism:

There are currently numerous documented hydrogen storage mechanisms including compressed gas storage; cryogenic liquified hydrogen storage; solid state storage (storage within another, solid material) [11]. Of these, compressed gas is the most mature, more efficient than liquid storage due to the cooling process [14], and suitable for stationary hydrogen delivery as was demonstrated by 80% of operational hydrogen fuelling stations around the world in 2010 [15].

4.1.3 Hydrogen power generation

Zhang, et al. [11] present numerous fuel cell technologies of which PEMFC is the most closely related to the aforementioned electrolyser technology, allowing for synergy in manufacturing at production sites in the UK. The report concludes with a discussion considering the uses of green hydrogen (produced with power from RES), though the discussion is centred around methanation – a reaction with CO_2 to produce methane. This project, however, will consider the direct expenditure of green hydrogen through PEMFC to produce electricity.

Well documented mathematical properties, relationships and behaviours associated with power to hydrogen systems along with component capacity constraints (e.g. thermal) have been analysed for a case study of a Scottish project; one of many existing demonstration systems [16].

Wang, et al. [17] present an optimised system incorporating multiple power generation and storage components in an arbitrary, microgrid location. The paper provides a detailed account of the management of a multiple technology system; however, the model is not centred around the UK.

4.2 Definition of RES

Given that this report is focussed on the potential of HES to facilitate the UK government's goal of a net zero power grid, the primary energy sources considered and modelled will be those with zero or close to zero emissions. For the purposes of a clear definition, Renewable Energy Sources (RES) will be defined as those with produce zero direct emissions during their operation.

Figure 2 presents data on all major electricity production technologies [18]. This information will educate the selection of RES.

Table A.III.2 | Emissions of selected electricity supply technologies (gCO_2eq/kWh)[i]

Options	Direct emissions	Infrastructure & supply chain emissions	Biogenic CO_2 emissions and albedo effect	Methane emissions	Lifecycle emissions (incl. albedo effect)
	Min/Median/Max	Typical values			Min/Median/Max
Currently Commercially Available Technologies					
Coal—PC	670/760/870	*9.6*	*0*	*47*	**740/820/910**
Gas—Combined Cycle	350/370/490	*1.6*	*0*	*91*	**410/490/650**
Biomass—cofiring	n.a. [ii]	–	–	–	**620/740/890**[iii]
Biomass—dedicated	n.a. [ii]	*210*	*27*	*0*	**130/230/420**[iv]
Geothermal	0	*45*	*0*	*0*	**6.0/38/79**
Hydropower	0	*19*	*0*	*88*	**1.0/24/2200**
Nuclear	0	*18*	*0*	*0*	**3.7/12/110**
Concentrated Solar Power	0	*29*	*0*	*0*	**8.8/27/63**
Solar PV—rooftop	0	*42*	*0*	*0*	**26/41/60**
Solar PV—utility	0	*66*	*0*	*0*	**18/48/180**
Wind onshore	0	*15*	*0*	*0*	**7.0/11/56**
Wind offshore	0	*17*	*0*	*0*	**8.0/12/35**
Pre-commercial Technologies					
CCS—Coal—Oxyfuel	14/76/110	*17*	*0*	*67*	**100/160/200**
CCS—Coal—PC	95/120/140	*28*	*0*	*68*	**190/220/250**
CCS—Coal—IGCC	100/120/150	*9.9*	*0*	*62*	**170/200/230**
CCS—Gas—Combined Cycle	30/57/98	*8.9*	*0*	*110*	**94/170/340**
Ocean	0	*17*	*0*	*0*	**5.6/17/28**

Notes:

i For a comprehensive discussion of methodological issues and underlying literature sources see Annex II, Section A.II.9.3. Note that input data are included in normal font type, output data resulting from data conversions are bolded, and intermediate outputs are italicized.

ii Direct emissions from biomass combustion at the power plant are positive and significant, but should be seen in connection with the CO_2 absorbed by growing plants. They can be derived from the chemical carbon content of biomass and the power plant efficiency. For a comprehensive discussion see Chapter 11, Section 11.13. For co-firing, carbon content of coal and relative fuel shares need to be considered.

iii Indirect emissions for co-firing are based on relative fuel shares of biomass from dedicated energy crops and residues (5-20%) and coal (80-95%).

iv Lifecycle emissions from biomass are for dedicated energy crops and crop residues. Lifecycle emissions of electricity based on other types of biomass are given in Chapter 7, Figure 7.6. For a comprehensive discussion see Chapter 11, Section 11.13.4. For a description of methodological issues see Annex II of this report.

Figure 2 – Emissions of Selected Electricity Supply Technologies (gCO_2e / kWh) [18]

Based on the information provided, energy sources categorised as RES will be limited to:

- Geothermal
- Hydropower
- Nuclear
- Concentrated Solar Power
- Solar Photovoltaics
- Wind (onshore and offshore)
- Tidal and shoreline wave power (not included in Figure 2)

The information given in this table will also support high-level estimations of emissions resulting of various power system configurations in 2035.

4.3 RES power output

The power outputs of some RES are dependent on atmospheric conditions, which are highly variable. For the case of solar PV, the amount of sunlight incident on a solar panel can vary depending on the time of day, weather, and the seasonal position of the sun in the sky. Similarly, the output of a wind turbine is affected by the speed and direction of the wind.

Other RES, such as hydropower from natural flow, may also vary seasonally and with environmental changes. However, the relative variation is assumed to be negligible when compared to the variability of atmospherically dependent RES (i.e., wind and solar PV). Hydropower plants can be affected by droughts or floods, which can cause changes in the amount of water available to generate power. However, these changes occur over a longer timescale than the variations in wind and solar PV output.

In summary, the variability of wind and solar PV output is significant and dependent on instantaneous atmospheric conditions, while the variability of other RES is assumed to be relatively constant. This distinction is important when designing and implementing energy systems that rely on renewable energy sources. For the purposes of this project, only wind and solar PV based RES will be considered to vary. The means by which they vary are discussed in the following sections.

4.3.1 Wind power

The UK, as an island neighbouring the Atlantic Ocean, has an excellent wind power resource. Wind power has been the leading form of RES in the UK since overtaking nuclear power in 2017 (considering installed capacity) [19].

Wind power is obtained via the conversion of the kinetic energy from wind incident on a system of aerodynamic blades which generate a lift force perpendicular to the wind direction. This force is used to drive a generator. The total power available to a turbine in wind travelling at a wind speed, v, is summarised in Equation 1.

Equation 1: Available Wind Power

$$P_{\max} = \frac{1}{2}\,\rho\,A\,v^3$$

($\rho =$ density of air, $A =$ swept area of turbine blades)

The real power output of modern wind turbines is equal to the product of available wind power, $P_{\max}$, the capacity factor (or load factor), and another factor introduced through the use of a control system. Control systems used to maintain acceptable loading on the components whilst achieving sufficient output, often achieved by rotating elements of the turbine such as the rotor itself and the blades. The capacity factor of a wind turbine generator is limited by Betz's law which states that the maximum kinetic energy of a fluid that can be extracted by an infinitely thin film rotor is equal to 16/27 (59.3%) of the total energy of the fluid [20].

4.3.2 Solar photovoltaics

Solar Photovoltaics rely on the incidence of solar energy (photons) on electrons within a semi-conductor material. The energy of an incident photon can cause an electron to jump to a higher valency (energy level) on collision and induce an electric current in the material [21]. The availability of solar energy for power generation is measured by solar irradiance (units $W.m^{-2}$) [22].

4.3.1 Environmental data

NASA's Prediction of Worldwide Energy Resources (POWER) database is a publicly available dataset providing an archive of historic global meteorological data [23]. The database is constructed using ground-based measurements, satellite observations, and weather model calculations. The data contained within the POWER database includes surface meteorology, solar irradiance, and climate model outputs, among other variables. The database contains useful information on both solar irradiance and wind speed at various altitudes which will be used for assessing and modelling relevant environmental factors for use in the simulation model.

4.4 UK Power Landscape Data

The UK power network is a large and complicated system, encompassing the assets and operations of many companies and local authorities. Wherever possible, official UK government energy statistics have been utilised. The UK government provides numerous regularly updated databases of energy statistics of which many utilised have been verified as *National Statistics*. This designation requires that the data and methodology employed to produce figures is of the highest quality. The major sources of energy statistics used in this project are identified in this section.

The Renewable Energy Planning Database, developed and maintained by BEIS, provides a comprehensive summary of renewable energy projects with undergoing or granted planning permission requests in the UK [24]. Included are details such as the location, technology used, and capacity of the installation. The database aims to increase transparency in the planning process for renewable energy projects and serves as a useful resource for studying the current state of renewable energy in the UK.

BEIS produce an annually updated Digest of UK Energy Statistics (DUKES), containing a number of tables pertaining to various aspects of power infrastructure, its use and power transmission. Two particular tables containing the annual installed plant capacity by energy source type from 1990 through 2021 (Table 5.7 [25]) and by grid connection (Table 5.12 [19]) were of particular use as an extension to the data available in the REPD.

Finally, the long-term trends dataset [26], containing data recorded between 1970 and 2020 is a broad reaching dataset containing data on the production, consumption and numerous other metrics regarding the UK power system.

4.4.1 Demand

Electricity is a vital component for almost all domestic and industrial tasks in the UK. Figure 3 presents a Sankey diagram illustrating the composition of electricity production and proportional consumption (demand) for each major consumer type in the UK in 2021.

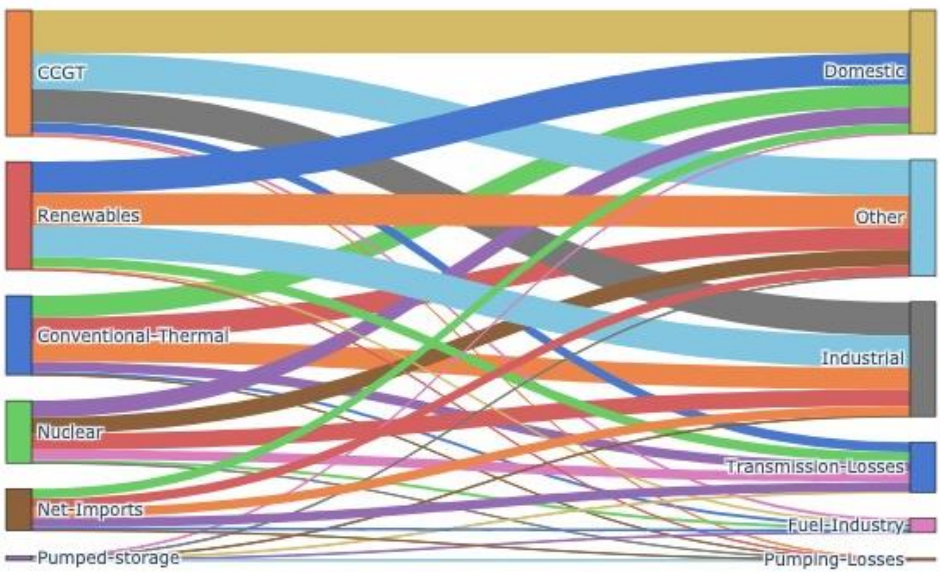

Figure 3 – Sankey Diagram of Electricity Generation and Consumption, UK 2021 [26][1]

[1] Sizes of interconnecting lines are proportioned to the relative sizes of the consumption type, not based on actual composition of power for each consumer.

[redacted]

The data show a roughly equal split between domestic, industrial and other (including retail and commercial, non-industrial consumption) categories. The data also give insight as to the transmission losses and losses associated with pumped storage, discussed further in sections 5.3.4.1 and 5.4.1 respectively.

To understand the changes in power demand over the course of a year, historic data for half-hour periods throughout the year of 2018, available from the National Grid Company [27], was employed.

Finally, the UK government provide annually updated databases detailing the total of the power demand for each local authority over the previous year in Great Britain [28] and Northern Ireland [29]. This data will be used to understand the geographical distribution of power demand in the UK.

4.4.2 Transmission

Data on the transmission system used to transfer electricity from sources to demand points is both more complex and far more challenging to obtain than environmental or power production data. Three companies own and operate the transmission network in Great Britain and another separate company operates the Northern Ireland newtork. The National Grid Company issue a regularly updated dataset containing shapefiles (.shp extension) which includes the geographical shape, names and technical details of all transmission and distribution cables in England and Wales [30]. Similar data may be obtained from both companies operating the Scottish networks through application and completion of relevant data sharing licenses.

4.4.3 Power in 2035

Forecasting the future power landscape is a challenging task and one that must encounter some assumptions. To provide a reasonable estimate without warranting excessive further calculations, it was decided that a single metric be coupled with the installed capacity of RES allowing extrapolation to 2035. The metric chosen was the Levelized cost of Electricity (LCOE).

LCOE is a metric that reflects the full cost of generating electricity over the lifecycle of a power plant, including initial investment capital and lifetime operating costs.

As renewable energy technologies continue to decline in cost [31], the LCOE of RES is becoming increasingly competitive with conventional power sources and driving a shift in the energy mix towards renewables [32]. LCOE has become a key driver for new power plant development decisions, and an important factor in determining the economic viability of new power plant developments [33].

The assumption that LCOE is a reasonable, correlated metric to the installed capacity of RES is based on the idea that cost is a significant factor in energy production decisions. If renewable energy sources have a lower LCOE than traditional fossil fuel sources, it stands to reason that they will be more attractive to investors and will therefore be deployed more widely.

There are issues, however, with relying solely on LCOE as a metric for renewable energy deployment. LCOE does not capture the full range of environmental and social benefits that RES can provide, such as reduced greenhouse gas emissions and increased energy access when using smaller, modular systems. LCOE is a useful metric for understanding the economic viability of renewable energy sources, though it should be considered alongside other factors, such as energy security, environmental impacts, and social benefits, when making decisions about energy development.

For the purposes of estimating the installed capacity of the relevant RES in 2035, this project will assume a relationship of inverse proportion between LCOE and installed capacity pertaining to a particular RES. A record of relevant values for LCOE have been obtained from IRENA [34] and are presented in Figure 33.

In order to improve the extrapolation of LCOE, information regarding the likely future cost of RES is required. Contracts for Difference are the UK Government's primary means for supporting RES projects by protecting operators from volatile prices with a guaranteed wholesale electricity price, known as the strike price [35]. As such, CfDs provide an indication of the future cost of some RES as they are prescribed by the strike price awarded. The outcomes of the first four CfD allocation rounds are publicly available on the UK governments website and provide detail on the awarded strike prices, the RES technologies to which they correspond, and the year in which they come into action. This data is a valuable resource and will be assumed to force the LCOE to meet the strike price at the time they occur.

5 Methodology

This section will introduce the proposed solution to meet the project objectives, their reasoning, and basis in literature, physical or mathematical principles. Methodology presentation occurs in a chronological order with preparatory stages introduced and developed prior to explanation of the full system.

5.1 Solution concept

The suitability of hydrogen for energy storage in the UK power system in 2035 is to be assessed by simulating the operation of the system both with and without an example of an HES system, through the generation of a computer-based model of the system. The model will consider both the quantities of power produced and consumed and the spatial arrangement of these factors over the course of the year 2035.

It is intended that the model power system be evaluated at one-hour intervals to provide regular insight into the power production and demand trends. Through evaluating the system at hourly intervals, the model can provide a detailed view of any residual power production or demand and identify the locations where these occur. This will help in prioritising locations requiring storage.

Furthermore, the model will illustrate how the need for storage may change throughout different periods of time (e.g., daily and monthly). This information will be valuable in determining the most effective and efficient storage solutions that can be deployed to meet the changing needs of the power system.

The decision to use a simulation model also provides a flexible approach that can be refined and extended over time to improve its accuracy and potential. This will allow for ongoing improvements to the model to ensure that it remains relevant and effective in addressing the evolving needs of the UK power system.

Overall, this simulation approach offers a more detailed and accurate view of the power system than basic annual calculations, allowing for more effective planning and deployment of storage solutions to meet the changing needs of the system and the goals of UK Government.

5.2 Program structure

Developing an effective program involves understanding the required outcomes before optimizing the algorithm through which this is achieved by minimizing the computational resources and time required for execution. To do this, it is crucial to plan the structure and objectives of each stage within the program. This process is often aided by the use of flow charts, which allow developers to visualize the structure and logical progression of the program. Flow charts use symbols to represent different program operations, such as inputs, outputs, and decision points, as detailed in Table 1.

Table 1 – Legend for Program Flow Chart

Symbol	Name	Implication
	Start / End Terminus	Program initiation or completion
	Process / Operation	Completion of a function
	Input / Output	Input: Data supplied in expected format Output: Results of recent operation printed to command window or output file
	Decision (Binary)	True or false test of mathematical expression based on known variables

After careful planning and development, a final program flow chart was selected for the construction of the simulation model. This flow chart, which is presented in Figure 4, illustrates the logical progression of the program and the connections between different stages. The final flow chart reflects the thorough planning and optimization efforts that were undertaken to develop an efficient and effective simulation model. With this flow chart as a guide, the program can be constructed and executed effectively.

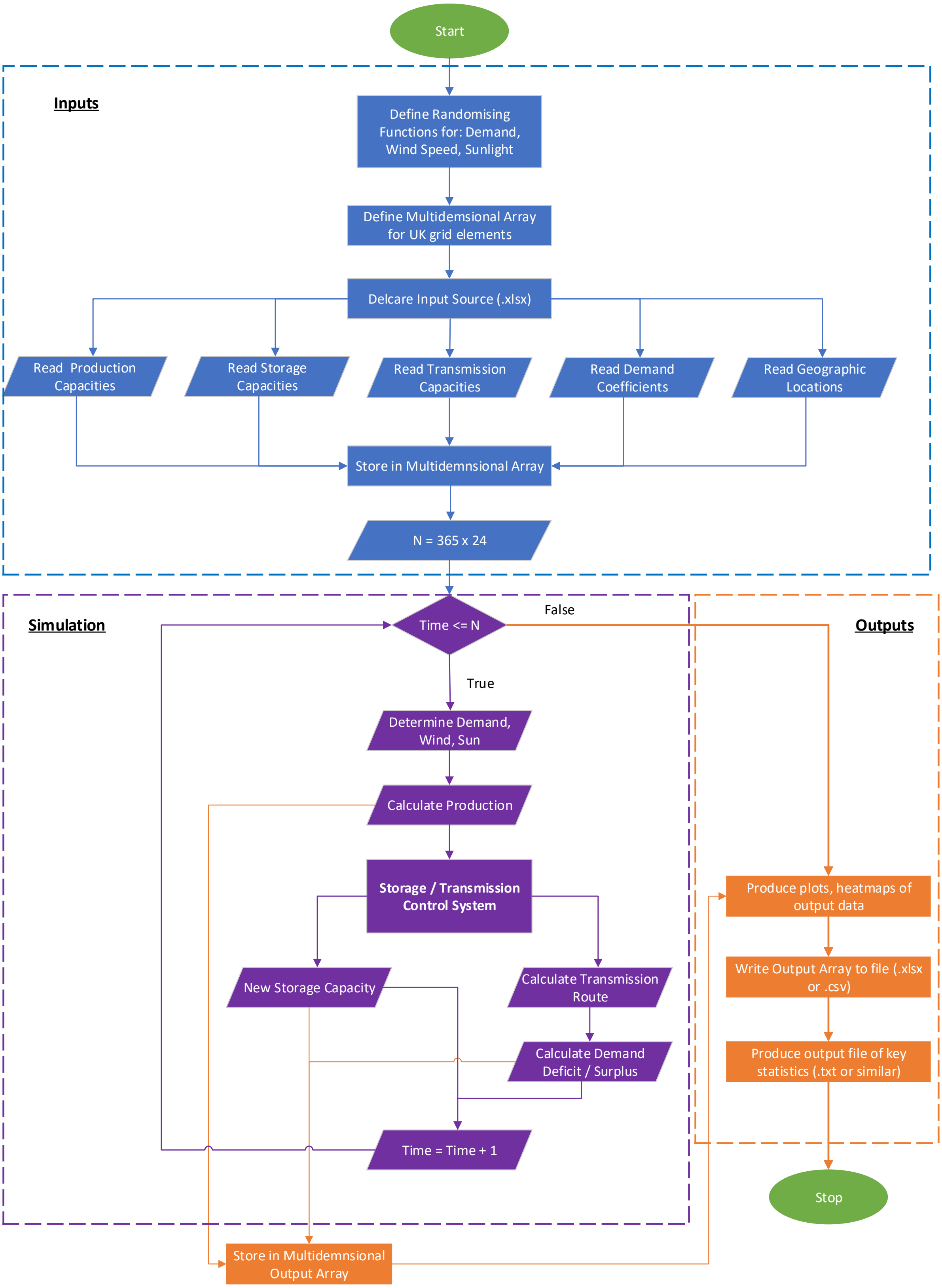

Figure 4 – Program Flow Chart

5.3 Understanding power in the UK

A major preparatory stage to simulating the operation of the UK power system is understanding and modelling the current power landscape. This project will consider three major factors making up the power system: power production, transmission, and demand (consumption).

For the simulation to represent the UK power system as a national network, the spatial distribution of its factors must be captured. Data sources have, therefore, been selected as those with the greatest detail and quantity of power system components.

This section will detail the procedure via which information regarding the current UK power system have been obtained, extracted, and manipulated for use in further calculations.

5.3.1 Power production

The UK relies on numerous sources for its power supply. Figure 5 presents the electricity generation sources used in the UK in 2021.

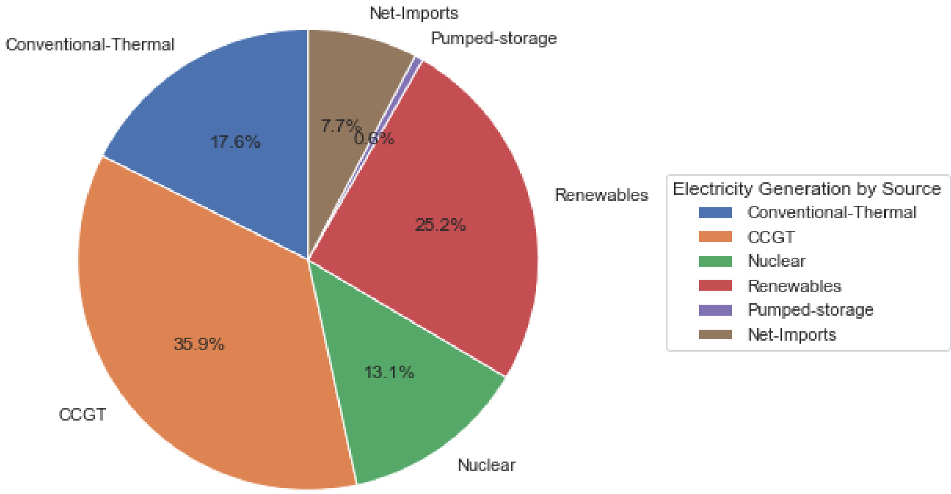

Figure 5 – Pie Chart of 2021 Power Generation Sources, UK [26]

A large shift from the current power production system, of which under 40% is currently from UK based RES, must be achieved before 2035 to meet the UK Government's net-zero target. Some power contribution will likely originate from neighbouring, interconnected countries (as apparent in 2021), however, <u>this model will analyse a scenario in which the UK is entirely self-sufficient and imports no power</u>. The remaining 60% (in the case of 2021) must therefore be sourced from new or optimised RES capacity.

The UK government provides a regularly updated database (Renewable Energy Planning Database, REPD) of existing RES plants, their locations, capacities and types [24]. The same information is also available for proposed, planned, and abandoned plants. These data are highly informative of the distributions of the various RES installed and planned in the UK.

A simple datum conversion from Ordinance Survey British National Grid 36 (OSBNG36) to WGS84 coordinates allows for geospatial plotting. Figure 6 presents these data on a map of the UK.

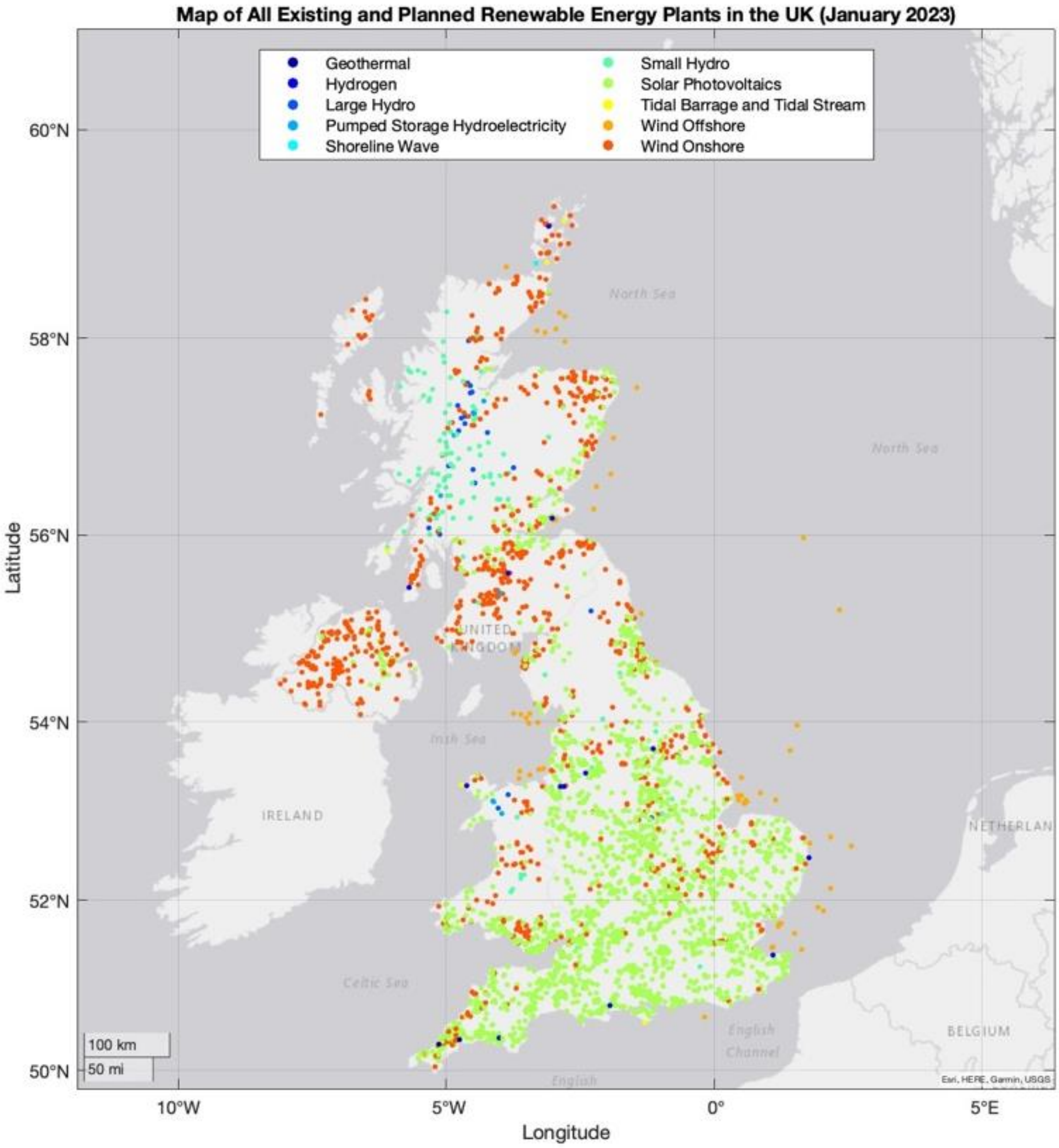

Figure 6 – Map of All RES Locations, UK [24]

A similar map showing the relative magnitudes of the installed capacity of each RES plant is given in Figure 23 in Appendix A – Figures.

The vast number of plants and their variety in capacity and type presents a challenge for corresponding calculations. It was therefore decided that the RES plant distribution be simplified by summing and grouping the planned and installed capacities of each power technology type within regions of a fixed size. These regions, herein referred to as nodes, can then be treated as single plants. This methodology allows for simplification of all three power system factors and easier visualisation of the analysis. This methodology is discussed further in section 5.3.1.

5.3.1.1 Nuclear power in the UK

Nuclear power, pertaining to far higher capacity per development and fewer total plants, is less complex. A government special feature details the full history and current future of nuclear plants in the UK, their capacities and service dates [36]. This information is shown in **Figure** 7.

Power Station	**Opening Date**	**Closure Date**	**Installed Capacity (MW)**	**Current Status**	**Reactor Type**
Calder Hall	1956	2003	220	Closed	Magnox
Chapelcross	1959	2004	196	Closed	Magnox
Berkeley	1962	1989	276	Closed	Magnox
Bradwell	1962	2002	242	Closed	Magnox
Hunterston A	1964	1989	180	Closed	Magnox
Dungeness A	1965	2006	450	Closed	Magnox
Trawsfynydd	1965	1991	470	Closed	Magnox
Hinkley Point A	1965	2000	500	Closed	Magnox
Sizewell A	1966	2006	420	Closed	Magnox
Oldbury	1967	2012	434	Closed	Magnox
Wylfa	1971	2015	980	Closed	Magnox
Hinkley Point B	1976	2023	1061	Operational	AGR
Hunterston B	1976	2023	1074	Operational	AGR
Hartlepool	1983	2024	1207	Operational	AGR
Heysham I	1983	2024	1179	Operational	AGR
Dungeness B	1983	2028	1120	Operational	AGR
Heysham II	1988	2030	1254	Operational	AGR
Torness	1988	2030	1250	Operational	AGR
Sizewell B	1995	2035[2]	1216	Operational	PWR
Hinkley Point C 1	2025	2086	1630	In construction	EPR
Hinkley Point C 2	2026	2087	1630	In construction	EPR
Sizewell C	*2030 - 2035*	*2090 - 2095*	*3340*	Proposed	Hualong One
Bradwell B	*2030 - 2035*	*2090 - 2095*	*2300*	Proposed	Hualong One

Figure 7 – Nuclear Power in the UK (1956 – 2035) [36]

The data show that only two plants intended to be operational in 2035 are currently under construction and a further two are proposed. This project will assume that both proposed nuclear reactors enter full operational service prior to 2035 and are thus to be included in the model. The existing Sizewell B reactor will also be included in the model as the site owner has applied to extend its service tenure to 2055 [36]. The total installed nuclear power production capacity to be included in the model is therefore 10.1 GW.

5.3.1 Model geometry

Various methods and levels of simplification were considered before opting for a grid of nodes spaced at 25km intervals over the area of the UK. The geographical geometry selected for implementation in the model is based on the geometry provided for a series of gridded meteorological datasets produced by the Met Office [37]. Each node is centred on a 625km^2 area into which all properties pertaining to the region are grouped.

Areas of this size were selected to give clear geographical differentiation whilst allowing for relatively simple calculations and the possibility for manual adjustments, if required.

Figure 8 shows an example of the 25km grid.

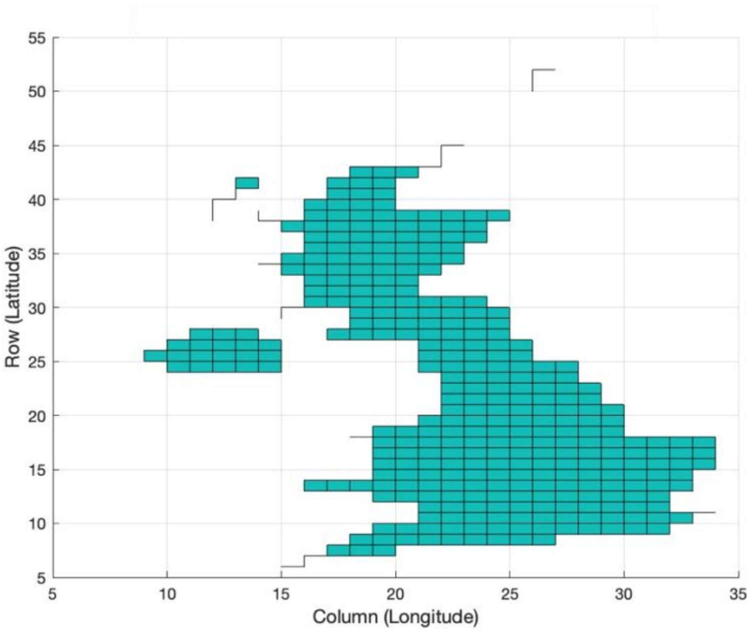

Figure 8 – Geometry [37]

The full area is encompassed by a 39 by 52 grid containing a total of 2028 nodes of which 436 represent areas of UK land.

By calculating the matrixial indices of the nearest node to the coordinates of an RES plant, the properties of the RES plant can be associated with a regional node on the grid. Iterating this process for all plants allows for the compilation of a simplified distribution of installed RES capacity for each technology type.

An example of the grouped capacity of RES plants, for Solar PV capacity, is given in Figure 9, with the function used for grouping power plants presented in Appendix B – MATLAB Scripts

Charts showing the grouped installed capacity, similar to Figure 9, for each RES type are shown in Figure 24 through to Figure 32 in Appendix A – Figures.

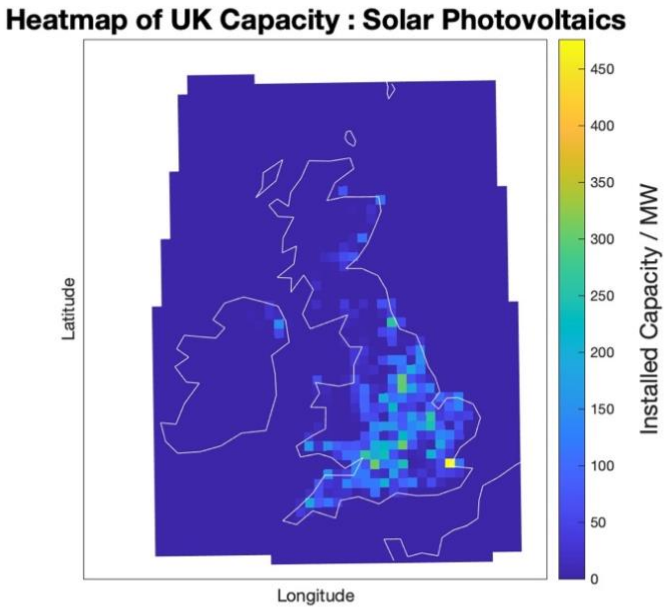

Figure 9 – Example of Grouped Installed Capacity, Solar PV [24]

[redacted]

5.3.2 Environmentally dependent RES

The output of some RES is dependent on instantaneous environmental conditions. This model will consider environmental variability of wind and solar PV output only – for further discussion, see section 4.3.

5.3.2.1 Wind power

For the purposes of the simulation described herein, the calculation of wind turbine output will be based on three components:

i) The installed capacity within a region. Taken to represent the maximum available wind power for all wind turbines within a region (e.g., the unfactored maximum output of wind turbine generators, neglecting Betz's coefficient).
ii) The average capacity factor of the installed wind turbine generators within a region. The product of the capacity factor and the installed capacity within a region is equal to the maximum actual output of the wind turbine generators in the region, thus the capacity factor must not exceed Betz's coefficient.
iii) The ratio of the wind speed over the rated wind speeds for maximum output. This ratio is to be equal to unity for all wind speeds in the rated wind speed range and zero for wind speeds above the rated wind speed range. Values of zero represent scenarios in which the wind turbine generators are configured in survival mode, withstanding high wind speed whilst producing no output.

The choice of rated wind speeds for all UK turbines is based on a review of available literature [38, 39] and results in minimum and maximum rated wind speeds of $14\ \mathrm{ms}^{-1}$ and $25\ \mathrm{ms}^{-1}$, respectively. The linear power curve resultant of component iii) is also justified by empirical results showing a largely linear profile due to the control system used [38, 39].

5.3.2.2 Wind speed

To enable accurate representation of wind power production, simulating a realistic variation in wind speeds across the UK is key. Using NASA's POWER database, wind speed data recorded between each day in 2020 through to the end of 2022 has been analysed [23]. The average wind speed for a 20 by 20 grid of locations is presented in Figure 10.

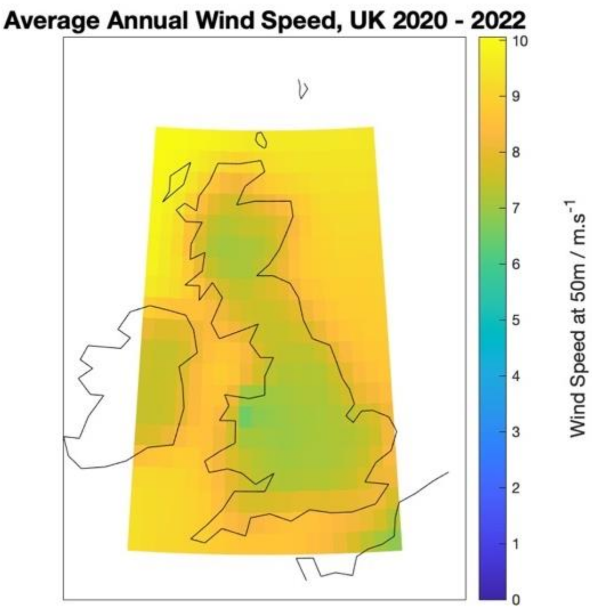

Figure 10 – Annual Average Wind Speed [23]

This, however, does not provide any detail on the range of wind speeds apparent at the locations. A histogram detailing the frequency of the full range of recorded wind speeds is presented in Figure 11.

[redacted]

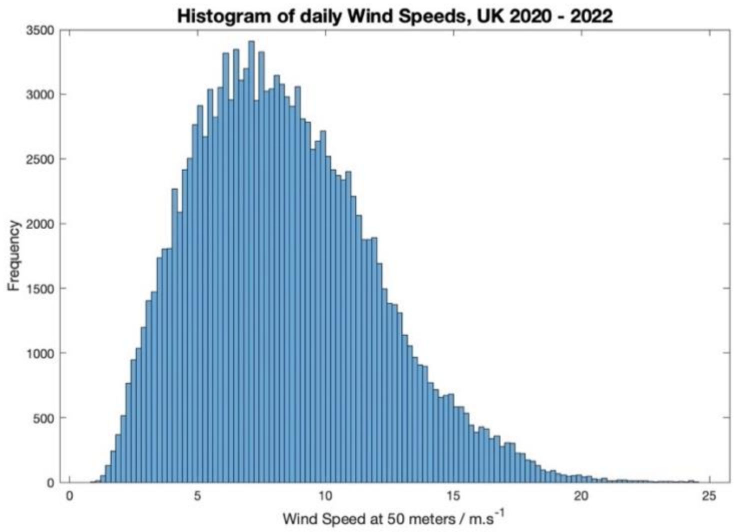

Figure 11 – Histogram, UK Wind Speeds

These data can be used to determine a reasonable and corresponding probability distribution function for the accurate simulation of day-to-day wind speed variation.

The Weibull distribution is a flexible distribution function often used for fitting recorded data to a probability function and has been frequently used for studies modelling wind power [40-42]. The two parameter Weibull distribution function is surmised in Equation 2 [42]:

Equation 2:
Weibull Distribution Function

$$W(v) = \frac{kv^{k-1}}{C^2} e^{-\frac{v^k}{C}}$$

Using MATLAB and it's *wblfit* function, shape (k) and scale (C) factors for each month and at each of the 20 by 20 grid points were determined from the recorded wind speed data (see Appendix B – MATLAB Scripts for the full code). This information was then grouped, as with the RES capacities in section 5.3.1, to match the selected model geometry. The data were then ready for use in generating random, correctly distributed wind speeds for all locations at each interval of the simulation.

5.3.2.3 Solar photovoltaics

The solar power generated at each interval of the model is equal to the product of the following three variables:

i) The installed capacity of solar PV panels within a region. Taken to represent the maximum, unfactored solar power available to the installed solar PV panels.
ii) The average capacity factor of the installed solar PV panels.
iii) The ratio of the apparent, simulated solar irradiance over the average rated irradiance for the solar PV panels. This ratio is to be equal to unity for all values of solar irradiance equal to and in excess of the rated value.

A rated solar irradiance of 0.8 kWh m^{-2} per day has been selected based on analysis of the solar resource in the UK [22]

5.3.2.4 Solar irradiance

NASA's POWER database also includes recordings of the daily incident solar irradiation [23] . The data are available and accessible in the same way as discussed for wind speed. The average solar irradiance data for the UK in 2020 through 2022 is presented in Figure 12. The chart shows a decreasing solar irradiance at higher latitudes in the UK. This lower solar resource is one reason why existing solar power plants are concentrated in southern regions of the UK, along with a greater population density; see Figure 6 and Figure 9.

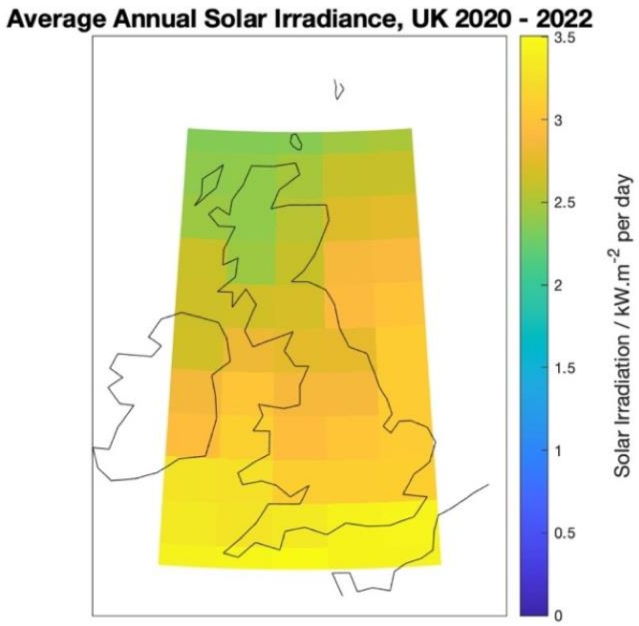

Figure 12 – Annual Average Solar Irradiation [23]

As with the wind speed data, a suitable probability distribution for each node must be determined from the solar irradiance data. To verify the use of the Weibull distribution, the monthly frequency distribution of solar irradiation recorded between 2020 and 2022 was inspected (presented in Figure 13).

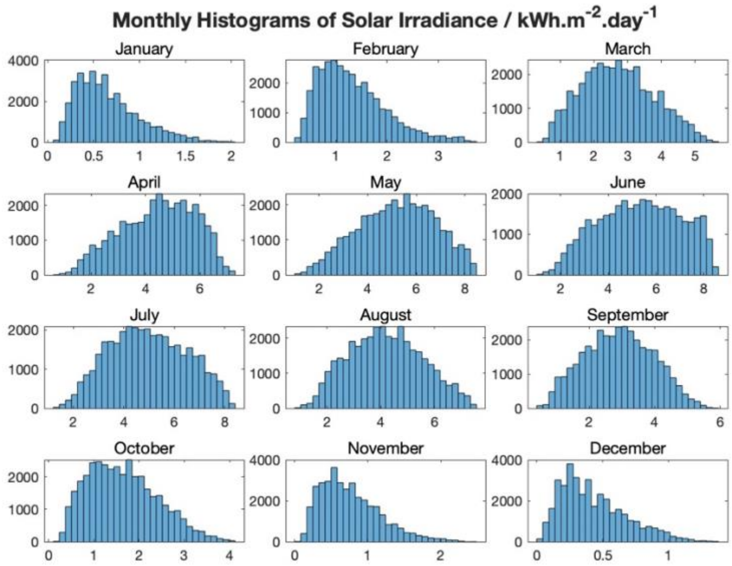

Figure 13 – Monthly Solar Irradiation Histograms

The distribution changes drastically between months and thus, a separate Weibull distribution function is required for each month as with the wind speed analysis.

Furthermore, the solar irradiance apparent at each point of the day is also dependent on the time (position of the sun), see Figure 14. This pattern is to be captured in the simulation by means of a 12 by 24 matrix of the ratios equal to the average solar irradiation at each hour of the day, and in each month, over the maximum hourly solar irradiation recorded during the year 2021. The use of this matrix ensures that the daily profile of solar power availability reflects the true data, with zero energy available for solar PV during night-time hours.

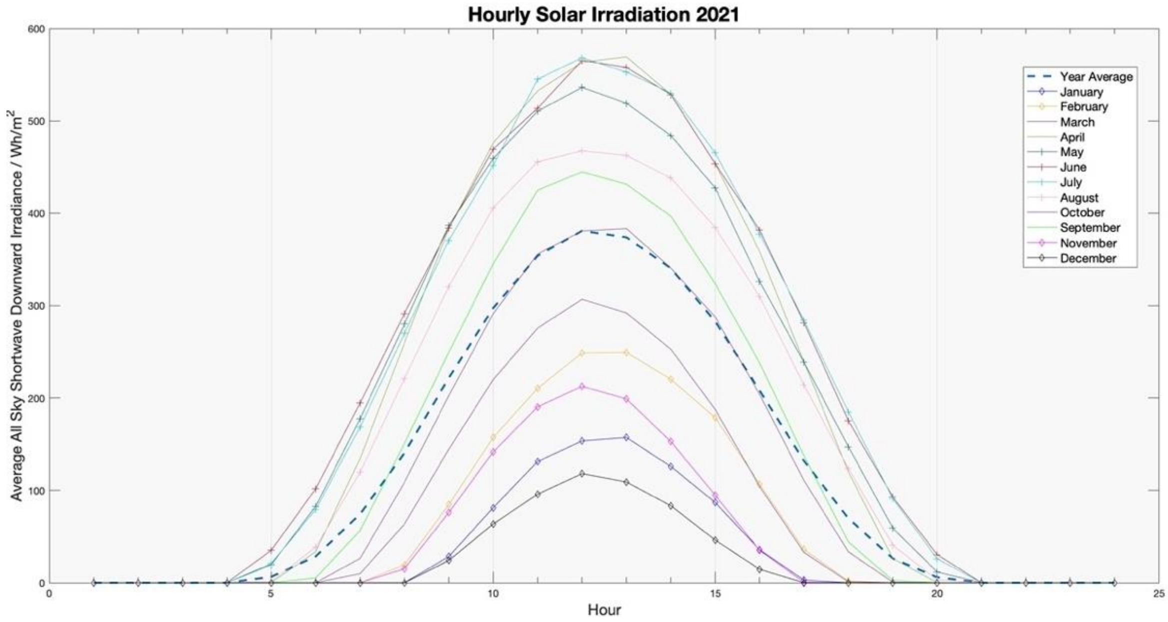

Figure 14 – Hourly Solar Irradiation Variation, UK 2021

5.3.3 Demand

To simulate a realistic variation in demand at each time interval, publicly available data on historic demand values was obtained from the National Grid Company [27]. The data provide a log of actual, total power demand at half-hour intervals throughout the full year of 2018. Averaging these data into one-hour intervals for each month results in the data presented in Figure 15.

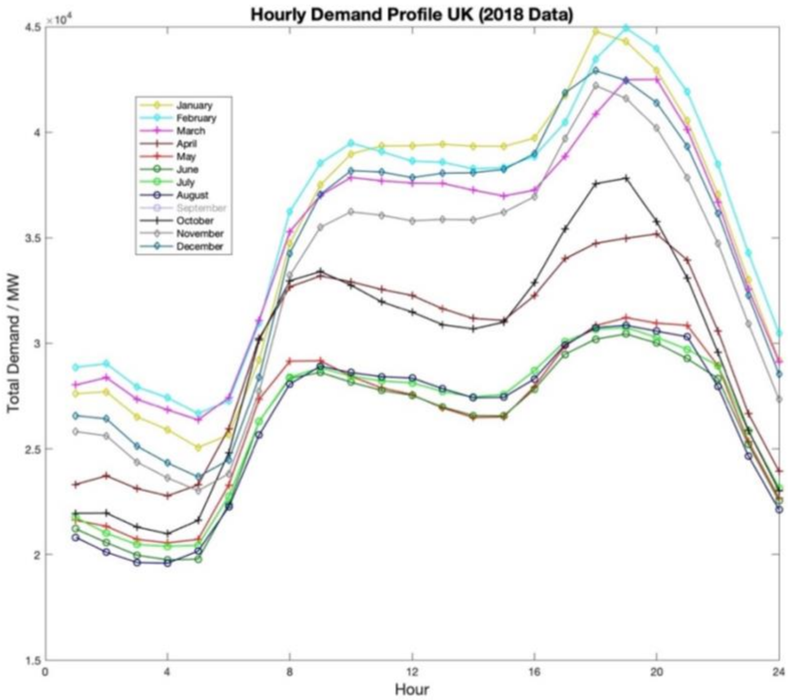

Figure 15 – Monthly Average Hourly Demand Profile, UK 2018 [27]

The data show a consistent daily profile with two peaks in power demand: one at around 08:00 and another between 18:00 and 20:00. Another pattern in the data is that demand is generally higher in the winter months.

Given the significant difference in magnitude of power demand between months, the demand variables considered in the simulations will be equal to those in Figure 15. The power demand profile will be assumed to match the monthly and hourly proportions identified using this data.

When considering the regional and spatial distribution of power demand, the UK government's information on subnational consumption of power for each year is particularly useful. The information contains a breakdown and total of the power demand for each local authority evaluated for a full year in Great Britain [28] and Northern Ireland [29].

To group these data onto the model geometry, the coordinate boundaries of each local authority were identified (using an Office for National Statistics database [43]) and the ratio of each local authority's annual power demand was associated with the relevant node. The resulting distribution of power demand across the UK is presented in Figure 16.

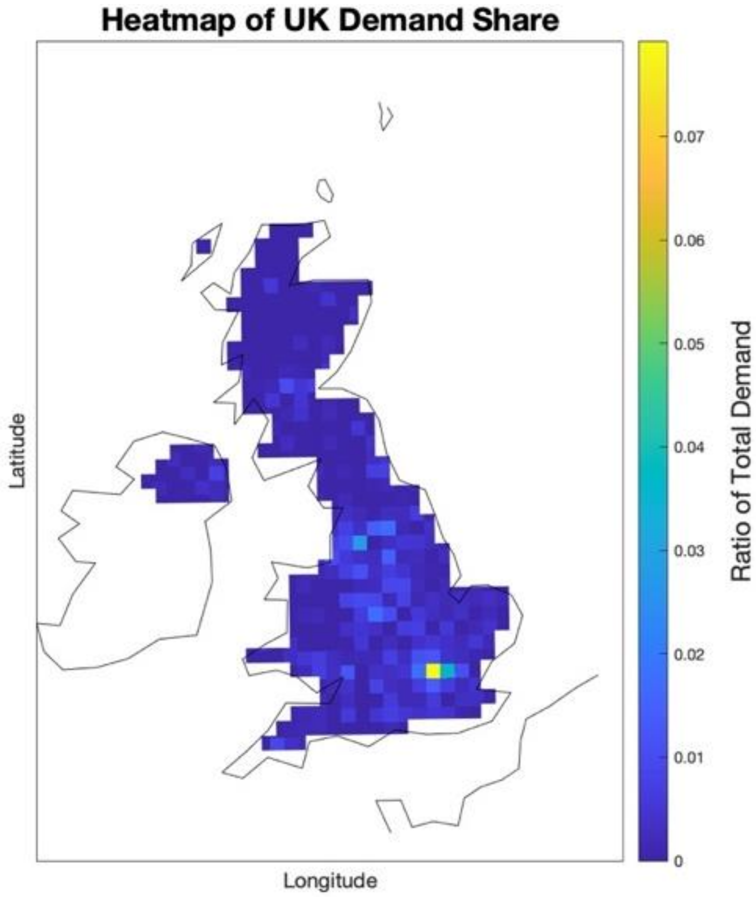

Figure 16 – Grouped Demand Ratios [28]

5.3.4 Transmission

With the UK transmission data obtained (see section 4.4.2 for details), the data were extracted using MATLAB. MATLAB's mapping toolbox contains various functions capable of handling data stored in shapefile format. Figure 17 is a map showing the transmission data obtained during this process. It was not possible to obtain data for Northern Ireland.

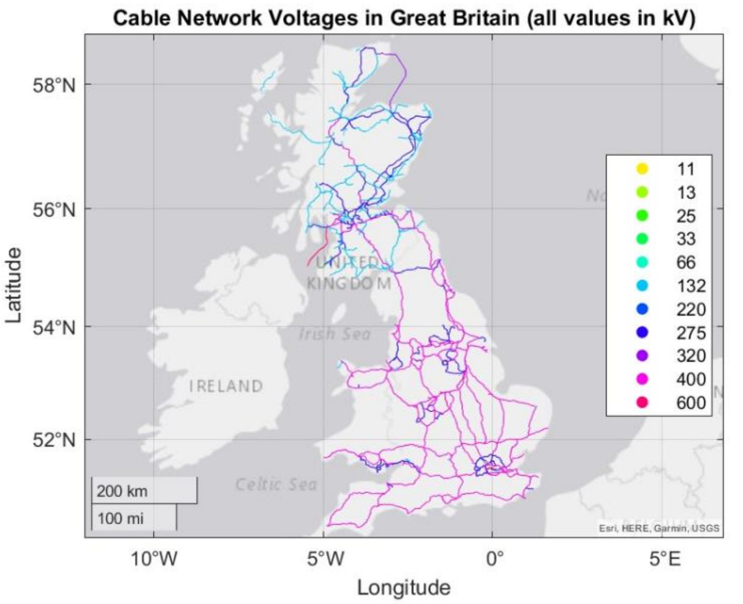

Figure 17 – Map of All Transmission Cables and Lines in Great Britain

It was intended that the transmission data obtained be decomposed into the nodal structure presented in section 5.3.1 of this report. This would allow for simplified simulation of power network balancing through the transmission network. Figure 18 presents the transmission data decomposed in the model geometry. Points marked with a circle are those where a transmission cable begins or ends with the black lines marking connections between nodes.

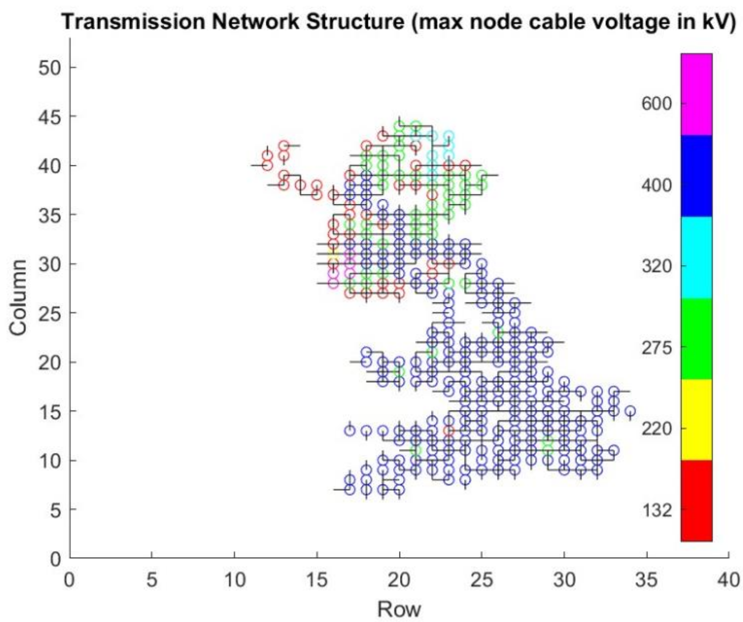

Figure 18 – Transmission Network Structure

The data presented in the figure are stored in a 2028 by 2028 matrix. Each element of the matrix represents a connection between two nodes on the grid. Only those elements which represent a real connection are assigned a value (equal to the voltage of the connecting cable) meaning it can be used for efficient analysis of complex transmission problems.

Iterations of the power system simulation may then involve a power balancing stage using the actual transmission structure identified through this exercise. This stage was found to be beyond the scope and time available for this project and is therefore discussed further in the Future Work section of this report.

5.3.4.1 Transmission Losses

Analysing the UK governments annual data on the sources and consumption of power between 1990 and 2021 reveals a consistent average of 8% of available power expended as losses during transmission [26]. This value is to be applied to all power produced during the simulation to account for any required transmission, in lieu of more complex analysis alluded to in in the Future Work section of this report, for the aforementioned reasons.

5.3.5 Data Extraction and Handling

Three main filetypes have been encountered during the preparatory analyses detailed in this section; spreadsheets, NetCDF4 files, and Shapefiles. This short section provides some key information regarding the extraction and manipulation of these file types using MATLAB.

5.3.5.1 Spreadsheets (.xls, .xlsx, .csv)

MATLAB provides the functions *readtable* and *readmatrix* to process spreadsheet data. *readtable* is used to read tabular data from spreadsheets in formats such as .xls, .xlsx, and .csv. The function returns a table object that contains the data in the file. *readmatrix*, on the other hand, reads numeric data from spreadsheets in formats such as .xls and .xlsx and returns a matrix.

5.3.5.2 NetCDF4 (.nc)

The NetCDF4 file format is commonly used in the scientific community to store large datasets and was encountered for numerous sources of environmental data. MATLAB provides the functions *ncdisp* and *ncread* to process NetCDF4 files. *ncdisp* is used to display the structure of a NetCDF4 file, including its dimensions, variables, and attributes. *ncread* is then used to read data from a NetCDF4 file. The function returns the data in a MATLAB array or structure, depending on the structure of the file.

5.3.5.3 Shapefiles (.shp, .shx)

Shapefiles are commonly used in Geographic Information Systems (GIS) to store spatial data. MATLAB provides the function *readgeotable* to process shapefiles. *readgeotable* reads the geometry and any other attribute data from a shapefile and returns a table containing both types of data. The function also sets the coordinate reference system (CRS) of the data, making it possible to translate, plot or perform calculations on the coordinate data within MATLAB.

5.4 Forecasting the power system of 2035

Following completion of data extraction and manipulation tasks, the focus of work moved to the extrapolation of the existing system to that expected in 2035. Given that the landscape of RES in the UK is expected and required to change drastically prior to 2035, the future is naturally uncertain. There is little available information on future RES beyond those planned and recorded in the REPD.

For certain technologies, utility scale plants are so large and complex that it is highly unlikely that developments not currently operational or under the scope of the REPD will be commission prior to 2035. Wave and tidal developments are currently very scarce and are therefore unlikely to make a significant contribution to the power landscape beyond their current influence due to the immaturity of the technology.

Therefore, it will be assumed that any expansion in capacity between now and 2035 for the following technologies is captured by the REPD or other UK Government information (e.g., [36] for Nuclear):

- Nuclear
- Wave and Tidal
- Hydro (Natural Flow)
- Pumped Storage Hydroelectric

For the remaining RES (i.e., wind and solar PV) it remained necessary to make assumptions and extrapolate to a reasonable installed capacity in 2035.

5.4.1 Factors affecting the uptake of RES

To do this, a suitable variable assumed to be correlated to the installed capacity of the RES was identified as levelized cost of electricity (LCOE). For further discussion on the use of LCOE, see section 4.4.3.

For the purposes of estimating the installed capacity of the relevant RES in 2035, this project will assume a relationship of inverse proportion between LCOE and installed capacity pertaining to a particular RES. A record of relevant values for LCOE have been obtained from IRENA [34] and are presented in Figure 33.

5.4.2 Installed Capacity Extrapolation

The calculations employed to extrapolate the installed capacity of a renewable energy source in the UK to a likely value in 2035 was based on an analysis of its levelized cost of electricity (LCOE) and the relationship between the two variables. To improve the accuracy of the future predictions, data on the CfD strike price for years beyond the IRENA data was used. CfD are the UK Government's primary means for supporting RES projects by protecting operators from volatile prices with a guaranteed wholesale electricity price, known as the strike price [35].

The LCOE data is transformed by taking the inverse of each value. A least-squares regression is then used to fit a line to the transformed data. The least-squares regression algorithm is iterated 20 times with varying settings to find the optimal fit.

The fitted line parameters are used to extrapolate the LCOE values to 2035. The resulting LCOE is then inverted back to its original scale for interpretation. An expression representing the best-fit linear relationship between installed capacity and inverse LCOE is then obtained by a second, least-squares regression. With this expression and the extrapolated LCOE values, an estimate for the installed capacity in 2035 is obtained. An example of the output of this analysis is presented in Figure 19.

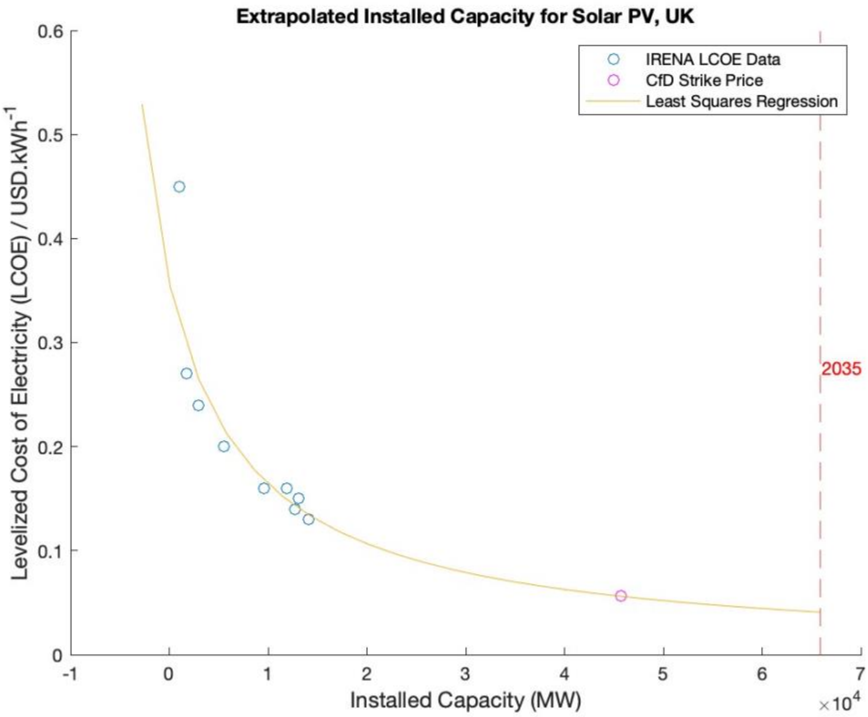

Figure 19 – Example of Extrapolated Installed Capacity, Solar PV

An example of the code used to implement these calculations is provided in Appendix B – MATLAB Scripts along with similar plots for the remaining RES presented in Figure 34 through to Figure 36.

5.4.1 Capacity Factor Extrapolation

The efficiencies or capacity factors for each RES was also considered. For those RES whose capacities were to remain the same as current values, their capacity factors will also remain stable. For those RES expecting increased capacity, however, a similar analysis to extrapolate the capacity factor based on historical values (also IRENA [34], see Figure 37) was performed.

For solar PV, the least-squares optimisation was used fit a negative exponential decay (general form in Equation 3:), representing convergence on a maximum possible capacity factor.

Equation 3: Negative Exponential Decay

$$f(x) = a - b \cdot e^{-c \cdot x}$$
$$a, b, c \in \mathbb{R}^+$$

For both onshore and offshore wind, the asymptote of the negative exponential decay was set to the Betz's coefficient as this is the theoretical, maximum capacity factor. The resulting relationships and extrapolation patterns are presented in Figure 38 through to Figure 40.

Table 2 contains a summary of the full installed capacity and capacity factor data discussed in this section for all RES considered.

Table 2 – Final Installed Capacities and Capacity Factors for 2035

Technology	**Growth Factor (beyond REPD)**	**Final Capacity / MW**	**Capacity Factor**	**Max Output / MW**
Wind Onshore	1.99	40,295.10	0.39	15,715.09
Wind Offshore	1.90	59,595.87	0.48	28,606.02
Large Hydro	2.51[1]	1,179.22	0.27 [2]	324.19
Pumped Storage Hydroelectricity	1.00	5,277.90	0.85 [3]	535.95
Shoreline Wave	1.00	24.00	0.44 [2]	10.57
Small Hydro	2.51[1]	444.99	0.27 [2]	122.34
Solar Photovoltaics	3.71	65,914.47	0.20	13,223.17
Tidal Barrage and Tidal Stream	1.00	377.90	0.44 [2]	166.43
Nuclear	1.00	10,116.00	0.75 [2]	7,592.04
ALL				**70,246.06**

1 Grid connected capacity recorded by BEIS Digest of UK Energy Statistics Table 5.12 exceeds that of the REPD and thus the higher capacity has been selected and corresponding growth factor applied [19]

2 Capacity factors determined via averaging of annual values between 2010 and 2020 from BEIS Digest of UK Energy Statistics Table 5.7 [25]

3 Capacity factor for discharging of 85% based on assumption that newly commission plants will be *state of the art* [44].

Any growth in installed capacity to an RES is applied as a scalar increase to the existing distribution of capacity identified in section 5.3.1.

5.5 Understanding the need for storage

Understanding the storage requirement for the UK in 2035 was achieved by running a preliminary stage of the simulation model including no storage capacity. The intention of this analysis is to reveal nodes which have a persistent, non-zero net power and identify them as regions of power surplus or deficiency.

The model was assembled to read-in the grouped RES capacities, extrapolate them to 2035 values based on the growth factors identified in Table 2 before performing power production calculations based on the methodology outlined in sections 5.3.1 and 5.3.2. Subtracting the demand for each region and applying the transmission loss to the remaining power then allows for a balancing calculation, whereby nodes with a power surplus transfer available power to meet residual demand of other nodes. This process occurs for each of 8,761 one-hour intervals evaluated during the 2035 simulation.

The MATLAB code used to complete this simulation is given in Appendix B – MATLAB Scripts .

Figure 20 presents the resulting net energy of each node for this preliminary simulation.

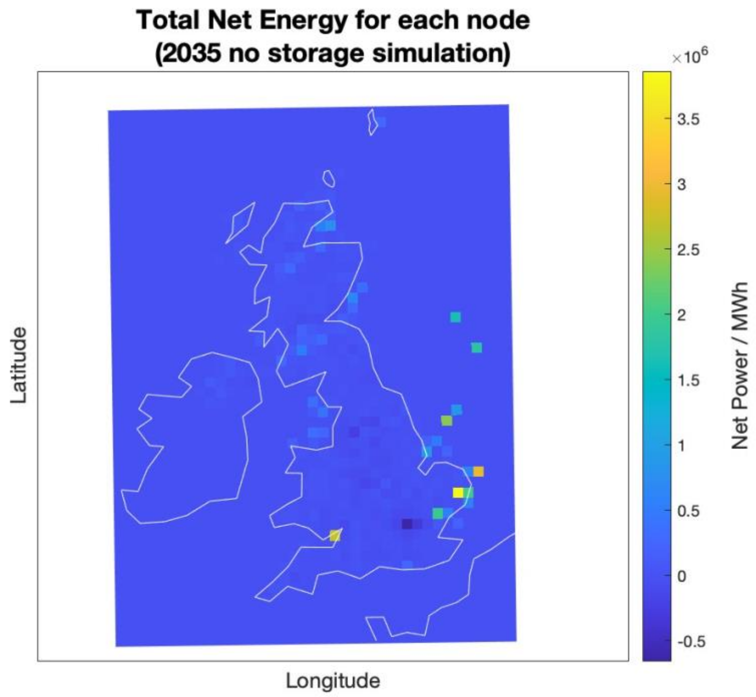

Figure 20 – Net energy locations following simulation with no storage

With this information, along with total net power at each interval obtained during the analysis, a reasonable size and distribution of HES plants can be determined. Graphs presenting the total production and demand, and a breakdown of production from each RES are given in Figure 41 and Figure 42 respectively.

5.5.1 A proposed HES network

To generate an HES network, a charging capacity and a storage capacity must be allocated. For the purposes of this project, HES plants have been located at all locations where there is a net surplus of power (i.e., as in Figure 20). The charging capacity of these plants has been selected based on the maximum instantaneous power surplus, following power balancing, identified during the no-storage simulation. The total storage capacity (in MWh) has been determined by taking the sum of all positive net energy nodes, subtracting any existing storage capacity (e.g., from pumped hydropower) and distributing the remaining storage capacity to each HES node in proportion with its charging capacity.

The resulting HES network distribution is presented in Figure 21.

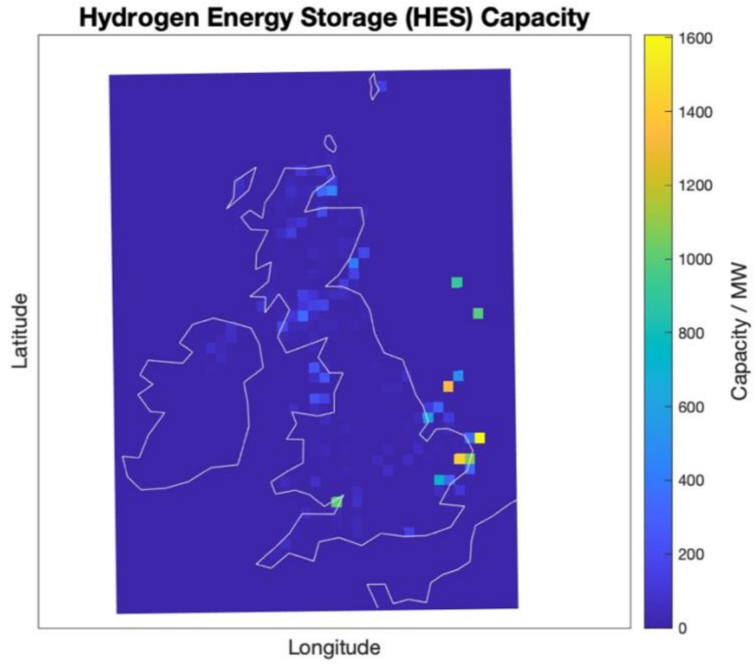

Figure 21 – Proposed HES Production Locations

5.5.2 Energy Storage Capacity

The UK currently has 8 existing or planned utility scale pumped hydropower plants. Details on these plants is presented in Table 3.

Table 3 – Details on Pumped Hydro Storage in the UK

Site Name	Installed Capacity (MW)	Storage Capacity (MWh)	Source
Dinorwig	1728.00	9,100.00	[45]
Ffestiniog	360.00	1,300.00	[45]
Cruachan	440.00	10,000.00	[45]
Foyers	300.00	6,300.00	[45]
Glyn Rhonwy (larger version)	99.90	700.00	[46]
Glenmuckloch Pumped Storage Hydro	400.00	1,600.00	[47]
Coire Glas	1500.00	30,000.00	[48]
Red John Pumped Storage	450.00	2,400.00	[49]

The analysis outlined in section 5.5.1 revealed a total storage requirement of 8.45 TWh and with a total pumped hydro storage capacity of 61.4 GWh, a hydrogen storage capacity of 8.39 TWh is to be modelled. This value is within the range predicted as required for the UK in 2035 in a report commission by the UK government [50]. Given the distributed nature of the proposed HES network, the maximum storage capacity of a single node is 640 GWh, achievable even for a single storage plant when considering salt cavern storage technologies [50].

5.6 Showing the influence of hydrogen energy storage

With an HES network designed and incorporated in the power system, two final stages remain to complete the HES demonstration. First, the efficiency of both energy storage technologies must be considered followed by completion of the model and the final simulation.

Table 4 – Details on the efficiencies of considered energy storage technologies

Storage Technology	Charging Efficiency	Discharging Efficiency	Round-trip efficiency
Pumped Hydro	88%	85%	75%
Hydrogen	65%	50%	32.5%

The pumped hydro storage roundtrip efficiency of 75% has been based on the assumption that the newly installed plants (equivalent to 55% of the modelled storage capacity) will bring up the lower roundtrip efficiencies of older plants to value closer to the 80% expected for *state of the art* plants [44].

The technologies chosen for use in the homogenous HES network are a PEM electrolyser (efficiency of 65% [51]) and a fuel cell (efficiency of 50% [52]). The resulting roundtrip efficiency of 32.5% is noted to be accurate for currently available technology, whilst being slightly pessimistic about improvement prior to 2035 [53].

The entire, complete code used for the final simulation is presented in Appendix B – MATLAB Scripts .

5.6.1 Carbon Dioxide Emissions Analysis

High level calculations regarding the emissions incurred through the use of fossil fuel technologies at periods of negative net power are based on the CO_2 emissions data presented in Figure 2, [18]. This analysis has been programmed into the post-processing stages of the simulation. The outcomes of this analysis are discussed in the following section.

6 Discussion

This section will discuss the results of the completed analyses and how they have met the objectives of the project, outlined in section 3.2.

6.1 Outcomes of the simulation analyses

The full analysis of 8,761 intervals is completed in under 5 minutes. The majority of output data are stored as matrices within MATLAB with only key information pertaining to the objectives of the study presented graphically or in the text output presented in Appendix C – Final Simulation Output.

This discussion will refer to a number of values in order to compare the scenarios analysed. These values are defined in the following equations:

Equation 4:
Net Power (Interval)

$$Net\ Power_i = \sum_{\substack{All\ Sources \\ All\ Nodes}} Production - Demand$$

Equation 5:
Total Net Energy

$$Total\ Net\ Energy. = \sum_{i=1}^{8761} Net\ Power_i$$

Equation 6:
Total Deficit

$$Total\ Deficit = \sum_{i=1}^{8761} Net\ Power_i\ [Net\ Power_i < 0]$$

Equation 7:
Total Curtailment

$$Total\ Curtailment = \sum_{i=1}^{8761} Net\ Power_i\ [Net\ Power_i > 0]$$

Considering a scenario with no storage, a combined total energy deficit of 6.1 TWh is apparent despite a total net energy of 35 TWh for the full year. These values reveal that ample power is produced by RES over the course of the year, however, there is both a huge amount of power curtailment (wastage) at times of insufficient demand and many (2,014) intervals during which the power production is less than the demand.

These findings were expected. The likelihood of excess power production from environmentally dependent RES is proportional to the installed capacity. With 90% of installed capacity in 2035 expected to be environmentally dependent (see Table 2), the lack of control is evident.

The inclusion of storage systems brings a huge benefit to both the energy deficit and curtailment issues. The data presented in Table 5 summarises the total energy deficit and the contributions from the two storage technologies considered, over the three scenarios.

Table 5 – Total energy deficit and energy storage contributions

Scenario	Total Energy Deficit / GWh	Storage Contribution / GWh	
		Pumped Hydro	Hydrogen
No storage	6,129	0	0
Pumped Hydro only	5,713	416	0
Full storage	1,065	416	4,647

The importance of the total energy deficit metric is that it represents the amount of energy required to be sourced from fossil fuel power sources or from imports. The latter of these two options is neither guaranteed to be net-zero, clean power nor available at the time of requirement. Therefore, this project has considered any energy deficit to be compensated by UK based, fossil fuel sources.

The data in Table 5 show that the 7% reduction in total energy deficit realised with the expected pumped hydro storage system is 12 times greater (83%) with the complimentary 8.4 TWh HES network proposed). The effects on the total net energy and curtailment between the no-storage and the full storage scenarios is illustrated in Figure 22.

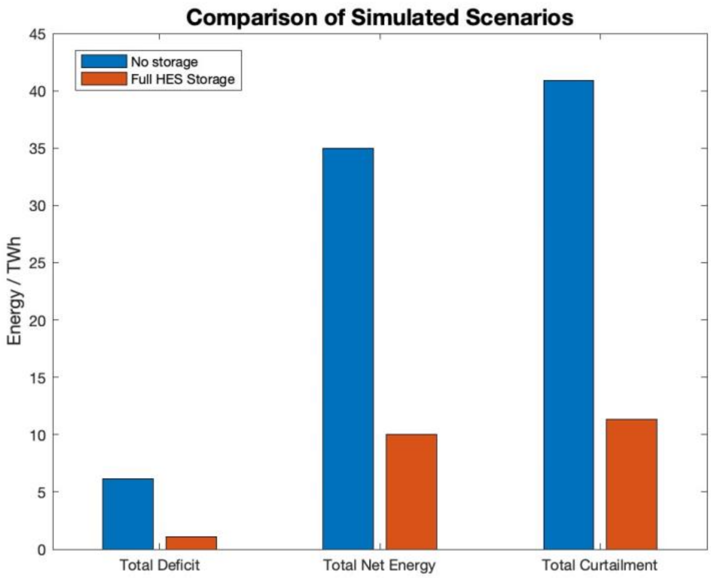

Figure 22 – Comparison of Storage Scenarios

6.1.1 CO_2 emissions

In 2021, the total emissions from power stations in the UK was 55.1 million tonnes of CO_2e (55.1 $MtCO_2e$) [54]. The simulations completed for this project have revealed expected emissions in the range of 0.17 – 4.97 $MtCO_2e$[2] with the full pumped hydro storage network and no HES network. With the proposed HES network these values are expected to fall to between 0.03 – 0.93 $MtCO_2e$. Table 6 contains a summary of the emissions required to compensate the total energy deficit for each scenario.

[2] Ranges of expected emissions are based on the maximum emissions scenario of 100% coal power and minimum emissions scenario of 100% combined-cycle gas turbines with carbon-capture storage. These power sources are used to meet the required energy deficit compensation.

[redacted]

Table 6 – Estimated CO_2e emissions from fossil fuel supplements

Scenario	Total Energy Deficit / GWh	Emission from fossil fuel supplement / $MtCO_2e$ [min / max]		
		Coal	CCGT	CCGT, with CCS
No storage	6,129	4.11 / 5.33	2.14 / 3.00	0.18 / 0.60
Full storage	1,065	0.71 / 0.93	0.37 / 0.52	0.03 / 0.10
Total Reduction	- 5,064	-4.82 / -6.26	-2.52 / -3.53	-0.22 / -0.71
Reduction from Pumped Hydro		-0.28 / -0.36	-0.15 / -0.20	-0.01 / -0.04
Reduction from HES		-3.11 / -4.04	-1.63 / -2.28	-0.14 / -0.46

In order to achieve its target of a net-zero power system, the UK government would have to offset the remaining emissions. Given that between 31 and 46 trees are required to offset 1 tonne of CO_2 [55], a minimum of over 5 million additional trees would have to be planted in the year 2035 for the no HES scenario. This is in contrast with just under 1 million with the proposed HES network in operation.

Finally, this analysis reveals that neither scenario results in a truly net-zero power system. Further analysis may be completed to determine the HES capacity required to meet this target.

6.2 HES network implementation

The proposed HES network modelled in this project features a nationwide, modular distribution of HES infrastructure. Each region is assumed to contain equal capacities of PEM electrolysis and PEMFC equipment for utility scale use. There are various factors to consider when developing a concept for an HES network including the locations, distribution and sizing of hydrogen production facilities, the mode of storage, any transportation of hydrogen and how it is expended. This section will discuss a few concepts, the conclusions that can be drawn from the simulation and the relative merits and downsides of each concept.

6.2.1 Network topologies

Three potential options for implementing HES infrastructure have been evaluated at a high level: offshore HES hubs, a hydrogen transportation network, and modular/regional grid-powered HES.

6.2.1.1 Offshore HES hubs

The offshore HES hubs option involves producing hydrogen offshore using excess power from offshore wind farms, shown to be a major source of net-power surplus (Figure 20). With the potential to reduce transmission losses prior to the HES stage and with far less stringent constraints on land availability offshore, this may be a very attractive option. Hydrogen can be stored on the seabed in tanks or underground in caverns and can be expended during times of low wind farm output. However, there remain important considerations. Firstly, the challenging nature of offshore construction may result in far higher capital requirement. Additionally, due to the use of existing grid infrastructure, transmission losses remain after the HES stage of the process as the power is transmitted to meet demand on land.

6.2.1.2 Hydrogen transportation network

The hydrogen transportation network option involves producing hydrogen at selected power generation sites (both onshore and offshore) and transporting it to modular storage near power demand areas via pipelines, vessels, tankers, or trucks. This option eliminates electrical transmission losses and promotes easier upgrades and maintenance to components, an inherent feature of a more modular system. There is also the potential for synergy with other non-power uses of hydrogen, such as domestic heating and cooking [56]. However, the use transport and distribution systems must be carefully considered and controlled to avoid further emissions. Additionally, storing hydrogen near populous areas could potentially be expensive and unsightly.

6.2.1.3 Modular/Regional grid powered HES

A modular/regional grid-powered HES system involves producing, storing, and expending hydrogen near demand areas using grid supplied electricity. This option offers the advantages of modular networks, similar to the transportation network, and a hydrogen production system decoupled from RES. The latter property may hold benefits in terms of the longevity and robustness of the HES system. The challenges associated with this option include the fact that transmission losses prior to the HES stage remain and a potential need for increased transmission network capacity due to exposure to higher power flow from RES. This property is not currently present due to the ability to curtail excess power and would not be a feature of either of the other HES topologies identified.

6.2.2 Further considerations

The development of an HES network in the UK by 2035 presents many challenges that must be addressed to ensure the safe and efficient operation of the system. There are risks associated with the production, storage, and transportation of hydrogen - leaks or explosions - which must be mitigated through the implementation of appropriate safety measures.

The maintenance of the HES network is critical to ensure its safe and efficient operation. Regular inspection and maintenance of the network components such as pipelines, tanks, and fuel cells are necessary to prevent leaks or failures, which could lead to significant safety hazards and system downtime which, in turn, could degrade the efficacy of the system.

The establishment of a robust manufacturing infrastructure requires significant investment in research and development, education and training, and regulatory frameworks to ensure the safe and efficient operation of the system.

Finally, the land requirement of any HES should also be considered. Societal effects can be minimized with offshore or underground placement of large components. However, the feasibility of such options is dependent on capital costs, environmental impact, and other technical considerations.

7 Conclusions

Simulation model

Having reviewed literature surrounding the use of hydrogen for energy storage in grid applications, a simulation approach was identified as optimal for modelling the operation of a hydrogen storage network. Various data sources were examined and processed for implementation into the model to be developed with MATLAB. The constructed model is founded in the current geographical distribution of RES in the UK and incorporates extrapolated predictions on future power production capacities. The operation of these systems is simulated and evaluated at one-hour intervals, allowing for calculations regarding the power and energy surplus over the course of the year 2035.

8.4TWh HES network

These calculations were employed to estimate the requirement for and propose a suitable HES network with a total unfactored charging capacity of 21 GW and a storage capacity of 8.4 TWh, verified to be within the range expected by reputable sources. Implementation of the proposed HES network and the expected pumped hydropower storage into the simulation revealed that HES has huge potential to reduce power system CO_2 emissions.

Reduced energy deficit

The total energy deficit over the year 2035 was shown to reduce by over 80% with the inclusion of HES, entailing a proportional reduction in the use of supplementary fossil-fuel based electricity.

Reduced emissions

Neither of the scenarios modelled indicated that the UK will have a truly net-zero power system in 2035. The expected emissions from a no HES power system are shown to be at least 90% lower than 2021 levels, with the inclusion of HES bringing this reduction up to (at least) 98%. The ability of HES to reduce reliance on unsustainable power sources is revealed to be both clear and scalable.

HES network implementation

Finally, three potential HES network topologies have been outlined and evaluated for suitability in the UK power system. Regionally, a blend of approaches to HES is both likely and should foster examples and precedent capable of informing future developments beyond 2035.

8 Future Work

There remain numerous tasks and questions that may consider further use of the simulation model. Further analysis may iteratively determine the hydrogen storage capacity required to meet the target of a net-zero power system in 2035. This work may be aided by refined extrapolation of RES capacities as more reliable data becomes available over the next decade.

Inclusion of power flow analyses to each evaluation interval would allow for integration of the transmission network as was found to be beyond the scope of this project. This would allow for a more detailed discussion on the siting of HES facilities and educate the selection of an optimal network topology.

Finally, sensitivity studies on variables within the model could be performed to further refine HES network designs. Alternative storage technologies, RES and more complex environmental dependencies should be considered for more detailed visualisation of power system behaviours.

This project has demonstrated the value of a simulation model in predicting the influence of modern storage solutions in sustainable power systems and may be used as a basis for further study.

9 References

[1] Earth Science Communications Team. "Global Climate Change: Vital Signs of the Planet." NASA's Jet Propulsion Laboratory. https://climate.nasa.gov/vital-signs (accessed 20/03/2023, 2023).

[2] G. Monbiot, "Heat : how to stop the planet burning / George Monbiot with research assistance from Matthew Prescott," 2007.

[3] DNV, "Energy Transition Outlook 2022," DNV, 13/10/2022 2022, vol. 6. [Online]. Available: https://www.dnv.com/energy-transition-outlook/download.html

[4] The Climate Change Committee, "The Sixth Carbon Budget: The UK's path to Net Zero," December 2020 2022. [Online]. Available: https://www.theccc.org.uk/wp-content/uploads/2020/12/The-Sixth-Carbon-Budget-The-UKs-path-to-Net-Zero.pdf

[5] Department for International Trade. *Hydrogen Investor Roadmap*. (2022). [Online]. Available: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/1114758/hydrogen-investor-roadmap.pdf

[6] Department for Business Energy and Industrial Strategy, "Impact Assessment for the sixth carbon budget," BEIS012(F)-21-CG, 16/04/2021 2021. [Online]. Available: https://www.legislation.gov.uk/ukia/2021/18/pdfs/ukia_20210018_en.pdf

[7] M. d. P. Argumosa, E. Chacon, and S. M. Schoenung, "Evaluation of integrated hydrogen systems: IEA Task 18," *International journal of hydrogen energy,* vol. 35, no. 18, pp. 10031-10037, 2010, doi: 10.1016/j.ijhydene.2010.04.151.

[8] P. Fu, D. Pudjianto, X. Zhang, and G. Strbac, "Integration of Hydrogen into Multi-Energy Systems Optimisation," *Energies (Basel),* vol. 13, no. 7, p. 1606, 2020, doi: 10.3390/en13071606.

[9] Lane Clark & Peacock, "Renewable curtailment and the role of long duration storage (Report for Drax)," 2022. [Online]. Available: https://www.drax.com/wp-content/uploads/2022/06/Drax-LCP-Renewable-curtailment-report.pdf

[10] N. Thomas, "National Grid warns Britons of blackouts on 'really cold' evenings," in *The Financial Times*, ed. London; UK, 2022.

[11] F. Zhang, P. Zhao, M. Niu, and J. Maddy, "The survey of key technologies in hydrogen energy storage," *International journal of hydrogen energy,* vol. 41, no. 33, pp. 14535-14552, 2016, doi: 10.1016/j.ijhydene.2016.05.293.

[12] R. Bhandari, C. A. Trudewind, and P. Zapp, "Life cycle assessment of hydrogen production via electrolysis – a review," *Journal of cleaner production,* vol. 85, pp. 151-163, 2014, doi: 10.1016/j.jclepro.2013.07.048.

[13] IEA, "Electrolysers," Paris, 2022. [Online]. Available: https://www.iea.org/reports/electrolysers

[14] M. Klell, "Thermodynamics of gaseous and liquid hydrogen storage," presented at the International Hydrogen Energy Congress and Exhibition, Istanbul, Turkey, 13/07/07, 2007.

[15] J. Zheng, X. Liu, P. Xu, P. Liu, Y. Zhao, and J. Yang, "Development of high pressure gaseous hydrogen storage technologies," *International journal of hydrogen energy,* vol. 37, no. 1, pp. 1048-1057, 2012, doi: 10.1016/j.ijhydene.2011.02.125.

[16] L. Valverde-Isorna, D. Ali, D. Hogg, and M. Abdel-Wahab, "Modelling the performance of wind–hydrogen energy systems: Case study the Hydrogen Office in Scotland/UK,"

Renewable & sustainable energy reviews, vol. 53, pp. 1313-1332, 2016, doi: 10.1016/j.rser.2015.08.044.

[17] F.-C. Wang, Y.-S. Hsiao, and Y.-Z. Yang, "The optimization of hybrid power systems with renewable energy and hydrogen generation," *Energies (Basel),* vol. 11, no. 8, p. 1948, 2018, doi: 10.3390/en11081948.

[18] S. Schlömer *et al.*, "Annex III: Technology-specific cost and performance parameters," in "Climate Change 2014: Mitigation of Climate Change. Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change," Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, 2014. [Online]. Available: https://www.ipcc.ch/site/assets/uploads/2018/02/ipcc_wg3_ar5_annex-iii.pdf#page=7

[19] Department for Business Energy and Industrial Strategy. *Digest of UK Energy Statistics. Table 5.12: Plant Installed Capacity, by connection*, Department for Business, Energy & Industrial Strategy. [Online]. Available: https://www.gov.uk/government/statistics/digest-of-uk-energy-statistics-dukes-2021

[20] J. F. Manwell, "Wind energy explained : theory, design and application / J. F. Manwell, J. G. McGowan and A. L. Rogers," 2002.

[21] A. P. Alivisatos and T. E. Hullar, "How does solar power work?," *Scientific American,* vol. 300, no. 1, pp. 108-108, 2009, doi: 10.1038/scientificamerican0109-108.

[22] D. Burnett, E. Barbour, and G. P. Harrison, "The UK solar energy resource and the impact of climate change," *Renewable energy,* vol. 71, pp. 333-343, 2014, doi: 10.1016/j.renene.2014.05.034.

[23] NASA. The Prediction of Worldwide Energy Resources (POWER) [Online] Available: https://power.larc.nasa.gov/data-access-viewer/

[24] Department for Business Energy and Industrial Strategy. Renewable Energy Planning Database [Online] Available: https://www.gov.uk/government/publications/renewable-energy-planning-database-monthly-extract

[25] Department for Business Energy and Industrial Strategy. *Digest of UK Energy Statistics. Table 5.7: Plant Capacity*, Department for Business, Energy & Industrial Strategy. [Online]. Available: https://www.gov.uk/government/statistics/electricity-chapter-5-digest-of-united-kingdom-energy-statistics-dukes

[26] Department for Business Energy and Industrial Strategy. *Annexes E-J and long-term trends dataset*, Department for Business, Energy & Industrial Strategy. [Online]. Available: https://www.gov.uk/government/statistics/digest-of-uk-energy-statistics-dukes-2021

[27] National Grid. Demand Data 2018 [Online] Available: https://www.nationalgrideso.com/document/109391/download

[28] Department for Business Energy and Industrial Strategy. *Subnational electricity consumption, Great Britain, 2005 - 2021*.

[29] Department for Business Energy and Industrial Strategy. *Subnational electricity consumption, Northern Ireland, 2009 - 2021*.

[30] © National Grid UK. "Network route maps." National Grid UK. https://www.nationalgrid.com/electricity-transmission/network-and-infrastructure/network-route-maps (accessed 02/11/2022.

[31] IRENA, "Renewable Power Generation Costs in 2022," International Renewable Energy Agency, Abu Dhabi, 2022. [Online]. Available: https://www.irena.org/-

/media/Files/IRENA/Agency/Publication/2019/May/IRENA_Renewable-Power-Generations-Costs-in-2018.pdf

[32] U.S. Energy Information Administration, "Levelized Costs of New Generation Resources in the Annual Energy Outlook 2022," in "Annual Energy Outlook 2022," U.S. Energy Information Administration,, 2022. [Online]. Available: https://www.eia.gov/outlooks/aeo/pdf/electricity_generation.pdf

[33] A. International Energy, "World energy outlook. [electronic resource]," *World energy outlook.,* 2021.

[34] IRENA, "Renewable Power Generation Costs in 2018," International Renewable Energy Agency, Abu Dhabi, 2019. [Online]. Available: https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2019/May/IRENA_Renewable-Power-Generations-Costs-in-2018.pdf

[35] Department for Business Energy and Industrial Strategy. "Policy Paper: Contracts for Difference." https://www.gov.uk/government/publications/contracts-for-difference/contract-for-difference (accessed 15/02/2023, 2023).

[36] (2018). *Nuclear electricity in the UK.* [Online] Available: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/789655/Nuclear_electricity_in_the_UK.pdf

[37] D. Hollis, M. McCarthy, M. Kendon, and T. Legg. *HadUK-Grid Gridded Climate Observations on a 25km grid over the UK, v1.1.0.0 (1836-2021)*, NERC EDS Centre for Environmental Data Analysis, doi: 10.5285/6f4ac352b19341eb8c5b26644845ac35.

[38] G. Sinden, "Characteristics of the UK wind resource: Long-term patterns and relationship to electricity demand," *Energy policy,* vol. 35, no. 1, pp. 112-127, 2007, doi: 10.1016/j.enpol.2005.10.003.

[39] N. Ø. Chemnitz, "The Orkney Power Curve," vol. 2023, ed. Copenhagen, Denmark: IT University of Copenhagen, 2020.

[40] R. N. Clark, "Small wind : planning and building successful installations / R. Nolan Clark," 2014.

[41] A. Allais, "Eco-friendly innovations in electricity transmission and distribution networks / edited by Jean-Luc Bessède contributors, A. Allais [and thirty-four others]," 2015.

[42] M. Kamran, "Microgrid and hybrid energy systems," in *Fundamentals of Smart Grid Systems*, M. Kamran Ed.: Academic Press, 2023, ch. 7, pp. 299-363.

[43] ONS Geography. Local Authority Districts (May 2022) UK BFC [Online] Available: https://geoportal.statistics.gov.uk/datasets/ons::local-authority-districts-may-2022-uk-bfc/about

[44] C.-J. Yang, "Pumped Hydroelectric Storage," pp. 25-38, 2016, doi: 10.1016/B978-0-12-803440-8.00002-6.

[45] J. R. Hillman and D. J. C. MacKay, "Sustainable Energy - Without the Hot Air," *Experimental agriculture,* vol. 46, no. 1, p. 117, 2010, doi: 10.1017/S0014479709990718.

[46] S. P. Hydro. "ABOUT GLYN RHONWY." https://www.snowdoniapumpedhydro.com/about (accessed 12/04/2023, 2023).

[47] Buccleuch. "LANDMARK INVESTMENT DEAL FOR GLENMUCKLOCH." https://www.buccleuch.com/landmark-investment-deal-for-glenmuckloch-pumped-storage-hydro-psh-and-windfarm-project-in-dumfries-and-galloway/ (accessed 12/04/2023, 2023).

[48] "A Social Cost Benefit Analysis of Grid-Scale Electrical Energy Storage Projects: Evaluating the Smarter Network Storage Project," *Social Cost Benefit Analysis of Grid-Scale Electrical Energy Storage Projects*.

[49] The Newsroom, "Meet Red John - the Highland hydro scheme that could be a gamechanger for renewables," in *The Scotsman*, ed, 2018.

[50] Frazer Nash Consultancy and Cornwall Insight, "Hydrogen Transportation and Storage Infrastructure: Assessment of Requirements up to 2035," Department of Business, Energy and Industrial Strategy, London, 2022, vol. 1. [Online]. Available: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/1123450/Hydrogen_infrastructure_requirements_up_to_2035_-_report.pdf

[51] A. Ursua, L. M. Gandia, and P. Sanchis, "Hydrogen Production From Water Electrolysis: Current Status and Future Trends," *Proceedings of the IEEE,* vol. 100, no. 2, pp. 410-426, 2012, doi: 10.1109/JPROC.2011.2156750.

[52] E. L. V. Eriksson and E. M. Gray, "Optimization and integration of hybrid renewable energy hydrogen fuel cell energy systems – A critical review," *Applied energy,* vol. 202, pp. 348-364, 2017, doi: 10.1016/j.apenergy.2017.03.132.

[53] F. Klumpp, "Comparison of pumped hydro, hydrogen storage and compressed air energy storage for integrating high shares of renewable energies—Potential, cost-comparison and ranking," *Journal of energy storage,* vol. 8, pp. 119-128, 2016, doi: 10.1016/j.est.2016.09.012.

[54] Department for Business Energy and Industrial Strategy, "2021 UK Greenhouse Gas Emissions, Final Figures," London, 07/02/2023 2023. [Online]. Available: https://www.gov.uk/government/statistics/provisional-uk-greenhouse-gas-emissions-national-statistics-2022

[55] encon. "Calculation of CO2 offsetting by trees." https://www.encon.eu/en/calculation-co2-offsetting-trees (accessed 21/04/2023.

[56] R. Castek and S. Harkin, "Evidence review for hydrogen for heat in buildings," 2021, doi: 10.7488/era/1239.

Appendix A – Figures

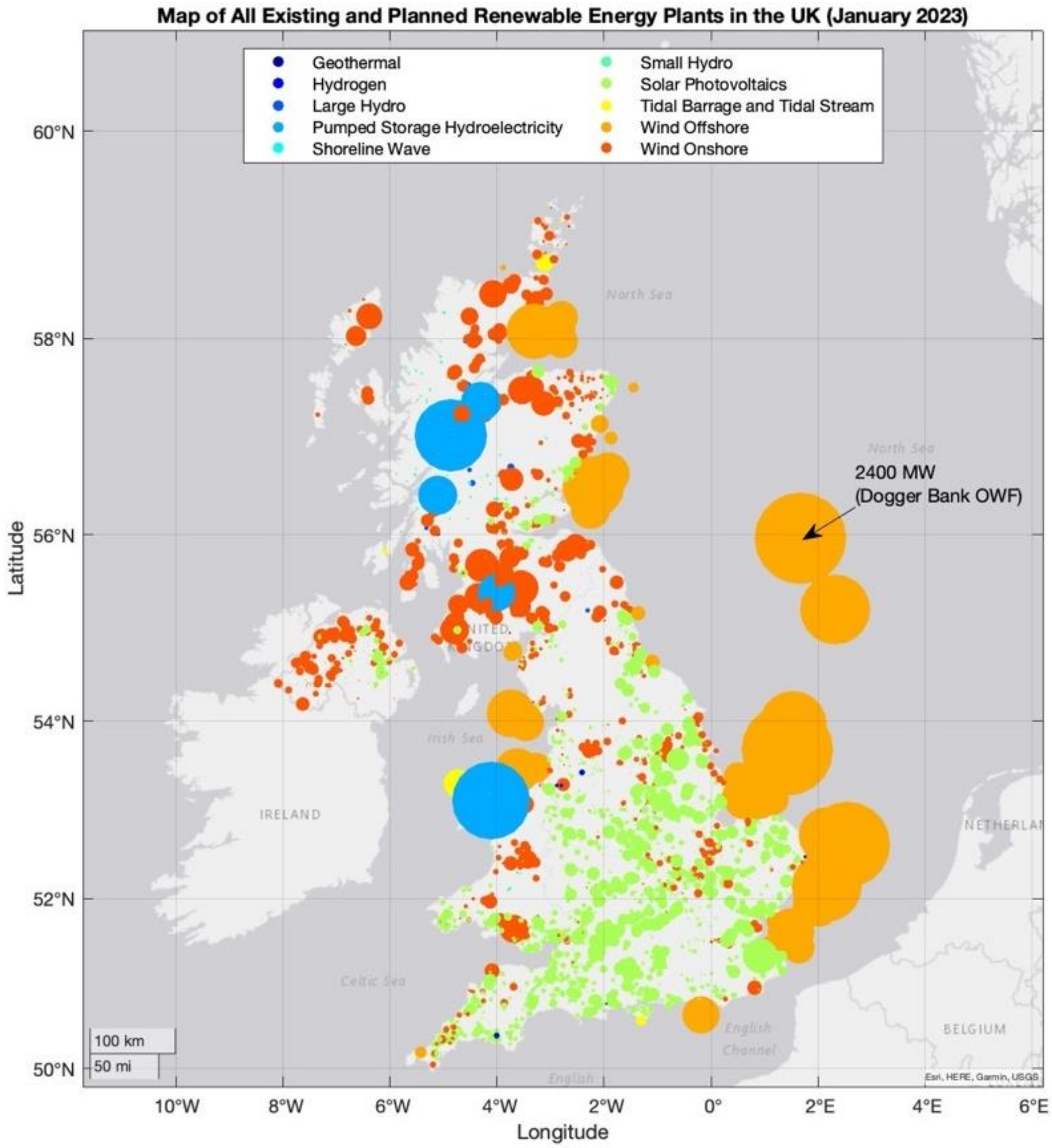

Figure 23 – Map of All RES Locations, UK, Sized by Plant Installed Capacity [24]

[redacted]

Grouped Installed Capacities of RES

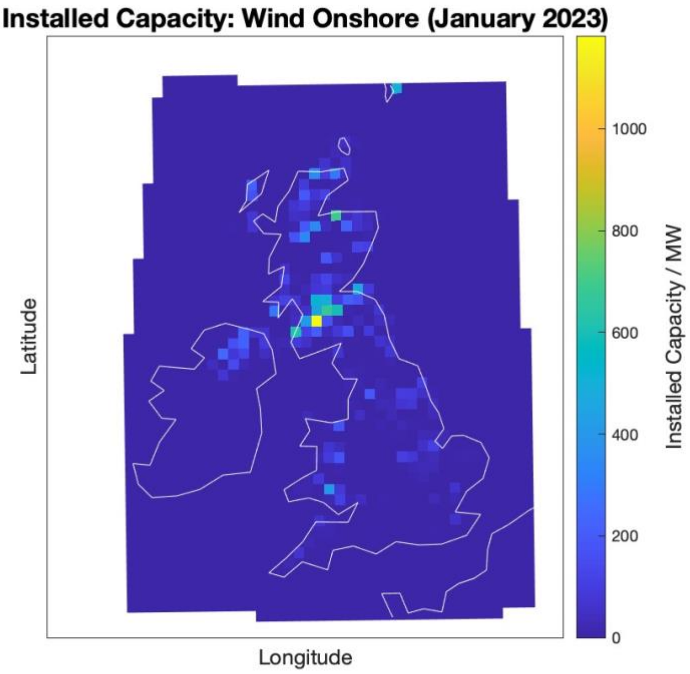

Figure 24 – Geographically Grouped Installed Capacity, Wind (Onshore) [24]

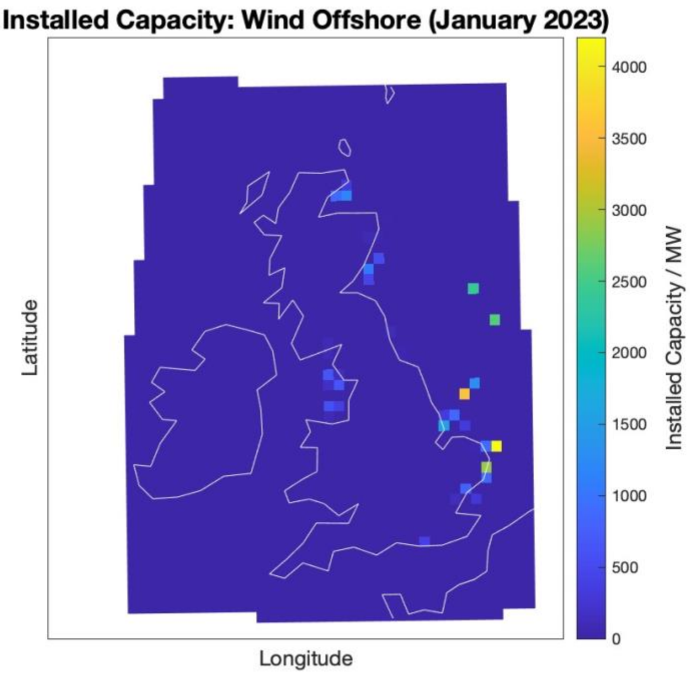

Figure 25 – Geographically Grouped Installed Capacity, Wind (Offshore) [24]

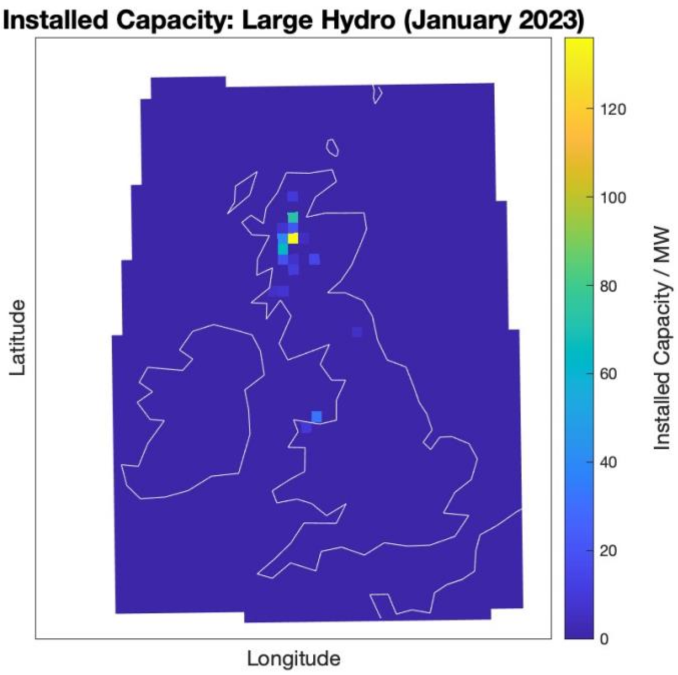

Figure 26 – Geographically Grouped Installed Capacity, Large Hydro (Natural Flow) [24]

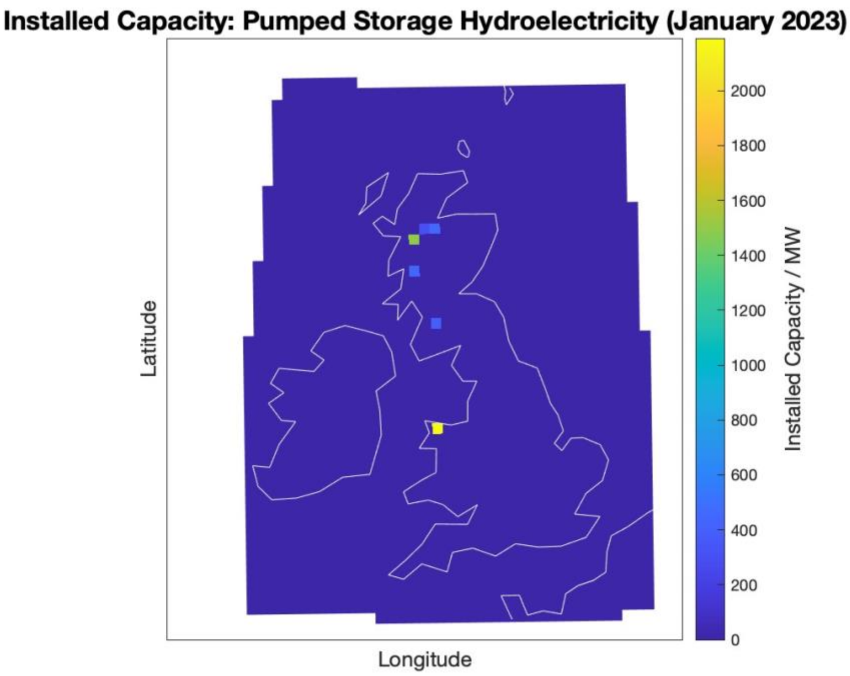

Figure 27 – Geographically Grouped Installed Capacity, Pumped Hydro Storage [24]

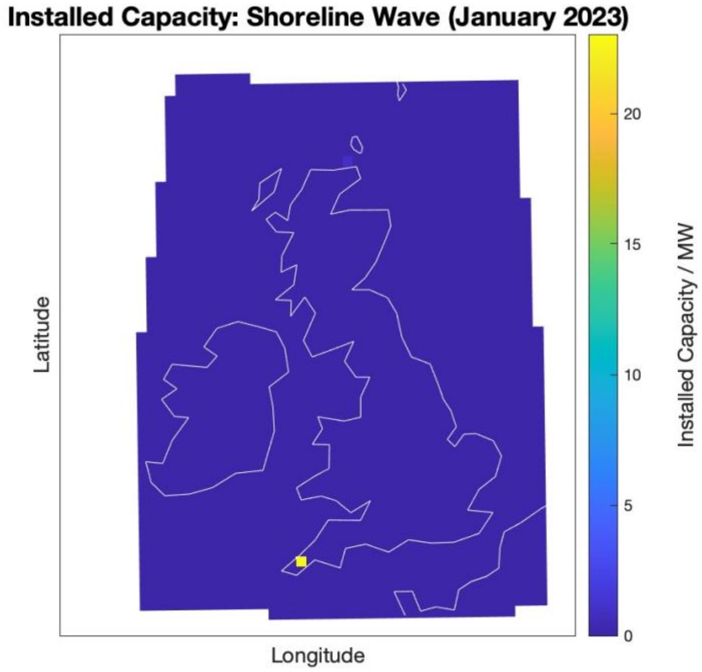

Figure 28 – Geographically Grouped Installed Capacity, Shoreline Wave [24]

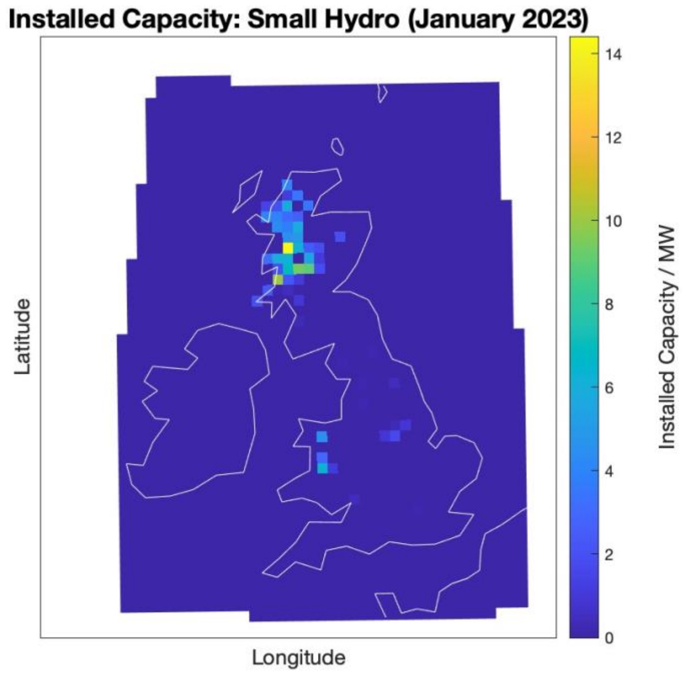

Figure 29 – Geographically Grouped Installed Capacity, Small Hydro (Natural Flow) [24]

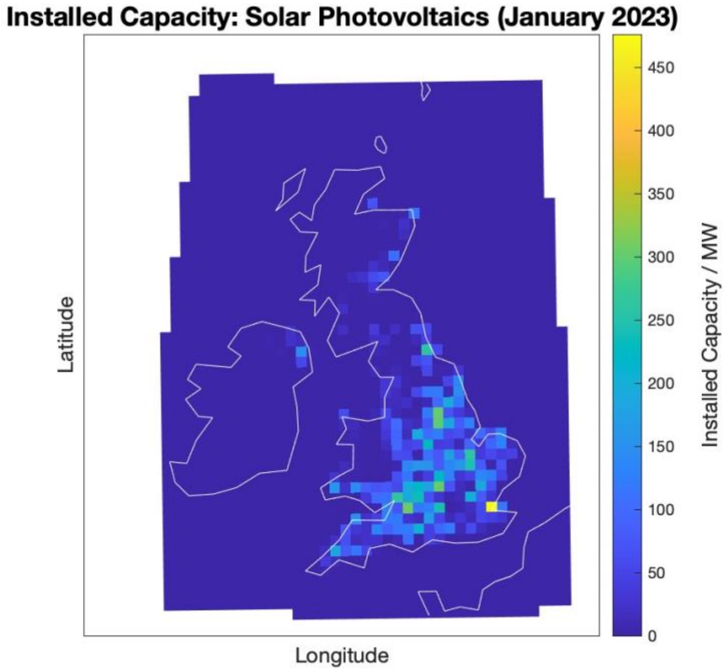

Figure 30 – Geographically Grouped Installed Capacity, Solar PV [24]

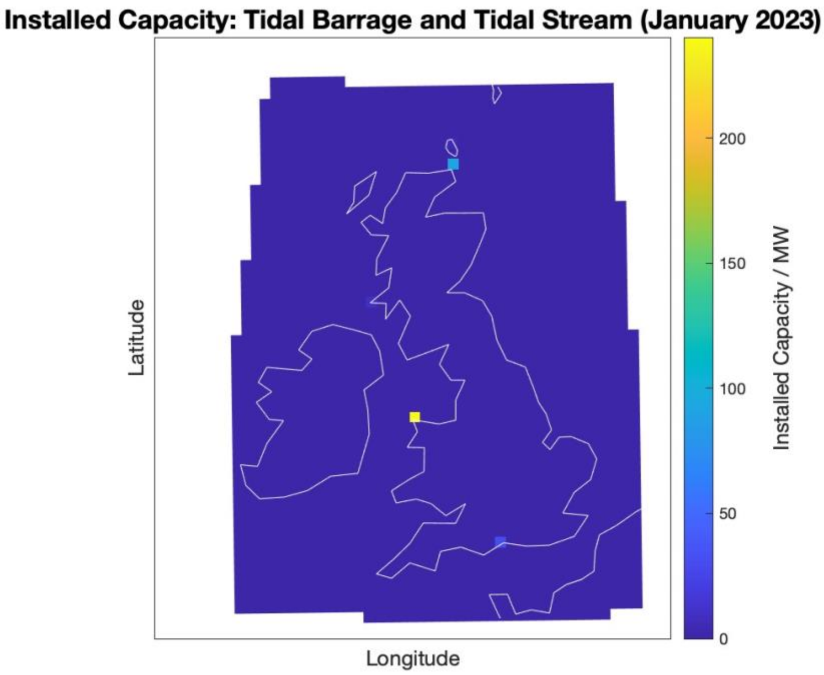

Figure 31 – Geographically Grouped Installed Capacity, Tidal Power [24]

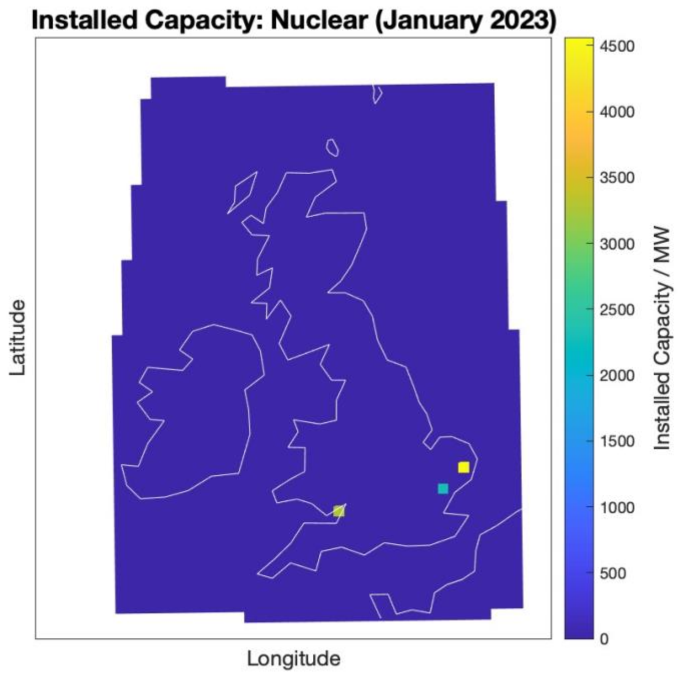

Figure 32 – Geographically Grouped Installed Capacity, Nuclear [24]

RES Capacity Extrapolation

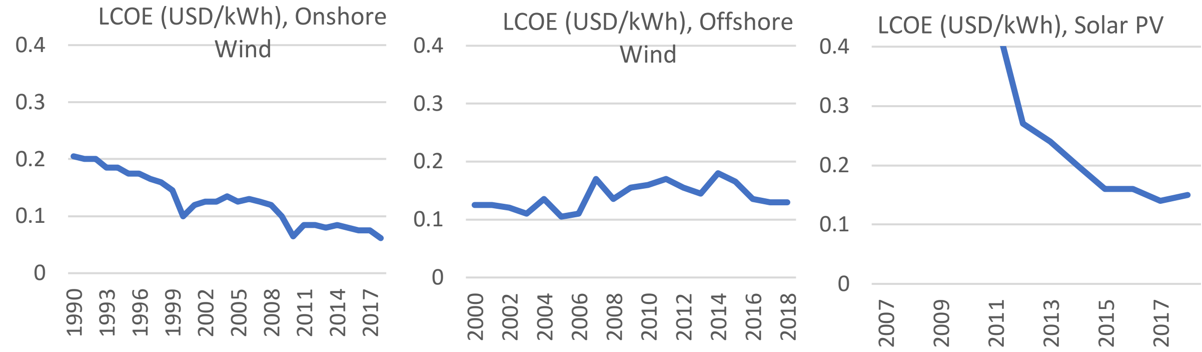

Figure 33 – Levelized Cost of Electricity Data [34]

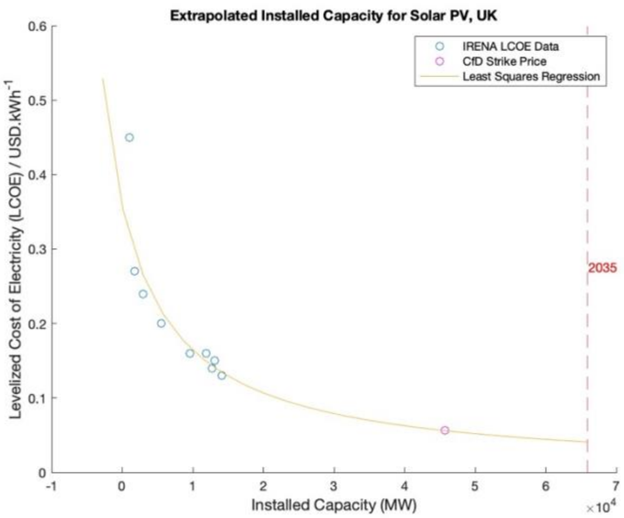

Figure 34 – Extrapolated Installed Capacity, Solar PV

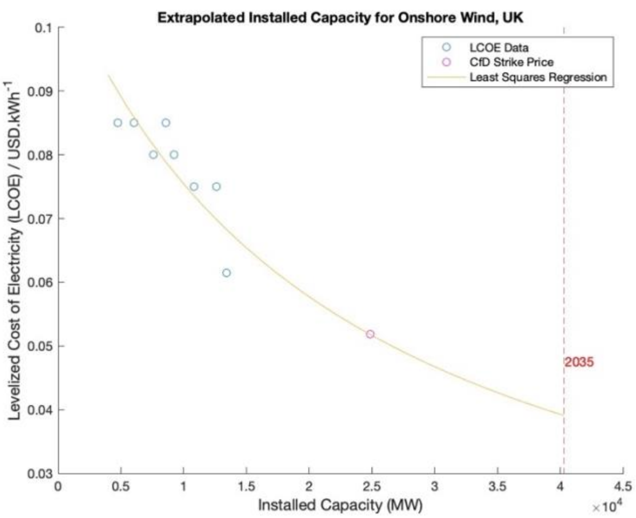

Figure 35 – Extrapolated Installed Capacity, Onshore Wind

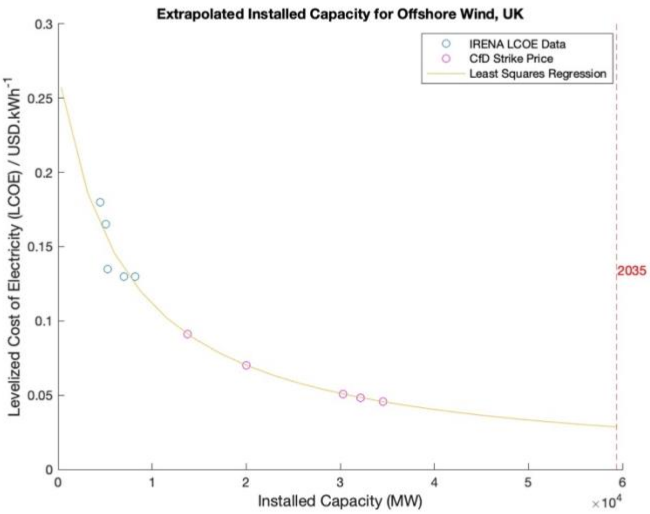

Figure 36 – Extrapolated Installed Capacity, Offshore Wind

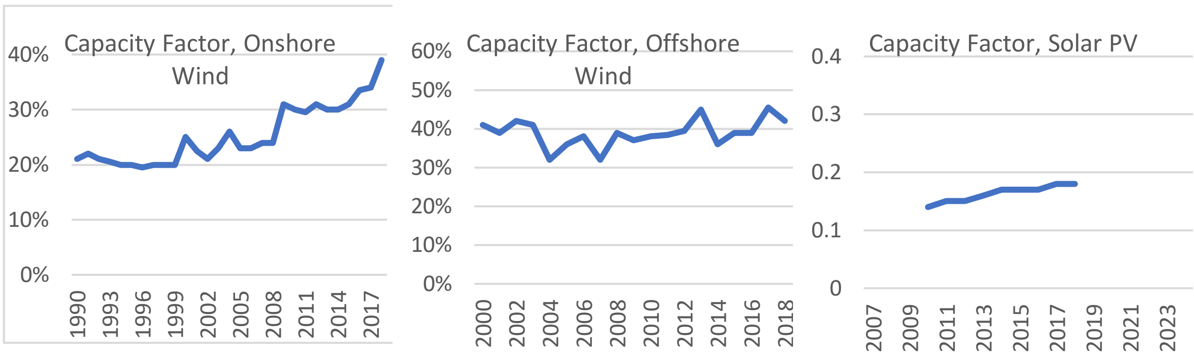

Figure 37 – Capacity Factor Data [34]

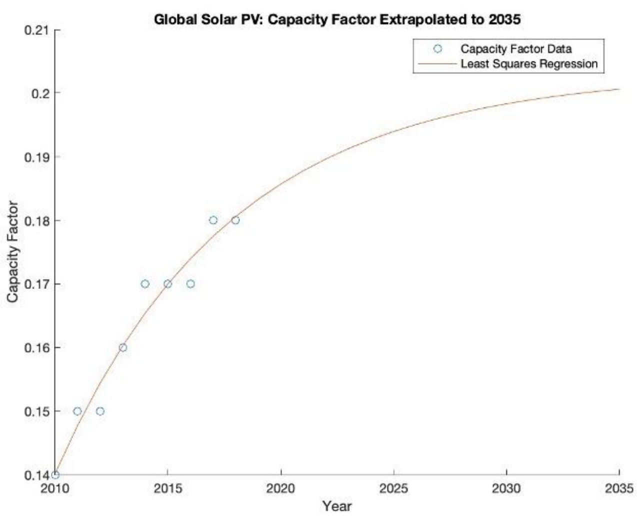

Figure 38 – Extrapolated Capacity Factor, Solar PV

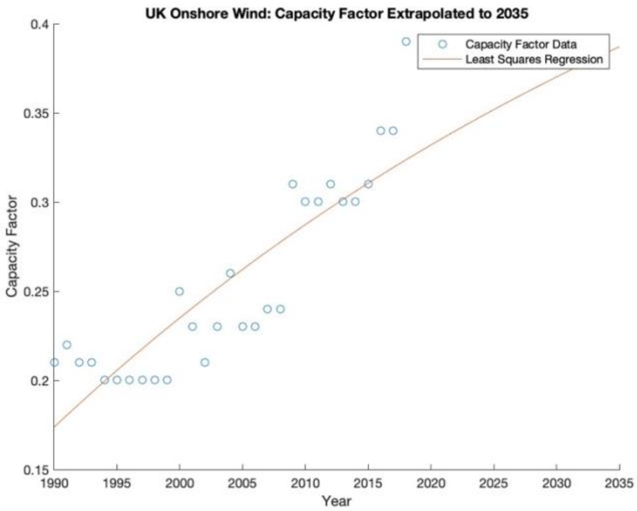

Figure 39 – Extrapolated Capacity Factor, Onshore Wind

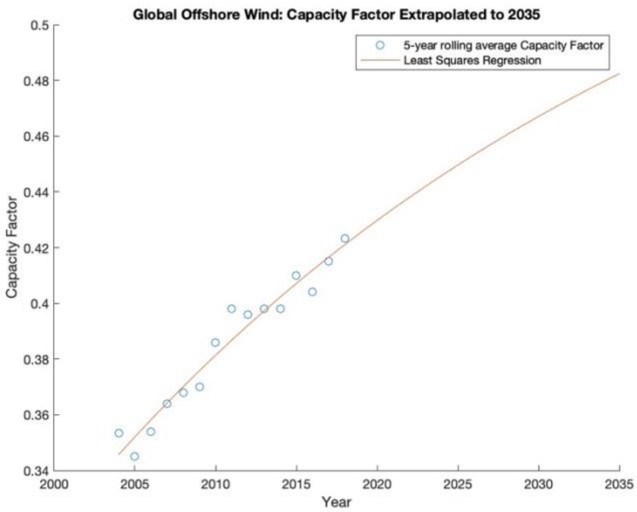

Figure 40 – Extrapolated Capacity Factor, Offshore Wind

RES Power Simulation (No Storage Scenario) Output Graphics

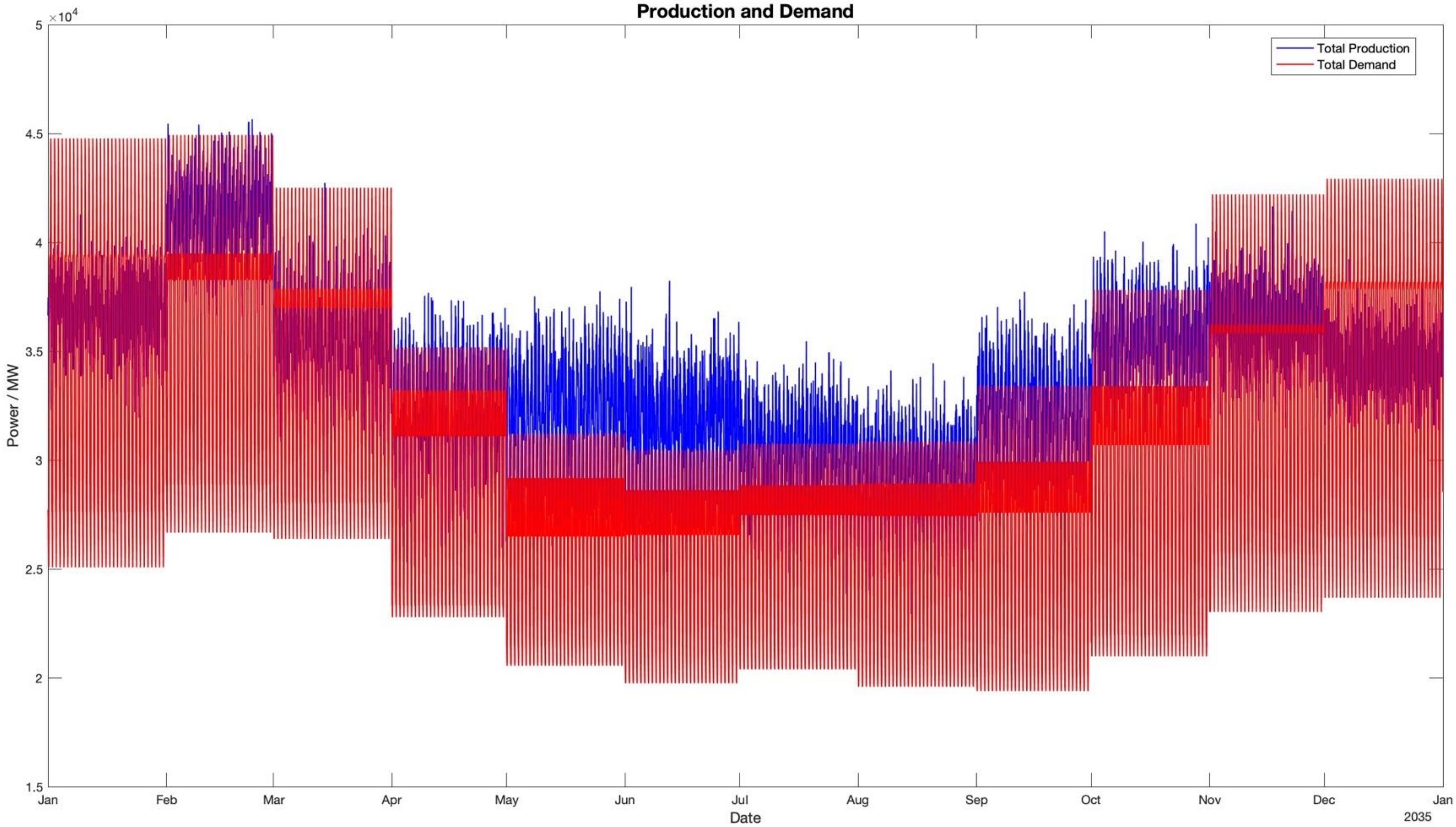

Figure 41 – Power and Demand, no storage simulation

[redacted]

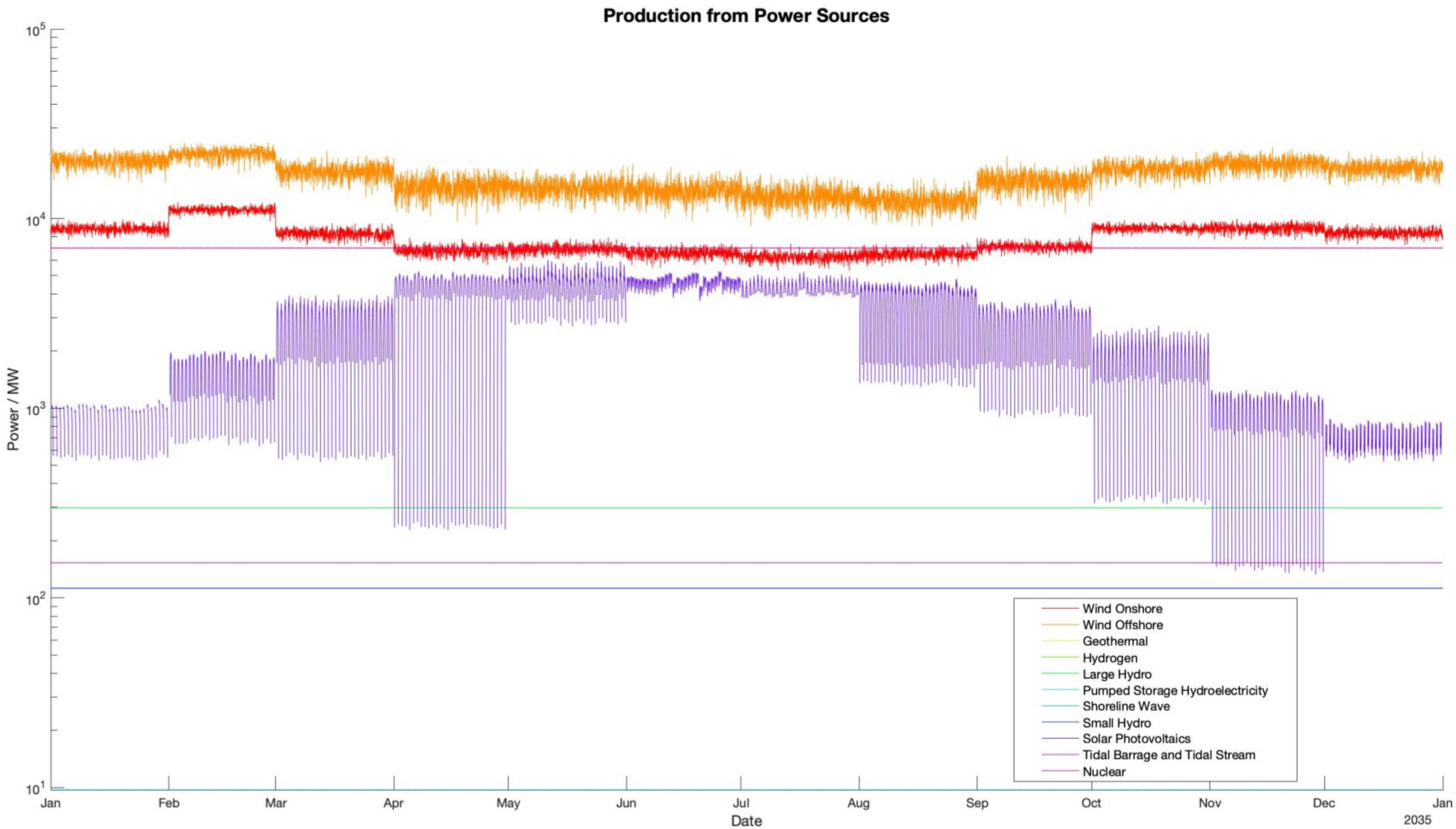

Figure 42 – Production from All Power Sources, no storage simulation

Full RES Power Simulation Output Graphics

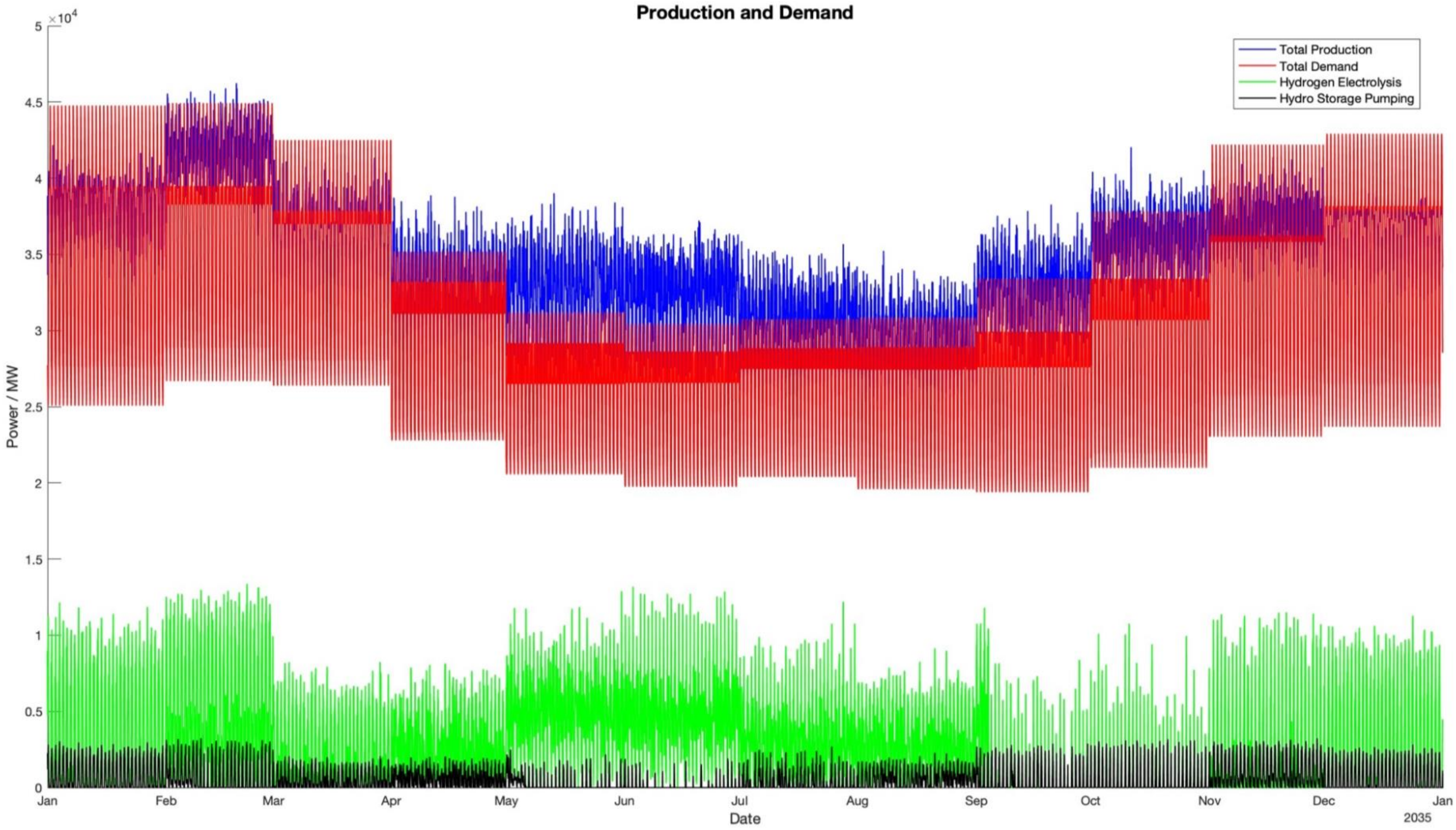

Figure 43 – Power and Demand, Full storage simulation

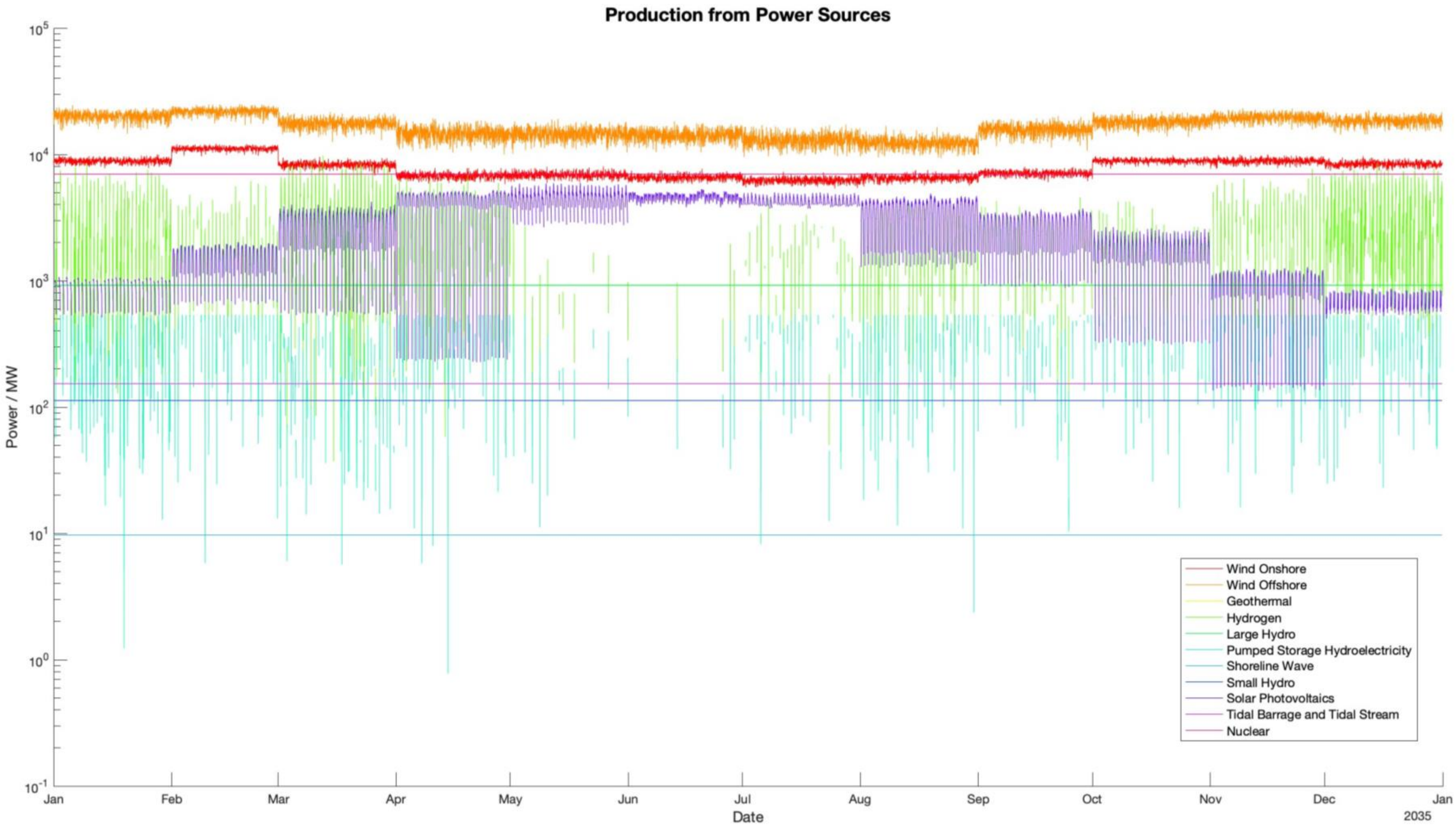

Figure 44 – Production from All Power Sources, Full storage simulation

Appendix B – MATLAB Scripts

Fitting Weibull Distribution Parameters (Wind Speed)

```
clear all;
close all;
clc;

% Aim of this programme is to determine the scale and shape factors for Weibull
Distribution wind speeds
% at different locations in the UK in each month.
% The discretisation function will then be applied to assign the Weibull parameters
to the relevant nodes on the 39x52 grid.

% Load geometries and nodal coordinates from geometry spreadsheet
geometry_file = 'Geometry.xlsx';
land_shape_sheet = 'LAND_SHAPE';

LONGITUDES = readmatrix(geometry_file,'Sheet',land_shape_sheet,'Range','A2:A40');
%1x39 double
LATITUDES = readmatrix(geometry_file,'Sheet',land_shape_sheet,'Range','B1:BA1');
%1x52 double

% Loop through NASA files
file_dir = 'Yearly_Data';
file_pattern = '*.nc';
files = dir(fullfile(file_dir, file_pattern));

Month_Comp = cell(12,1) ;
Weibull_Settings = NaN(12,2,39,52);

for i = 1:length(files)
    filename = fullfile(file_dir, files(i).name);
    year_str = regexp(files(i).name, '^\d+(?=_)', 'match');
    year = str2double(year_str);
    wind_data = ncread(filename,'WS50M');
    lat = ncread(filename,'lat');
    lon = ncread(filename,'lon');

    time_series_int = ncread(filename,'time');
    start_date = datetime(strcat(year_str,'-01-01')) ;
    end_date = datetime(strcat(year_str,'-12-31')) ;
    datetime_array = start_date:days(1):end_date ;

    for day = 1:length(time_series_int)

Month_Comp{month(datetime_array(day)),1}(size(Month_Comp{month(datetime_array(day))
},1)+1,:,:) = wind_data(:,:,day) ;
    end
end

for month = 1:size(Month_Comp,1)
    Weibull_Settings(month,:,:,:) = fit_and_discretise(Month_Comp{month,1}(:,:,:),
lat, lon, LATITUDES, LONGITUDES);
    Monthly_Av(month,:,:) = squeeze(mean(Month_Comp{month,1},1,"omitmissing"));
    if length(Monthly_Av(month,isnan(Monthly_Av(month,:,:)))) > 0
        E = squeeze(Monthly_Av(month,:,:));
        error('NaN properties')
    end
end

Annual_Av = mean(Monthly_Av,[1]);

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%% Map Plotting Code %%%%%%%%%%%%
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Convert the latitude and longitude arrays to meshgrid
[lat_mesh, lon_mesh] = meshgrid(LATITUDES, LONGITUDES);

latlim = [min(LATITUDES) max(LATITUDES)];
lonlim = [min(LONGITUDES) max(LONGITUDES)];

load coastlines;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% For Feb Weibull Scale data:
[la_mesh, lo_mesh] = meshgrid(lat, lon);

F =   figure('Name','Annual Average Wind Speed');
% Generate fixed scale for colorbar
[maxValue, maxInd] = max(squeeze(Annual_Av), [], "all", "linear") ; % find the
maximum value
colorlim = [0 maxValue]; % set color scale for plots to range from 0 to the maximum
value

plot_name = sprintf('Average Annual Wind Speed, UK 2020 - 2022', i);
ax = axesm('MapProjection', 'utm', 'MapLatLimit', latlim, 'MapLonLimit', lonlim);
h = geoshow(la_mesh, lo_mesh, squeeze(Annual_Av), 'DisplayType', 'texturemap');
set(h, 'CData', squeeze(Annual_Av));
plotm(coastlat, coastlon, 'Color', 'k');
a = colorbar ; % add a colorbar to the plot
caxis(colorlim) ; % set the color scale for the plot
a.Label.String = 'Wind Speed at 50m / m.s^{-1}';
a.Label.Position(1) = 3;
a.Label.FontSize = 12;
t = title(plot_name);
t.FontSize = 15;

saveas(F,'Annual_Solar_Averages.jpg');
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% For Feb Weibull Scale data:
F1 =   figure('Name','Weibull Scale Factors');

% Generate fixed scale for colorbar
[maxValue, maxInd] = max(Weibull_Settings(:,1,:,:), [], "all", "linear") ; % find
the maximum value
colorlim = [0 maxValue]; % set color scale for plots to range from 0 to the maximum
value

for i = 1:12
  plot_name = sprintf('Heatmap of Weibull Scale Factors for Wind Speed, Month %d',
i);
  ax = axesm('MapProjection', 'utm', 'MapLatLimit', latlim, 'MapLonLimit', lonlim);
  h = geoshow(lat_mesh, lon_mesh, squeeze(Weibull_Settings(i,1,:,:)),
'DisplayType', 'texturemap');
  set(h, 'CData', squeeze(Weibull_Settings(i,1,:,:)));
  plotm(coastlat, coastlon, 'Color', 'w');
  a = colorbar ; % add a colorbar to the plot
  caxis(colorlim) ; % set the color scale for the plot
  a.Label.String = 'Weibull Scale Factor Parameter';
  a.Label.Position(1) = 3;
  a.Label.FontSize = 12;
  t = title(plot_name);
  t.FontSize = 15;

  frame = getframe(F1); % capture the current plot as a frame
  scale_im{i} = frame2im(frame); % store the frame in the 'im' cell array
end

F2 =   figure('Name','Weibull Shape Factors');

% Generate fixed scale for colorbar
```

```
[maxValue, maxInd] = max(Weibull_Settings(:,2,:,:), [], "all", "linear") ; % find
the maximum value
colorlim = [0 maxValue]; % set color scale for plots to range from 0 to the maximum
value

for i = 1:12
  plot_name = sprintf('Heatmap of Weibull Shape Factors for Wind Speed, Month %d',
i);
  ax = axesm('MapProjection', 'utm', 'MapLatLimit', latlim, 'MapLonLimit', lonlim);
  h = geoshow(lat_mesh, lon_mesh, squeeze(Weibull_Settings(i,2,:,:)),
'DisplayType', 'texturemap');
  set(h, 'CData', squeeze(Weibull_Settings(i,2,:,:)));
  plotm(coastlat, coastlon, 'Color', 'w');
  a = colorbar ; % add a colorbar to the plot
  caxis(colorlim) ; % set the color scale for the plot
  a.Label.String = 'Weibull Shape Factor Parameter';
  a.Label.Position(1) = 3;
  a.Label.FontSize = 12;
  t = title(plot_name);
  t.FontSize = 15;

  frame = getframe(F2); % capture the current plot as a frame
  shape_im{i} = frame2im(frame); % store the frame in the 'im' cell array
end

F3 = figure('Name','Compiled Plots'); % create a new figure
% Loop through each month to display the plot for that month
for idx = 1:12
    subplot(1,2,1); % create a subplot for shape and scale plots
    imshow(scale_im{idx}); % display the plots for the current month
    subplot(1,2,2); % create a subplot for shape and scale plots
    imshow(shape_im{idx}); % display the plots for the current month

    frame = getframe(F3); % capture the current plot as a frame
    combined_im{idx} = frame2im(frame); % store the frame in the 'im' cell array
end


filename = "Weibull Scale Factors.gif"; % Specify the output file name
% Loop through each month to save the plot for that month as a frame in the .gif
file
for idx = 1:12
    [A,map] = rgb2ind(scale_im{idx},256);
    if idx == 1
        imwrite(A,map,filename,"gif","LoopCount",Inf,"DelayTime",1); % create the
.gif file and add
    else
        imwrite(A,map,filename,"gif","WriteMode","append","DelayTime",1); % create
the .gif file and add
    end
end
filename = "Weibull Shape Factors.gif"; % Specify the output file name
% Loop through each month to save the plot for that month as a frame in the .gif
file
for idx = 1:12
    [A,map] = rgb2ind(shape_im{idx},256);
    if idx == 1
        imwrite(A,map,filename,"gif","LoopCount",Inf,"DelayTime",1); % create the
.gif file and add
    else
        imwrite(A,map,filename,"gif","WriteMode","append","DelayTime",1); % create
the .gif file and add
    end
end
filename = "Weibull Fitted Factors.gif"; % Specify the output file name
% Loop through each month to save the plot for that month as a frame in the .gif
file
for idx = 1:12
```

```
    [A,map] = rgb2ind(combined_im{idx},256);
    if idx == 1
        imwrite(A,map,filename,"gif","LoopCount",Inf,"DelayTime",1); % create the
.gif file and add
    else
        imwrite(A,map,filename,"gif","WriteMode","append","DelayTime",1); % create
the .gif file and add
    end
end
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%%%%%%%%%%%%% Functions %%%%%%%%%%%%%%%%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function output_array = fit_and_discretise(Month_Data, lat, lon, LATITUDES,
LONGITUDES)

    % 1. Apply Weibull Distribution and return parameters
    [~, width, height] = size(Month_Data);
    monthly_weibull = zeros(1,2,39,52);

    shape_array = zeros(width,height);
    scale_array = zeros(width,height);
    try
        for i = 1:width
            for j = 1:height
              loc_data = squeeze(Month_Data(:,i,j));
              % Identify the NaN values
              nan_idx = isnan(loc_data);
              % Remove the NaN values and shorten the double
              loc_data = loc_data(~nan_idx);

              params = wblfit(loc_data);
              a = params(1);          % scale parameter
              b = params(2);          % shape parameter

              scale_array(i,j) = a;
              shape_array(i,j) = b;
            end
        end
    catch
        disp(size(loc_data));
        disp(loc_data);
    end
    for i = 1:length(LATITUDES)
        for j = 1:length(LONGITUDES)
          [~, lat_index] = min(abs(lat - LATITUDES(i)));
          [~, lon_index] = min(abs(lon - LONGITUDES(j)));

          monthly_weibull(1, 1, j, i) = scale_array(lat_index,lon_index) ;    %
assign scale parameter
          monthly_weibull(1, 2, j, i) = shape_array(lat_index,lon_index) ;    %
assign shape parameter
        end
    end

    output_array = monthly_weibull;
    end
```

Published with MATLAB® R2023a

Extrapolate Installed Capacity based on LCOE (Offshore Wind)

```
clear all;
close all;
clc;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

data = [2014     4501.3   0.18     ;
        2015     5093.4   0.165    ;
        2016     5273.4   0.135    ;
        2017     6987.85  0.13     ;
        2018     8180.5   0.13];

cfd_strike_price = [2021  0.091195 ;
                    2022  0.07015  ;
                    2023  0.048373 ;
                    2024  0.05076542       ;
                    2026  0.045567 ];

years = data(1:end,1);
years_full = years;
installed_capacity = data(1:end,2);
LCOE_hist = data(1:end,3);
LCOE_full = LCOE_hist;

for i = 1:length(cfd_strike_price(:,2))
  LCOE_full(length(LCOE_hist)+i) = cfd_strike_price(i,2);
  years_full(length(years)+i) = cfd_strike_price(i,1);
end


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% inv transform the LCOE data
LCOE_inv = LCOE_hist.^(-1);

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Show plot of inverse relationship, unoptimised
figure('Name','Inverse Relationship')
scatter(installed_capacity, LCOE_inv)
xlabel('Installed Capacity (MW)')
ylabel('Levelized Cost of Energy (LCOE)^{-1} / kWh*USD^{-1}')
hold on ;


LCOE_inv_full = LCOE_full.^(-1);
% Fit a line to the inv transformed data
f = @(q,x) q(1) * x + q(2);
q0 = [1.4 -290];

filename = 'Least Squares Optimisation, Offshore wind.gif';

for i = 1:20
    options = optimoptions("lsqcurvefit","PlotFcn",{'optimplotfirstorderopt',
@optimplotresnorm}, "MaxIterations",i, "FunctionTolerance",1e-
20,"StepTolerance",0);
    % Call lsqcurvefit with predefined inputs and options
    [q,resnorm,~,~,output] = lsqcurvefit(f, q0, years_full, LCOE_inv_full, [], [],
options);

    % Get handles to the figures created by optimoptions PlotFcn
    fig = findobj('Type', 'Figure', 'Name', 'Optimization Plot Function');
    set(gca, 'YScale', 'log','YLim',[10e+00 10e+08]);

    frame = getframe(fig);
```

[redacted]

```
    im{i} = frame2im(frame);
    [imind,cm] = rgb2ind(im{i},256);
    if i == 1
        % If it's the first frame, create the gif file
        imwrite(imind,cm,filename,'gif','DelayTime',0.4,'Loopcount',inf);
    else
        % If it's not the first frame, append the frame to the gif file
        imwrite(imind,cm,filename,'gif','DelayTime',0.4,'WriteMode','append');
    end
end

% Call lsqcurvefit with predefined inputs and options
options2 = optimoptions("lsqcurvefit", "Display", "final", "MaxIterations",20,
"FunctionTolerance",1e-20,"StepTolerance",0);
[q,resnorm,~,~,output] = lsqcurvefit(f, q0, years_full, LCOE_inv_full, [],
[],options2);

% Get the iteration history and function values from the output struct
iter = output.iterations;

% Calculate the R-squared value
yfit = f(q,years_full);
yresid = LCOE_inv_full - yfit;
SSresid = sum(yresid.^2);
SStotal = (length(LCOE_inv_full)-1) * var(LCOE_inv_full);
rsq = 1 - SSresid/SStotal;

% Print the fitted parameters, residual norm, and R-squared value
fprintf('Fitted parameters: a=%f, b=%f\n',q(1),q(2));
fprintf('Residual norm: %f\n',resnorm);
fprintf('R-squared value: %f\n',rsq);

% Extract the parameters
a = q(1);
b = q(2);

% Plot the inv transformed data and the fitted line
figure('Name','Linear Fit');
hold on;
plot(years, LCOE_inv, 'o');
plot(cfd_strike_price(:,1), cfd_strike_price(:,2).^(-1), 'mo') % add strike price
data
x = linspace(min(years_full), max(years_full), 100);
y = a * x + b;
plot(x, y, '-');
xlabel('Year');
ylabel('(LCOE)^{-1}');
legend('LCOE data','CfD Strike Price','Fitted Curve')
title('Linear Fit of (LCOE)^{-1} / kWh*USD^{-1}');
hold off;

% Transform the fitted line back to LCOE
LCOE_fit = (a * x + b).^(-1);

% Extrapolate the LCOE to 2035
years = min(years):1:2035;
LCOE_extrap = (a .* years + b).^(-1);
LCOE_2035 = LCOE_extrap(length(LCOE_extrap));
fprintf('The LCOE in 2035 is %0.2f\n', LCOE_2035);

% Plot the extraplated LCOE data and the fitted LCOE
years = min(years):1:2035;
figure('Name','Extrapolated LCOE');
plot(years, LCOE_extrap);
xlabel('Year');
ylabel('LCOE / USD*kWh^{-1}');
title('LCOE Extrapolated to 2035');
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Fit a polynomial regression model
p = polyfit(installed_capacity, LCOE_hist.^(-1), 1);

installed_capacity_extrap = ((LCOE_extrap).^(-1)-(p(2)))/(p(1));
installed_capacity_2035 =
installed_capacity_extrap(length(installed_capacity_extrap));
fprintf('The Installed Capacity in 2035 is %0.2f\n', installed_capacity_2035);

cfd_capacity = ((cfd_strike_price(:,2)).^(-1)-(p(2)))/(p(1));

% Plot the data and the regression line
figure('Name','Power Regression')
hold on;
scatter(installed_capacity, LCOE_hist, 'o')
scatter(cfd_capacity, cfd_strike_price(:,2), 'mo')
xlabel('Installed Capacity (MW)')
ylabel('Levelized Cost of Energy (LCOE) / / USD*kWh^{-1}')
plot(installed_capacity_extrap, LCOE_extrap)
legend('LCOE Data', 'CfD Strike Price', 'Least Squares Regression')
title('Extrapolated Installed Capacity for Offshore Wind, UK')
title.FontSize = 16;

xline(installed_capacity_2035, 'r--', 'HandleVisibility', 'off');
text(installed_capacity_2035+100, max(LCOE_extrap)/2, '2035',
'HorizontalAlignment', 'left', 'VerticalAlignment', 'bottom', 'Color', 'r');
hold off ;
```

Published with MATLAB® R2023a

Example of Installed Capacity Extrapolation (Offshore Wind)

```
clear all;
close all;
clc;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

data = [2014     4501.3   0.18     ;
        2015     5093.4   0.165    ;
        2016     5273.4   0.135    ;
        2017     6987.85  0.13     ;
        2018     8180.5   0.13];

cfd_strike_price = [2021  0.091195 ;
                    2022  0.07015  ;
                    2023  0.048373 ;
                    2024  0.05076542       ;
                    2026  0.045567 ];

years = data(1:end,1);
years_full = years;
installed_capacity = data(1:end,2);
LCOE_hist = data(1:end,3);
LCOE_full = LCOE_hist;

for i = 1:length(cfd_strike_price(:,2))
  LCOE_full(length(LCOE_hist)+i) = cfd_strike_price(i,2);
  years_full(length(years)+i) = cfd_strike_price(i,1);
end


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% inv transform the LCOE data
LCOE_inv = LCOE_hist.^(-1);

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Show plot of inverse relationship, unoptimised
figure('Name','Inverse Relationship')
scatter(installed_capacity, LCOE_inv)
xlabel('Installed Capacity (MW)')
ylabel('Levelized Cost of Energy (LCOE)^{-1} / kWh*USD^{-1}')
hold on ;


LCOE_inv_full = LCOE_full.^(-1);
% Fit a line to the inv transformed data
f = @(q,x) q(1) * x + q(2);
q0 = [1.4 -290];

filename = 'Least Squares Optimisation, Offshore wind.gif';

for i = 1:20
    options = optimoptions("lsqcurvefit","PlotFcn",{'optimplotfirstorderopt',
@optimplotresnorm}, "MaxIterations",i, "FunctionTolerance",1e-
20,"StepTolerance",0);
    % Call lsqcurvefit with predefined inputs and options
    [q,resnorm,~,~,output] = lsqcurvefit(f, q0, years_full, LCOE_inv_full, [], [],
options);

    % Get handles to the figures created by optimoptions PlotFcn
    fig = findobj('Type', 'Figure', 'Name', 'Optimization Plot Function');
    set(gca, 'YScale', 'log','YLim',[10e+00 10e+08]);

    frame = getframe(fig);
    im{i} = frame2im(frame);
```

```
    [imind,cm] = rgb2ind(im{i},256);
    if i == 1
        % If it's the first frame, create the gif file
        imwrite(imind,cm,filename,'gif','DelayTime',0.4,'Loopcount',inf);
    else
        % If it's not the first frame, append the frame to the gif file
        imwrite(imind,cm,filename,'gif','DelayTime',0.4,'WriteMode','append');
    end
end

% Call lsqcurvefit with predefined inputs and options
options2 = optimoptions("lsqcurvefit", "Display", "final", "MaxIterations",20,
"FunctionTolerance",1e-20,"StepTolerance",0);
[q,resnorm,~,~,output] = lsqcurvefit(f, q0, years_full, LCOE_inv_full, [],
[],options2);

% Get the iteration history and function values from the output struct
iter = output.iterations;

% Calculate the R-squared value
yfit = f(q,years_full);
yresid = LCOE_inv_full - yfit;
SSresid = sum(yresid.^2);
SStotal = (length(LCOE_inv_full)-1) * var(LCOE_inv_full);
rsq = 1 - SSresid/SStotal;

% Print the fitted parameters, residual norm, and R-squared value
fprintf('Fitted parameters: a=%f, b=%f\n',q(1),q(2));
fprintf('Residual norm: %f\n',resnorm);
fprintf('R-squared value: %f\n',rsq);

% Extract the parameters
a = q(1);
b = q(2);

% Plot the inv transformed data and the fitted line
figure('Name','Linear Fit');
hold on;
plot(years, LCOE_inv, 'o');
plot(cfd_strike_price(:,1), cfd_strike_price(:,2).^(-1), 'mo') % add strike price
data
x = linspace(min(years_full), max(years_full), 100);
y = a * x + b;
plot(x, y, '-');
xlabel('Year');
ylabel('(LCOE)^{-1}');
legend('LCOE data','CfD Strike Price','Fitted Curve')
title('Linear Fit of (LCOE)^{-1} / kWh*USD^{-1}');
hold off;

% Transform the fitted line back to LCOE
LCOE_fit = (a * x + b).^(-1);

% Extrapolate the LCOE to 2035
years = min(years):1:2035;
LCOE_extrap = (a .* years + b).^(-1);
LCOE_2035 = LCOE_extrap(length(LCOE_extrap));
fprintf('The LCOE in 2035 is %0.2f\n', LCOE_2035);

% Plot the extraplated LCOE data and the fitted LCOE
years = min(years):1:2035;
figure('Name','Extrapolated LCOE');
plot(years, LCOE_extrap);
xlabel('Year');
ylabel('LCOE / USD*kWh^{-1}');
title('LCOE Extrapolated to 2035');

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
% Fit a polynomial regression model
p = polyfit(installed_capacity, LCOE_hist.^(-1), 1);

installed_capacity_extrap = ((LCOE_extrap).^(-1)-(p(2)))/(p(1));
installed_capacity_2035 =
installed_capacity_extrap(length(installed_capacity_extrap));
fprintf('The Installed Capacity in 2035 is %0.2f\n', installed_capacity_2035);

cfd_capacity = ((cfd_strike_price(:,2)).^(-1)-(p(2)))/(p(1));

% Plot the data and the regression line
figure('Name','Power Regression')
hold on;
scatter(installed_capacity, LCOE_hist, 'o')
scatter(cfd_capacity, cfd_strike_price(:,2), 'mo')
xlabel('Installed Capacity (MW)')
ylabel('Levelized Cost of Energy (LCOE) / / USD*kWh^{-1}')
plot(installed_capacity_extrap, LCOE_extrap)
legend('LCOE Data', 'CfD Strike Price', 'Least Squares Regression')
title('Extrapolated Installed Capacity for Offshore Wind, UK')
title.FontSize = 16;

xline(installed_capacity_2035, 'r--', 'HandleVisibility', 'off');
text(installed_capacity_2035+100, max(LCOE_extrap)/2, '2035',
'HorizontalAlignment', 'left', 'VerticalAlignment', 'bottom', 'Color', 'r');
hold off ;
```

```
Fitted parameters: a=1.483243, b=-2983.362591

Residual norm: 34.506228

R-squared value: 0.907784

The LCOE in 2035 is 0.03

The Installed Capacity in 2035 is 59395.86
```

Published with MATLAB® R2023a

RES Power System Simulation (No Storage)

```
% [redacted] - 20167060 - BEng Individual Project
% RES Power System Script (Version 0)

% This script will:
%   1. Read variables stored in the '6. Input File' directory
%   (/Users/cameronyell/OneDrive - The University of Nottingham/M3/Third
%   Year/BEng Project/4. Modelling/6. Input Files)
%   2. Calculate the power produced by various utilities at 1-hour
%   intervals throughout the year 2035
%   3. Balance out negative net power until all positive net power nodes
%   are 0
%   4. Analyse the power production data and present relevant plots
clear;
close all;
clc;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Declare input file directory

filepath = "/Users/cameronyell/OneDrive - The University of Nottingham/M3/Third
Year/BEng Project/4. Modelling/6. Input Files" ;
% Load variables as 1x1 struct
folder = "CAPACITY" ;
UNFORMATTED_CAPACITY_ARRAYS_UK = load(fullfile(filepath, folder,
strcat("CAPACITY_ARRAYS_UK",".mat"))) ;
GROWTH_FACTORS = load(fullfile(filepath, folder, strcat("GROWTH_FACTORS",".mat")))
;
CAPACITY_FACTORS = load(fullfile(filepath, folder,
strcat("CAPACITY_FACTORS",".mat"))) ;

folder = "DEMAND" ;
DEMAND_ARRAY = load(fullfile(filepath, folder, strcat("DEMAND_ARRAY",".mat"))) ;
MONTHLY_HOURLY_DEMAND = load(fullfile(filepath, folder,
strcat("MONTHLY_HOURLY_DEMAND",".mat"))) ;

folder = "ENVIRONMENTAL" ;
HOURLY_IRRADIANCE_RATIOS = load(fullfile(filepath, folder,
strcat("HOURLY_IRRADIANCE_RATIOS",".mat"))) ;
WEIBULL_SETTINGS_SOLAR = load(fullfile(filepath, folder,
strcat("WEIBULL_SETTINGS_SOLAR",".mat"))) ;
WEIBULL_SETTINGS_WIND = load(fullfile(filepath, folder,
strcat("WEIBULL_SETTINGS_WIND",".mat"))) ;
WIND_K_FACTORS = load(fullfile(filepath, folder, strcat("WIND_K_FACTORS",".mat")))
;
SOLAR_AREA_COEFFICIENTS = load(fullfile(filepath, folder,
strcat("SOLAR_AREA_COEFFICIENTS",".mat"))) ;

folder = "GEOMETRY" ;
LAND_SHAPE = load(fullfile(filepath, folder, strcat("LAND_SHAPE",".mat"))) ;
LATITUDES_39x52 = load(fullfile(filepath, folder,
strcat("LATITUDES_39x52",".mat"))) ;
LONGITUDES_39x52 = load(fullfile(filepath, folder,
strcat("LONGITUDES_39x52",".mat"))) ;

% Re-format variables if required
UNFORMATTED_CAPACITY_ARRAYS_UK =
squeeze(struct2cell(UNFORMATTED_CAPACITY_ARRAYS_UK)) ;
GROWTH_FACTORS = squeeze(cell2mat(struct2cell(GROWTH_FACTORS))) ;
CAPACITY_FACTORS = squeeze(cell2mat(struct2cell(CAPACITY_FACTORS))) ;
DEMAND_ARRAY = squeeze(cell2mat(struct2cell(DEMAND_ARRAY))) ;
MONTHLY_HOURLY_DEMAND = squeeze(cell2mat(struct2cell(MONTHLY_HOURLY_DEMAND))) ;
HOURLY_IRRADIANCE_RATIOS = squeeze(cell2mat(struct2cell(HOURLY_IRRADIANCE_RATIOS)))
;
WEIBULL_SETTINGS_SOLAR = squeeze(cell2mat(struct2cell(WEIBULL_SETTINGS_SOLAR))) ;
WEIBULL_SETTINGS_WIND = squeeze(cell2mat(struct2cell(WEIBULL_SETTINGS_WIND))) ;
WIND_K_FACTORS = squeeze(struct2cell(WIND_K_FACTORS)) ;
```

```
SOLAR_AREA_COEFFICIENTS = squeeze(cell2mat(struct2cell(SOLAR_AREA_COEFFICIENTS))) ;
LAND_SHAPE = squeeze(cell2mat(struct2cell(LAND_SHAPE))) ;
LATITUDES_39x52 = squeeze(cell2mat(struct2cell(LATITUDES_39x52))) ;
LONGITUDES_39x52 = squeeze(cell2mat(struct2cell(LONGITUDES_39x52))) ;
%VARIABLE = squeeze(cell2mat(struct2cell(VARIABLE))) ;

TRANSMISSION_LOSS_FACTOR = 0.08 ;        % 8% of power available lost in
transmission
RATED_WIND_SPEED = 14 ;                  % m/s^2
RATED_IRRADIANCE = 0.8 ;                   % kW / m^2

for i = 1:size(UNFORMATTED_CAPACITY_ARRAYS_UK{1,1},2)
    CAPACITY_ARRAYS_UK{i} = UNFORMATTED_CAPACITY_ARRAYS_UK{1,1}{1,i} *
GROWTH_FACTORS(i) ;
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Define names of power sources
POWER_SOURCES = ["Wind Onshore"                         %1
                 "Wind Offshore"                        %2
                 "Geothermal"                           %3
                 "Hydrogen"                             %4
                 "Large Hydro"                          %5
                 "Pumped Storage Hydroelectricity"      %6
                 "Shoreline Wave"                       %7
                 "Small Hydro"                          %8
                 "Solar Photovoltaics"                  %9
                 "Tidal Barrage and Tidal Stream"       %10
                 "Nuclear"] ;                           %11

% assign lists for indecies of utility types
BASELOAD_INDEX = [5, 7, 8, 10, 11] ;
WIND_INDEX = [1, 2] ;
SOLAR_INDEX = 9 ;
STORAGE_INDEX = [4, 6] ;
% Geothermal not used

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Prepare output arrays
start_date = datetime('2035-01-01 00:00:00') ;
end_date = datetime('2035-12-31 23:00:00') ;
datetime_array = start_date:hours(1):end_date ; % create a datetime series for each
interval

N_INTERVALS = length(datetime_array) ; % calculate number of intervals and store as
a variable
[rows, cols] = size(LAND_SHAPE) ;

PRODUCTION = cell(N_INTERVALS,length(POWER_SOURCES)) ;  % pre-define production
output variable
COMBINED_PRODUCTION = zeros(N_INTERVALS, rows, cols) ;  % initialise variable for
sum of power production at each interval
DEMAND = zeros(N_INTERVALS, rows, cols) ;               % initialise variable for
demand data
NET_POWER = zeros(N_INTERVALS, rows, cols) ;            % initialise variable for
calculation of net power (eg. COMBINED_PRODUCTION - DEMNAND)
ENVIRO_DATA = cell(N_INTERVALS,2);                      % initialise variable for
wind speeds and solar irradiances

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Simulation

for interval = 1:N_INTERVALS
    tic;    % start timer
    fprintf('Simulation Running...\t\t%0.2f%%\n',(interval/N_INTERVALS)*100) ;  %
display percentage progress through iterations

    % display an estimate of time until completion based on recorded
```

```
    % iteration times
    if interval <= 300
        fprintf('Estimated time until completion:\tcalculating') ;
    else
        fprintf('Estimated time until
completion:\t%02.0f:%02.0f\n',floor(((N_INTERVALS -
interval)*mean(interval_time))/60),floor(((N_INTERVALS -
interval)*mean(interval_time))-(60*floor(((N_INTERVALS -
interval)*mean(interval_time))/60)))) ;
    end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    % PRODUCTION
    % Baseload production / MW
    for source = 1:length(BASELOAD_INDEX)
        production = CAPACITY_ARRAYS_UK{BASELOAD_INDEX(source)} *
CAPACITY_FACTORS(BASELOAD_INDEX(source)) ;    % constant output from baseload
sources
        PRODUCTION{interval,BASELOAD_INDEX(source)} = production ;
    end

    % Wind production / MW
    wind_speeds = Hourly_Wind_Speed_Generator(WEIBULL_SETTINGS_WIND,
month(datetime_array(interval))) ;     % generate windspeeds using
Hourly_Wind_Speed_Generator function
    ENVIRO_DATA{interval,1} = wind_speeds ;     % record wind speeds in ENVIRO_DATA
    K_RATIO = zeros(39,52) ;    % define / clear K_RATIO variable
    for i = 1:rows
        for j = 1:cols
            if round(wind_speeds(i,j)) <= 13
                K_RATIO(i,j) = wind_speeds(i,j) / RATED_WIND_SPEED ;
            elseif (round(wind_speeds(i,j)) >= 14) && (round(wind_speeds(i,j)) <=
25)
                K_RATIO(i,j) = 1 ;
            else
                K_RATIO(i,j) = 0 ;
            end
        end
    end
    for source = 1:length(WIND_INDEX)
        production = CAPACITY_ARRAYS_UK{WIND_INDEX(source)} .* K_RATIO *
CAPACITY_FACTORS(WIND_INDEX(source));
        PRODUCTION{interval,WIND_INDEX(source)} = production ;
    end

    % Solar production / MW
    if (hour(datetime_array(interval))+1 == 1)
        daily_solar_irrad = Hourly_Solar_Irrad_Generator(WEIBULL_SETTINGS_SOLAR,
month(datetime_array(interval))) ;   % in kWh/m^2/day
        solar_irrad = (daily_solar_irrad .*
HOURLY_IRRADIANCE_RATIOS(hour(datetime_array(interval))+1,
month(datetime_array(interval)))) ./
sum(HOURLY_IRRADIANCE_RATIOS(:,month(datetime_array(interval))),'all') ; % in
kWh/m^2
        ENVIRO_DATA{interval,2} = solar_irrad ;
        if ((solar_irrad / RATED_IRRADIANCE) >= 1)
            production = CAPACITY_ARRAYS_UK{SOLAR_INDEX} .*
CAPACITY_FACTORS(SOLAR_INDEX) ;
        else
            production = CAPACITY_ARRAYS_UK{SOLAR_INDEX} .* (solar_irrad /
RATED_IRRADIANCE)  .* CAPACITY_FACTORS(SOLAR_INDEX) ;
        end
        PRODUCTION{interval,SOLAR_INDEX} = production ;  % in MW
    else
        solar_irrad = (daily_solar_irrad .*
HOURLY_IRRADIANCE_RATIOS(hour(datetime_array(interval))+1,
month(datetime_array(interval)))) ./
```

```
sum(HOURLY_IRRADIANCE_RATIOS(:,month(datetime_array(interval))),'all') ; % in
kWh/m^2
        ENVIRO_DATA{interval,2} = solar_irrad ;
        if ((solar_irrad / RATED_IRRADIANCE) >= 1)
            production = CAPACITY_ARRAYS_UK{SOLAR_INDEX} .*
CAPACITY_FACTORS(SOLAR_INDEX) ;
        else
            production = CAPACITY_ARRAYS_UK{SOLAR_INDEX} .* (solar_irrad /
RATED_IRRADIANCE)  .* CAPACITY_FACTORS(SOLAR_INDEX) ;
        end
        PRODUCTION{interval,SOLAR_INDEX} = production ;  % in MW
    end

    % TRANSMISSION LOSS
    for source = 1:size(PRODUCTION,2)
        PRODUCTION{interval,source} = PRODUCTION{interval,source} * (1 -
TRANSMISSION_LOSS_FACTOR) ;
    end

    % DEMAND
    DEMAND(interval,:,:) = MONTHLY_HOURLY_DEMAND(month(datetime_array(interval)),
hour(datetime_array(interval))+1) * DEMAND_ARRAY ;   % in MW

    % COMBINED POWER PRODUCTION
    for source = 1:length(POWER_SOURCES)
        if size(PRODUCTION{interval,source},1) ~= 0
            COMBINED_PRODUCTION(interval,:,:) =
squeeze(COMBINED_PRODUCTION(interval,:,:)) + PRODUCTION{interval,source} ;
        end
    end
    NET_POWER(interval,:,:) = COMBINED_PRODUCTION(interval,:,:) -
DEMAND(interval,:,:) ;

    TOTAL_NET_POWER(interval) = sum(NET_POWER(interval,:,:),'all') ;
    TNP = TOTAL_NET_POWER(interval);

    % BALANCING
    POWER = squeeze(NET_POWER(interval,:,:));
    SINKS = POWER;
    SUPPLY = POWER;
    SINKS(POWER >= 0) = 0;
    SUPPLY(POWER <= 0) = 0;

    % validity check
    if (max(max(SINKS)) > 0) || (min(min(SUPPLY)) < 0)
        error("SUPPLY / SINKS contain value of wrong sign");
    end

    % Scale SINKS / SUPPLY to account for full net power
    if (TNP > 0)
        SINKS = SINKS .* 0;
        SUPPLY = (SUPPLY ./ sum(SUPPLY,"all")) .* TNP;
    elseif (TNP < 0)
        SUPPLY = SUPPLY .* 0;
        SINKS = (SINKS ./ sum(SINKS,"all")) .* TNP;
    else
        SINKS = SINKS .* 0;
        SUPPLY = SUPPLY .* 0;
    end

    NET_POWER(interval,:,:) = SINKS + SUPPLY;

    clc;
    interval_time(interval) = toc;
end
```

Full RES Power System Simulation

```
% [redacted] - 20167060 - BEng Individual Project
% RES Power System Script (Version 0)

% This script will:
%   1. Read variables stored in the '6. Input File' directory
%   (/Users/cameronyell/OneDrive - The University of Nottingham/M3/Third
%   Year/BEng Project/4. Modelling/6. Input Files)
%   2. Calculate the power produced by various utilities at 1-hour
%   intervals throughout the year 2035
%   3. Balance out negative net power until all positive net power nodes
%   are 0
%   4. Analyse the power production data and present relevant plots
clear;
close all;
clc;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Declare input file directory

filepath = "/Users/cameronyell/OneDrive - The University of Nottingham/M3/Third
Year/BEng Project/4. Modelling/6. Input Files" ;
% Load variables as 1x1 struct

folder = "CAPACITY" ;
CAPACITY_FACTORS = load(fullfile(filepath, folder,
strcat("CAPACITY_FACTORS",".mat"))) ;
HYDRO_STORAGE_CAPACITIES = load(fullfile(filepath, folder,
"HYDRO_STORAGE_CAPACITIES.mat")) ;
HYDROGEN_STORAGE_CAPACITIES = load(fullfile(filepath, folder,
"HYDROGEN_STORAGE_CAPACITIES.mat")) ;

folder = "DEMAND" ;
DEMAND_ARRAY = load(fullfile(filepath, folder, strcat("DEMAND_ARRAY",".mat"))) ;
MONTHLY_HOURLY_DEMAND = load(fullfile(filepath, folder,
strcat("MONTHLY_HOURLY_DEMAND",".mat"))) ;

folder = "ENVIRONMENTAL" ;
HOURLY_IRRADIANCE_RATIOS = load(fullfile(filepath, folder,
strcat("HOURLY_IRRADIANCE_RATIOS",".mat"))) ;
WEIBULL_SETTINGS_SOLAR = load(fullfile(filepath, folder,
strcat("WEIBULL_SETTINGS_SOLAR",".mat"))) ;
WEIBULL_SETTINGS_WIND = load(fullfile(filepath, folder,
strcat("WEIBULL_SETTINGS_WIND",".mat"))) ;
WIND_K_FACTORS = load(fullfile(filepath, folder, strcat("WIND_K_FACTORS",".mat")))
;
SOLAR_AREA_COEFFICIENTS = load(fullfile(filepath, folder,
strcat("SOLAR_AREA_COEFFICIENTS",".mat"))) ;

folder = "GEOMETRY" ;
LAND_SHAPE = load(fullfile(filepath, folder, strcat("LAND_SHAPE",".mat"))) ;
LATITUDES_39x52 = load(fullfile(filepath, folder,
strcat("LATITUDES_39x52",".mat"))) ;
LONGITUDES_39x52 = load(fullfile(filepath, folder,
strcat("LONGITUDES_39x52",".mat"))) ;

filepath = "/Users/cameronyell/OneDrive - The University of Nottingham/M3/Third
Year/BEng Project/4. Modelling/7. Analysis/1. Script Development" ;
% Load variables as 1x1 struct

folder = "3. Storage Calculation" ;
CAPACITY_ARRAYS_UK = load(fullfile(filepath, folder,
strcat("CAPACITY_ARRAYS_UK",".mat"))) ;

% Re-format variables if required
CAPACITY_ARRAYS_UK = squeeze(struct2cell(CAPACITY_ARRAYS_UK)) ;
CAPACITY_ARRAYS_UK = CAPACITY_ARRAYS_UK{1,1} ;
```

```
CAPACITY_FACTORS = squeeze(cell2mat(struct2cell(CAPACITY_FACTORS))) ;
HYDRO_STORAGE_CAPACITIES = squeeze(cell2mat(struct2cell(HYDRO_STORAGE_CAPACITIES)))
;
HYDROGEN_STORAGE_CAPACITIES =
sparse(squeeze(cell2mat(struct2cell(HYDROGEN_STORAGE_CAPACITIES)))) ;
DEMAND_ARRAY = squeeze(cell2mat(struct2cell(DEMAND_ARRAY))) ;
MONTHLY_HOURLY_DEMAND = squeeze(cell2mat(struct2cell(MONTHLY_HOURLY_DEMAND))) ;
HOURLY_IRRADIANCE_RATIOS = squeeze(cell2mat(struct2cell(HOURLY_IRRADIANCE_RATIOS)))
;
WEIBULL_SETTINGS_SOLAR = squeeze(cell2mat(struct2cell(WEIBULL_SETTINGS_SOLAR))) ;
WEIBULL_SETTINGS_WIND = squeeze(cell2mat(struct2cell(WEIBULL_SETTINGS_WIND))) ;
WIND_K_FACTORS = squeeze(struct2cell(WIND_K_FACTORS)) ;
SOLAR_AREA_COEFFICIENTS = squeeze(cell2mat(struct2cell(SOLAR_AREA_COEFFICIENTS))) ;
LAND_SHAPE = squeeze(cell2mat(struct2cell(LAND_SHAPE))) ;
LATITUDES_39x52 = squeeze(cell2mat(struct2cell(LATITUDES_39x52))) ;
LONGITUDES_39x52 = squeeze(cell2mat(struct2cell(LONGITUDES_39x52))) ;

TRANSMISSION_LOSS_FACTOR = 0.08 ;       % 8% of power available lost in
transmission
RATED_WIND_SPEED = 14 ;                 % m/s^2
RATED_IRRADIANCE = 0.8 ;                % kW / m^2

CHARGING_EFFICIENCY = [0.65, 0.88] ;
HES_DISCHARGE_EFFICIENCY = 0.5 ;        % mass fuel cell plant efficiency
PUMPED_HYDRO_DISCHARGE_EFFICIENCY = 0.85 ;
CAPACITY_FACTORS(4) = HES_DISCHARGE_EFFICIENCY ;
CAPACITY_FACTORS(5) = PUMPED_HYDRO_DISCHARGE_EFFICIENCY ;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Define names of power sources
POWER_SOURCES = ["Wind Onshore"                         %1
                "Wind Offshore"                         %2
                "Geothermal"                            %3
                "Hydrogen"                              %4
                "Large Hydro"                           %5
                "Pumped Storage Hydroelectricity"       %6
                "Shoreline Wave"                        %7
                "Small Hydro"                           %8
                "Solar Photovoltaics"                   %9
                "Tidal Barrage and Tidal Stream"        %10
                "Nuclear"] ;                            %11

% assign lists for indecies of utility types
BASELOAD_INDEX = [5, 7, 8, 10, 11] ;
WIND_INDEX = [1, 2] ;
SOLAR_INDEX = 9 ;
STORAGE_INDEX = [4, 6] ;
% Geothermal not used

STORAGE_CAPACITIES = zeros(2,39,52) ;
STORAGE_CAPACITIES(1,:,:) = HYDROGEN_STORAGE_CAPACITIES ;
STORAGE_CAPACITIES(2,:,:) = HYDRO_STORAGE_CAPACITIES ;

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Prepare output arrays
start_date = datetime('2035-01-01 00:00:00') ;
end_date = datetime('2035-12-31 23:00:00') ;
datetime_array = start_date:hours(1):end_date ; % create a datetime series for each
interval

N_INTERVALS = length(datetime_array) ; % calculate number of intervals and store as
a variable
[rows, cols] = size(LAND_SHAPE) ;

PRODUCTION = cell(N_INTERVALS,length(POWER_SOURCES)) ;  % pre-define production
output variable
COMBINED_PRODUCTION = zeros(N_INTERVALS, rows, cols) ;  % initialise variable for
sum of power production at each interval
```

```
DEMAND = zeros(N_INTERVALS, rows, cols) ;                 % initialise variable for
demand data
NET_POWER = zeros(N_INTERVALS, rows, cols) ;              % initialise variable for
calculation of net power (eg. COMBINED_PRODUCTION - DEMNAND)
NET_POWER_NO_STORAGE  = zeros(N_INTERVALS, rows, cols) ;
NET_POWER_STORAGE  = zeros(N_INTERVALS, rows, cols) ;
ENVIRO_DATA = cell(N_INTERVALS,2);                        % initialise variable for
wind speeds and solar irradiances

STORED_ENERGY = zeros(N_INTERVALS, 2, rows, cols) ;      % initialise variable for
energy stored
CHARGING = zeros(N_INTERVALS, 2, rows, cols) ;           % initialise variable for
power going to storage

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Simulation

for interval = 1:N_INTERVALS
    tic;    % start timer
    fprintf('Simulation Running...\t\t%0.2f%%\n',(interval/N_INTERVALS)*100) ;  %
display percentage progress through iterations

    % display an estimate of time until completion based on recorded
    % iteration times
    if interval <= 300
        fprintf('Estimated time until completion:\tcalculating') ;
    else
        fprintf('Estimated time until
completion:\t%02.0f:%02.0f\n',floor(((N_INTERVALS -
interval)*mean(interval_time))/60),floor(((N_INTERVALS -
interval)*mean(interval_time))-(60*floor(((N_INTERVALS -
interval)*mean(interval_time))/60)))) ;
    end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    % PRODUCTION
    % Baseload production / MW
    for source = 1:length(BASELOAD_INDEX)
        production = CAPACITY_ARRAYS_UK{BASELOAD_INDEX(source)} *
CAPACITY_FACTORS(BASELOAD_INDEX(source)) ;     % constant output from baseload
sources
        PRODUCTION{interval,BASELOAD_INDEX(source)} = production ;
    end

    % Wind production / MW
    wind_speeds = Hourly_Wind_Speed_Generator(WEIBULL_SETTINGS_WIND,
month(datetime_array(interval))) ;      % generate windspeeds using
Hourly_Wind_Speed_Generator function
    ENVIRO_DATA{interval,1} = wind_speeds ;      % record wind speeds in ENVIRO_DATA
    K_RATIO = zeros(39,52) ;     % define / clear K_RATIO variable
    for i = 1:rows
        for j = 1:cols
            if round(wind_speeds(i,j)) <= 13
                K_RATIO(i,j) = wind_speeds(i,j) / RATED_WIND_SPEED ;
            elseif (round(wind_speeds(i,j)) >= 14) && (round(wind_speeds(i,j)) <=
25)
                K_RATIO(i,j) = 1 ;
            else
                K_RATIO(i,j) = 0 ;
            end
        end
    end
    for source = 1:length(WIND_INDEX)
        production = CAPACITY_ARRAYS_UK{WIND_INDEX(source)} .* K_RATIO *
CAPACITY_FACTORS(WIND_INDEX(source));
        PRODUCTION{interval,WIND_INDEX(source)} = production ;
    end
```

```
    % Solar production / MW
    if (hour(datetime_array(interval))+1 == 1)
        daily_solar_irrad = Hourly_Solar_Irrad_Generator(WEIBULL_SETTINGS_SOLAR,
month(datetime_array(interval))) ;   % in kWh/m^2/day
        solar_irrad = (daily_solar_irrad .*
HOURLY_IRRADIANCE_RATIOS(hour(datetime_array(interval))+1,
month(datetime_array(interval)))) ./
sum(HOURLY_IRRADIANCE_RATIOS(:,month(datetime_array(interval))),'all') ; % in
kWh/m^2
        ENVIRO_DATA{interval,2} = solar_irrad ;
        if ((solar_irrad / RATED_IRRADIANCE) >= 1)
            production = CAPACITY_ARRAYS_UK{SOLAR_INDEX} .*
CAPACITY_FACTORS(SOLAR_INDEX) ;
        else
            production = CAPACITY_ARRAYS_UK{SOLAR_INDEX} .* (solar_irrad /
RATED_IRRADIANCE)  .* CAPACITY_FACTORS(SOLAR_INDEX) ;
        end
        PRODUCTION{interval,SOLAR_INDEX} = production ;  % in MW
    else
        solar_irrad = (daily_solar_irrad .*
HOURLY_IRRADIANCE_RATIOS(hour(datetime_array(interval))+1,
month(datetime_array(interval)))) ./
sum(HOURLY_IRRADIANCE_RATIOS(:,month(datetime_array(interval))),'all') ; % in
kWh/m^2
        ENVIRO_DATA{interval,2} = solar_irrad ;
        if ((solar_irrad / RATED_IRRADIANCE) >= 1)
            production = CAPACITY_ARRAYS_UK{SOLAR_INDEX} .*
CAPACITY_FACTORS(SOLAR_INDEX) ;
        else
            production = CAPACITY_ARRAYS_UK{SOLAR_INDEX} .* (solar_irrad /
RATED_IRRADIANCE)  .* CAPACITY_FACTORS(SOLAR_INDEX) ;
        end
        PRODUCTION{interval,SOLAR_INDEX} = production ;  % in MW
    end

    % TRANSMISSION LOSS
    for source = 1:size(PRODUCTION,2)
        PRODUCTION{interval,source} = PRODUCTION{interval,source} * (1 -
TRANSMISSION_LOSS_FACTOR) ;
    end

    % DEMAND
    DEMAND(interval,:,:) = MONTHLY_HOURLY_DEMAND(month(datetime_array(interval)),
hour(datetime_array(interval))+1) * DEMAND_ARRAY ;   % in MW

    % COMBINED POWER PRODUCTION
    for source = 1:length(POWER_SOURCES)
        if size(PRODUCTION{interval,source},1) ~= 0
            COMBINED_PRODUCTION(interval,:,:) =
squeeze(COMBINED_PRODUCTION(interval,:,:)) + PRODUCTION{interval,source} ;
        end
    end
    NET_POWER(interval,:,:) = COMBINED_PRODUCTION(interval,:,:) -
DEMAND(interval,:,:) ;

    TOTAL_NET_POWER(interval) = sum(NET_POWER(interval,:,:),'all') ;
    TNP = TOTAL_NET_POWER(interval);

    % BALANCING
    POWER = squeeze(NET_POWER(interval,:,:));
    SINKS = POWER;
    SUPPLY = POWER;
    SINKS(POWER >= 0) = 0;
    SUPPLY(POWER <= 0) = 0;

    % validity check
    if (max(max(SINKS)) > 0) || (min(min(SUPPLY)) < 0)
        error("SUPPLY / SINKS contain value of wrong sign");
```

```
    end

    % Scale SINKS / SUPPLY to account for full net power
    if (TNP > 0)
        SINKS = SINKS .* 0;
        SUPPLY = (SUPPLY ./ sum(SUPPLY,"all")) .* TNP;
    elseif (TNP < 0)
        SUPPLY = SUPPLY .* 0;
        SINKS = (SINKS ./ sum(SINKS,"all")) .* TNP;
    else
        SINKS = SINKS .* 0;
        SUPPLY = SUPPLY .* 0;
    end

    NET_POWER_NO_STORAGE(interval,:,:) = squeeze(SINKS) + squeeze(SUPPLY);

    % STORAGE
    if (interval == 1)
        STORED_ENERGY(interval,:,:,:) = zeros(2,39,52) ;
    else
        STORED_ENERGY(interval,:,:,:) = STORED_ENERGY(interval-1,:,:,:) ;
    end

    % if there is a power surplus (ie. charging available)
    if (TNP > 0)
        % determine ratio of available power to maximum charging capacity
        charging_ratio = TNP / (sum(CAPACITY_ARRAYS_UK{STORAGE_INDEX(1)},"all") +
sum(CAPACITY_ARRAYS_UK{STORAGE_INDEX(2)},"all")) ;
        charging_ratio(charging_ratio > 1) = 1 ;     % if available power is greater
than the maximum charging capacity, set the charging capacity ratio to 1
        % iterate through the two storage technologies, HES and Pumped
        % Hydropower
        for technology = 1:size(STORED_ENERGY,2)
            % calculate the full power used for charging of storage
            CHARGING(interval, technology,:,:) =
CAPACITY_ARRAYS_UK{STORAGE_INDEX(technology)} .* charging_ratio;
            % evaluate the new matrix of stored energy by adding the new energy
supplied to the previous stored energy
            NEW_STORED_ENERGY = squeeze(STORED_ENERGY(interval, technology, :, :))
+ (squeeze(CHARGING(interval, technology, :, :)) .*
CHARGING_EFFICIENCY(technology))  ;
            % create a new variable for the storage capacity of the
            % technology considered in this iteration
            STORAGE_CAPACITY = squeeze(STORAGE_CAPACITIES(technology,:,:));
            % set 0 values of STORAGE_CAPACITY to NaN for division
            % operations
            STORAGE_CAPACITY(STORAGE_CAPACITY == 0) = NaN ;
            % generate an array of ratios to check if storage capacity
            % is exceeded
            capacity_flags = NEW_STORED_ENERGY ./ STORAGE_CAPACITY ;
            capacity_flags(isnan(capacity_flags)) = 0 ;

            if (max(max(capacity_flags)) <= 1) % if the new storage amount is less
than the capacity for all storage nodes
                % proceed to add the NEW_STORED_ENERGY to the overall
                % storage matrix
                STORED_ENERGY(interval, technology, :, :) = NEW_STORED_ENERGY ;
                %disp("1");

            else % if the the new storage amount exceeds the capacity for any
storage nodes
                % redefine NaN values to zero for storage capacities to
                % allow for multiplication
                STORAGE_CAPACITY(isnan(STORAGE_CAPACITY)) = 0 ;
                % re-allocate ratios exceeding 1 to 1
                capacity_flags(capacity_flags > 1) = 1 ;
                % re-calculate NEW_STORED_ENERGY based on capacity_flags
                NEW_STORED_ENERGY = capacity_flags .* STORAGE_CAPACITY ;
```

```
                % re-calculate the charging based on the new values for
NEW_STORED_ENERGY
                CHARGING(interval, technology,:,:) = (NEW_STORED_ENERGY -
squeeze(STORED_ENERGY(interval, technology, :, :))) ./
CHARGING_EFFICIENCY(technology);
                % proceed to add the NEW_STORED_ENERGY to the overall
                % storage matrix
                STORED_ENERGY(interval, technology, :, :) = NEW_STORED_ENERGY;
                %disp("2");
            end
        end
        % adjust the SUPPLY matrix to only include any surplus power
        SUPPLY = squeeze(SUPPLY) .* (1 - (sum((CHARGING(interval,
:,:,:)),"all")/TNP)) ;
        SUPPLY((SUPPLY<0)&(SUPPLY>-0.1)) = 0;

    % if there is a power deficit (ie. discharging required)
    elseif (TNP < 0)
        % if there is sufficient stored energy for maximum discharge
        % calculate the ratio of power required over maximum unfactored
        % output
        discharging_ratio = abs(TNP) /
(sum(CAPACITY_ARRAYS_UK{STORAGE_INDEX(1)},"all") +
sum(CAPACITY_ARRAYS_UK{STORAGE_INDEX(2)},"all")) ;
        % rescale ratio values exceeding 1
        discharging_ratio(discharging_ratio > 1) = 1 ;
        % iterate through the two storage technologies, HES and Pumped
        % Hydropower
        for technology = 1:size(STORED_ENERGY,2)
            % determine the factored ratio of power required over
            % output (so that power required is met)
            discharging_ratio_tech = discharging_ratio ./
CAPACITY_FACTORS(STORAGE_INDEX(technology)) ;
            % rescale ratio values exceeding 1
            discharging_ratio_tech(discharging_ratio_tech > 1) = 1 ;
            % calculate the production of power from storage
            production = CAPACITY_ARRAYS_UK{STORAGE_INDEX(technology)} .*
discharging_ratio_tech .* CAPACITY_FACTORS(STORAGE_INDEX(technology));
            % record the production in the overall production array
            PRODUCTION{interval,STORAGE_INDEX(technology)} = production ;
            % calculate the new stored energy by subtracting
            NEW_STORED_ENERGY = squeeze(STORED_ENERGY(interval, technology, :, :))
- (production ./ CAPACITY_FACTORS(STORAGE_INDEX(technology)));

            % generate an array of ratios to check if storage capacity
            % is exceeded
            capacity_flags = zeros(39,52);
            capacity_flags(NEW_STORED_ENERGY < 0) = 1 ;

            if (max(max(capacity_flags)) < 1) % if the new storage amount is less
than the capacity for all storage nodes
                % proceed to add the NEW_STORED_ENERGY to the overall
                % storage matrix
                STORED_ENERGY(interval, technology, :, :) = NEW_STORED_ENERGY ;
                %disp("3");

            else % if the the new storage amount is negative for any storage nodes
                % re-allocate ratios exceeding 1 to 1
                capacity_flags(capacity_flags > 1) = 1 ;
                % re-calculate NEW_STORED_ENERGY based on capacity_flags
                NEW_STORED_ENERGY(capacity_flags == 1) = 0 ;
                % re-calculate the production based on the new values for
NEW_STORED_ENERGY
                production = (squeeze(STORED_ENERGY(interval, technology, :, :)) -
NEW_STORED_ENERGY) .* CAPACITY_FACTORS(STORAGE_INDEX(technology));
                % record the production in the overall production array
                PRODUCTION{interval,STORAGE_INDEX(technology)} = production ;
                % proceed to add the NEW_STORED_ENERGY to the overall
```

```
                % storage matrix
                STORED_ENERGY(interval, technology, :, :) = NEW_STORED_ENERGY;
                %disp("4");
            end
        end
        % adjust the SINKS matrix to only include any remaining deficit power
        SINKS = squeeze(SINKS) .* (1 - (sum(PRODUCTION{interval,STORAGE_INDEX(1)} +
PRODUCTION{interval,STORAGE_INDEX(2)},"all")/abs(TNP))) ;
    end

    NET_POWER_STORAGE(interval,:,:) = squeeze(SINKS) + squeeze(SUPPLY);

    if sum(STORED_ENERGY(interval, :, :, :),"all") == 0
        %disp("zero stored energy");
    elseif min(min(min(STORED_ENERGY(interval, :, :, :)))) < 0
        error("negative stored energy")
    elseif isnan(sum(STORED_ENERGY(interval, :, :, :),"all"))
        error("NaN stored energy")
    elseif abs(sum(NET_POWER_NO_STORAGE(interval,:,:),"all")) <
abs(sum(NET_POWER_STORAGE(interval,:,:),"all"))
        error("power imbalance")
    end

    COMBINED_PRODUCTION(interval,:,:) = zeros(39,52) ; % re-set so storage
production can be included
    for source = 1:length(POWER_SOURCES)
        if size(PRODUCTION{interval,source},1) ~= 0
            COMBINED_PRODUCTION(interval,:,:) =
squeeze(COMBINED_PRODUCTION(interval,:,:)) + PRODUCTION{interval,source} ;
        end
    end

    clc;
    interval_time(interval) = toc;
end

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Output Data
delete("ANALYSIS_OUTPUT.txt") % clear any existing diary file
diary("ANALYSIS_OUTPUT.txt") ;  % record output data in a text file

% Environmental Analysis
max_enviros = zeros(size(ENVIRO_DATA,1),3);
mean_enviros = zeros(size(ENVIRO_DATA,1),3);
for i = 1:size(ENVIRO_DATA,1)
    land_wind = ENVIRO_DATA{i,1}.*LAND_SHAPE;
    max_enviros(i,1) = max(max(ENVIRO_DATA{i,1}));
    max_enviros(i,2) = max(max(land_wind));
    max_enviros(i,3) = max(max(ENVIRO_DATA{i,2}));
    mean_enviros(i,1) = mean(ENVIRO_DATA{i,1},'all');
    mean_enviros(i,2) = mean(land_wind,'all','omitmissing');
    mean_enviros(i,3) = mean(ENVIRO_DATA{i,2},'all');
end
fprintf("\nMaximum Wind Speed:\t\t\t%0.2f m.s^{-1}\n", ...
        max(max_enviros(:,1)));
fprintf("Average Wind Speed:\t\t\t%0.2f m.s^{-1}\n", ...
        mean(mean_enviros(:,1)));
fprintf("Maximum Wind Speed (Land only):\t\t%0.2f m.s^{-1}\n", ...
        max(max_enviros(:,2)));
fprintf("Average Wind Speed (Land only):\t\t%0.2f m.s^{-1}\n", ...
        mean(mean_enviros(:,2)));
fprintf("Maximum Solar Irrad:\t\t\t%0.2f kW.m^{-2}\n", ...
        max(max_enviros(:,3)));
fprintf("Average Solar Irrad:\t\t\t%0.2f kW.m^{-2}\n\n", ...
        mean(mean_enviros(:,3)));
disp("UK-Wide Average at peak...")
fprintf("Wind Speed:\t\t\t\t%0.2f m.s^{-1}\n", ...
        max(mean_enviros(:,1)));
```

```
fprintf("Wind Speed (Land only):\t\t\t%0.2f m.s^{-1}\n", ...
        max(mean_enviros(:,2)));
fprintf("Solar Irrad:\t\t\t\t%0.2f kW.m^{-2}\n", ...
        max(mean_enviros(:,3)));

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Final Data

PRODUCTION_totals = zeros(11,1) ;
for i = 1:length(PRODUCTION_totals)
    subtotal = 0;
    for j = 1:length(PRODUCTION)
        subtotal = subtotal + sum(PRODUCTION{j,i},'all');
    end
    PRODUCTION_totals(i) = subtotal;
end
PRODUCTION_series = cell(11,1);
for i = 1:size(PRODUCTION,1)
    for j = 1:size(PRODUCTION,2)
        PRODUCTION_series{j,1}{i,1} = sum(PRODUCTION{i,j},'all') ;
    end
end

% No storage statistics
NET_POWER_NO_STORAGE_year = sum(NET_POWER_NO_STORAGE,1) ;
NET_POWER_NO_STORAGE_series = sum(NET_POWER_NO_STORAGE,[2,3]);
NET_POWER_NO_STORAGE_cumulative = cumsum(sum(NET_POWER_NO_STORAGE,[2,3]),1) ;
DEMAND_year = sum(DEMAND,[2,3]);
POWER_DEFICIT_NO_STORAGE = NET_POWER_NO_STORAGE_series(NET_POWER_NO_STORAGE_series
< 0) ;
POWER_CURTAILMENT_NO_STORAGE =
NET_POWER_NO_STORAGE_series(NET_POWER_NO_STORAGE_series > 0) ;
fprintf("\n*********** NO STORAGE SCENARIO  ***********\n\n") ;
fprintf("Total residual net power of:\t%0.0f
TWh\n",sum(NET_POWER_NO_STORAGE,'all')/(10^6)) ;

fprintf("\nPower Deficit Statistics...\n")
fprintf("Total energy deficit:\t\t%0.0f
GWh\n",sum(POWER_DEFICIT_NO_STORAGE,"all")/10^3) ;
fprintf("Number of deficit periods:\t%0.0f
hours\n",length(POWER_DEFICIT_NO_STORAGE)) ;
fprintf("Maximum power deficit of:\t\t%0.0f
GW\n",abs(min(POWER_DEFICIT_NO_STORAGE))/10^3) ;

fprintf("\nPower Curtailment Statistics...\n")
fprintf("Total energy surplus:\t\t%0.0f
GWh\n",sum(POWER_CURTAILMENT_NO_STORAGE,"all")/10^3) ;
fprintf("Number of curtailment periods:\t%0.0f
hours\n",length(POWER_CURTAILMENT_NO_STORAGE)) ;
fprintf("Maximum power curtailed of:\t%0.0f
GW\n",abs(max(POWER_CURTAILMENT_NO_STORAGE))/10^3) ;
fprintf("\n*******************************************\n") ;

% HES storage statistics
NET_POWER_STORAGE_year = sum(NET_POWER_STORAGE,1) ;
NET_POWER_STORAGE_series = sum(NET_POWER_STORAGE,[2,3]);
NET_POWER_STORAGE_cumulative = cumsum(sum(NET_POWER_STORAGE,[2,3]),1) ;
STORED_ENERGY_series = sum(STORED_ENERGY, [2,3,4]);
POWER_DEFICIT_STORAGE = NET_POWER_STORAGE_series(NET_POWER_STORAGE_series < 0) ;
POWER_CURTAILMENT_STORAGE = NET_POWER_STORAGE_series(NET_POWER_STORAGE_series > 0)
;

fprintf("\n*********** HES STORAGE SCENARIO ***********\n\n") ;
fprintf("Total residual net power of:\t%0.0f
TWh\n",sum(NET_POWER_STORAGE,'all')/(10^6)) ;

fprintf("\nPower Deficit Statistics...\n")
```

```
fprintf("Total energy deficit:\t\t%0.0f
GWh\n",sum(POWER_DEFICIT_STORAGE,"all")/(10^3)) ;
fprintf("Number of deficit periods:\t%0.0f
hours\n",length(POWER_DEFICIT_STORAGE(POWER_DEFICIT_STORAGE < -0.1))) ;
fprintf("Maximum power deficit of:\t\t%0.0f
GW\n",abs(min(POWER_DEFICIT_STORAGE))/(10^3)) ;

fprintf("\nPower Curtailment Statistics...\n")
fprintf("Total energy surplus:\t\t%0.0f
GWh\n",sum(POWER_CURTAILMENT_STORAGE,"all")/(10^3)) ;
fprintf("Number of curtailment periods:\t%0.0f
hours\n",length(POWER_CURTAILMENT_STORAGE)) ;
fprintf("Maximum power curtailed of:\t%0.0f
GW\n",abs(max(POWER_CURTAILMENT_STORAGE))/(10^3)) ;

fprintf("\nStorage Statistics...\n")
fprintf("Maximum Hydrogen Storage:\t\t%0.0f
GWh\n",max(max(max(STORED_ENERGY(:,1,:,:))))/10^3) ;
fprintf("Total Hydrogen Energy Supplied:\t%0.0f
GWh\n",PRODUCTION_totals(STORAGE_INDEX(1))/10^3) ;
fprintf("Maximum Pumped Storage:\t\t%0.0f
GWh\n",max(max(max(STORED_ENERGY(:,2,:,:))))/10^3) ;
fprintf("Total Hydro Energy Supplied:\t%0.0f
GWh\n",PRODUCTION_totals(STORAGE_INDEX(2))/10^3) ;
fprintf("\n*******************************************\n") ;

EMMISSIONS = cell(3) ;
% Direct Emissions (tonnes of CO2 eq per GWh)
% ./ 10^3 is the conversion from g per kWh into tonnes per GWh
EMMISSIONS{1,1} = [670, 760, 870] ./ 10^6;   % coal (min, median, max)
EMMISSIONS{2,1} = [350, 370, 490] ./ 10^6;   % gas
EMMISSIONS{3,1} = [30, 57, 98] ./ 10^6;   % gas CCS
% Lifecycle Emissions (g of CO2 eq per kWh)
EMMISSIONS{1,2} = [740, 820, 910] ./ 10^6;   % coal (min, median, max)
EMMISSIONS{2,2} = [410, 490, 650] ./ 10^6;   % gas
EMMISSIONS{3,2} = [94, 170, 340] ./ 10^6;   % gas CCS
% Fossil Fuel power source names
% Lifecycle Emissions (g of CO2 eq per kWh)
EMMISSIONS{1,3} = "Coal";   % coal (min, median, max)
EMMISSIONS{2,3} = "Gas CCGT";   % gas
EMMISSIONS{3,3} = "Gas CCGT + CCS";   % gas CCS

SCENARIO_EMMISSIONS = cell(4,3,2) ;
fprintf("\n************ EMMISIONS ANALYSIS ************\n\n") ;
fprintf("\n************ Direct Emmissions *************\n\n(Values in 1000 metric
tonnes of CO2 or equivailent)\n") ;
fprintf("\nScenario\t|\t\t\tFossil Fuel Power Source\t\t|") ;
fprintf("\n\t\t| %s\t\t\t| %s\t\t|
%s\t|",EMMISSIONS{1,3},EMMISSIONS{2,3},EMMISSIONS{3,3}) ;
fprintf("\n-------------------------------------------------------------------------
-") ;
fprintf("\nNo Storage\t|")
for i = 1:3
    SCENARIO_EMMISSIONS{1,i,1} = - EMMISSIONS{i,1} .*
sum(POWER_DEFICIT_NO_STORAGE,"all");
    fprintf(" %0.0f/%0.0f/%0.0f\t|",SCENARIO_EMMISSIONS{1,i,1}(1),
SCENARIO_EMMISSIONS{1,i,1}(2), SCENARIO_EMMISSIONS{1,i,1}(3)) ;
end
fprintf("\n-------------------------------------------------------------------------
-") ;
fprintf("\nFull Storage\t|")
for i = 1:3
    SCENARIO_EMMISSIONS{2,i,1} = - EMMISSIONS{i,1} .*
sum(POWER_DEFICIT_STORAGE,"all");
    fprintf(" %0.0f/%0.0f/%0.0f\t\t|",SCENARIO_EMMISSIONS{2,i,1}(1),
SCENARIO_EMMISSIONS{2,i,1}(2), SCENARIO_EMMISSIONS{2,i,1}(3)) ;
end
```

```
fprintf("\n-----------------------------------------------------------------------
-") ;
fprintf("\nTotal Reduction\t|")
for i = 1:3
    REDUCTION_EMMISSIONS{i,1} = -(SCENARIO_EMMISSIONS{2,i,1} - (EMMISSIONS{i,1} .*
sum(POWER_DEFICIT_NO_STORAGE,"all")));
    fprintf(" %0.0f/%0.0f/%0.0f\t|",REDUCTION_EMMISSIONS{i,1}(1),
REDUCTION_EMMISSIONS{i,1}(2), REDUCTION_EMMISSIONS{i,1}(3)) ;
end
fprintf("\n-----------------------------------------------------------------------
-") ;
fprintf("\nPumped Storage\t|")
for i = 1:3
    SCENARIO_EMMISSIONS{3,i,1} = - EMMISSIONS{i,1} .*
PRODUCTION_totals(STORAGE_INDEX(2));
    fprintf(" %0.0f/%0.0f/%0.0f\t|",SCENARIO_EMMISSIONS{3,i,1}(1),
SCENARIO_EMMISSIONS{3,i,1}(2), SCENARIO_EMMISSIONS{3,i,1}(3)) ;
end
fprintf("\nHydrogen \t|")
for i = 1:3
    SCENARIO_EMMISSIONS{4,i,1} = - EMMISSIONS{i,1} .*
PRODUCTION_totals(STORAGE_INDEX(1));
    fprintf(" %0.0f/%0.0f/%0.0f\t|",SCENARIO_EMMISSIONS{4,i,1}(1),
SCENARIO_EMMISSIONS{4,i,1}(2), SCENARIO_EMMISSIONS{4,i,1}(3)) ;
end

SCENARIO_EMMISSIONS = cell(4,3,2) ;

fprintf("\n*********** Lifecycle Emmissions ***********\n\n(Values in 1000 metric
tonnes of CO2 or equivailent)\n") ;
fprintf("\nScenario\t|\t\t\tFossil Fuel Power Source\t\t|") ;
fprintf("\n\t\t| %s\t\t\t| %s\t\t|
%s\t|",EMMISSIONS{1,3},EMMISSIONS{2,3},EMMISSIONS{3,3}) ;
fprintf("\n-----------------------------------------------------------------------
-") ;
fprintf("\nNo Storage\t|")
for i = 1:3
    SCENARIO_EMMISSIONS{1,i,2} = -EMMISSIONS{i,2} .*
sum(POWER_DEFICIT_NO_STORAGE,"all");
    fprintf(" %0.0f/%0.0f/%0.0f\t|",SCENARIO_EMMISSIONS{1,i,2}(1),
SCENARIO_EMMISSIONS{1,i,2}(2), SCENARIO_EMMISSIONS{1,i,2}(3)) ;
end
fprintf("\n-----------------------------------------------------------------------
-") ;
fprintf("\nFull Storage\t|")
for i = 1:3
    SCENARIO_EMMISSIONS{2,i,2} = -EMMISSIONS{i,2} .*
sum(POWER_DEFICIT_STORAGE,"all");
    fprintf(" %0.0f/%0.0f/%0.0f\t\t|",SCENARIO_EMMISSIONS{2,i,2}(1),
SCENARIO_EMMISSIONS{2,i,2}(2), SCENARIO_EMMISSIONS{2,i,2}(3)) ;
end
fprintf("\n-----------------------------------------------------------------------
-") ;
fprintf("\nTotal Reduction\t|")
for i = 1:3
    REDUCTION_EMMISSIONS{i,1} = -(SCENARIO_EMMISSIONS{2,i,2} - (EMMISSIONS{i,1} .*
sum(POWER_DEFICIT_NO_STORAGE,"all")));
    fprintf(" %0.0f/%0.0f/%0.0f\t|",REDUCTION_EMMISSIONS{i,1}(1),
REDUCTION_EMMISSIONS{i,1}(2), REDUCTION_EMMISSIONS{i,1}(3)) ;
end
fprintf("\n-----------------------------------------------------------------------
-") ;
fprintf("\nPumped Storage\t|")
for i = 1:3
    SCENARIO_EMMISSIONS{3,i,2} = - EMMISSIONS{i,2} .*
PRODUCTION_totals(STORAGE_INDEX(2));
    fprintf(" %0.0f/%0.0f/%0.0f\t|",SCENARIO_EMMISSIONS{3,i,2}(1),
SCENARIO_EMMISSIONS{3,i,2}(2), SCENARIO_EMMISSIONS{3,i,2}(3)) ;
```

```
end
fprintf("\nHydrogen \t|")
for i = 1:3
    SCENARIO_EMMISSIONS{4,i,2} = - EMMISSIONS{i,2} .*
PRODUCTION_totals(STORAGE_INDEX(1));
    fprintf(" %0.0f/%0.0f/%0.0f\t|",SCENARIO_EMMISSIONS{4,i,2}(1),
SCENARIO_EMMISSIONS{4,i,2}(2), SCENARIO_EMMISSIONS{4,i,2}(3)) ;
end
fprintf("\n-----------------------------------------------------------------------------------
-\n") ;
diary off ;  % finish writing to text file

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Data Visualisation

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Production vs Demand

plot_name = sprintf('Production and Demand');
F1 = figure('Name','Production and Demand') ;
hold on;
plot(datetime_array,sum(COMBINED_PRODUCTION,[2,3]),'b-
',datetime_array,sum(DEMAND,[2,3]),'r-','LineWidth',1.0) ;
plot(datetime_array,sum(CHARGING(:,1,:,:),[3 4]),'g-
',datetime_array,sum(CHARGING(:,2,:,:),[3 4]),'k-','LineWidth',1.0) ;
lgd = legend({'Total Production','Total Demand','Hydrogen Electrolysis','Hydro
Storage Pumping'},"Location","best","FontSize",10) ;
t = title(plot_name);
t.FontSize = 15;
% Add x- and y-axis labels
xlabel('Date','FontSize',12);
ylabel('Power / MW','FontSize',12);

saveas(F1,'Power and Demand Lines','jpeg');

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Power Sources

plot_name = sprintf('Production from Power Sources');
F2 = figure('Name','Power Source');
set(F2,'defaultAxesColorOrder',hsv(11));
hold on;
for i = 1:size(PRODUCTION_series,1)
    h(i) = plot(datetime_array,cell2mat(PRODUCTION_series{i}));
end
lgd = legend(h,POWER_SOURCES,"Location","best","FontSize",10);
t = title(plot_name);
t.FontSize = 15;
% Add x- and y-axis labels
xlabel('Date','FontSize',12);
ylabel('Power / MW','FontSize',12);
hold off;

saveas(F2,'Power Sources','jpeg');
```

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Appendix C – Final Simulation Output

```
Maximum Wind Speed:                 34.79  m.s^{-1}
Average Wind Speed:                 8.55   m.s^{-1}
Maximum Wind Speed (Land only):     32.58  m.s^{-1}
Average Wind Speed (Land only):     7.93   m.s^{-1}
Maximum Solar Irrad:                0.89   kW.m^{-2}
Average Solar Irrad:                0.12   kW.m^{-2}

UK-Wide Average at peak...
Wind Speed:                         12.46  m.s^{-1}
Wind Speed (Land only):             12.18  m.s^{-1}
Solar Irrad:                        0.42   kW.m^{-2}

*********** NO STORAGE SCENARIO  ***********

Total residual net power of:        35     TWh

Power Deficit Statistics...
Total energy deficit:               6129   GWh
Number of deficit periods:          2014   hours
Maximum power deficit of:           12     GW

Power Curtailment Statistics...
Total energy surplus:               40906  GWh
Number of curtailment periods:      6746   hours
Maximum power curtailed of:         18     GW

********************************************

*********** HES STORAGE SCENARIO ***********

Total residual net power of:        10     TWh

Power Deficit Statistics...
Total energy deficit:               1065   GWh
Number of deficit periods:          1145   hours
Maximum power deficit of:           11     GW

Power Curtailment Statistics...
Total energy surplus:               11320  GWh
Number of curtailment periods:      4730   hours
Maximum power curtailed of:         18     GW

Storage Statistics...
Maximum Hydrogen Storage:           640    GWh
Total Hydrogen Energy Supplied:     4647   GWh
Maximum Pumped Storage:             30     GWh
Total Hydro Energy Supplied:        416    GWh

********************************************
```

************ EMMISIONS ANALYSIS ************

************ Direct Emissions *************
(Values in 1000 metric tonnes of CO2 or equivalent)

Scenario	Fossil Fuel Power Source		
	Coal	Gas CCGT	Gas CCGT + CCS
No Storage	4106/4658/5332	2145/2268/3003	184/349/601
Full Storage	714/810/927	373/394/522	32/61/104
Total Reduction	-4820/-5467/-6259	-2518/-2662/-3525	-216/-410/-705
Pumped Storage	-279/-316/-362	-146/-154/-204	-12/-24/-41
Hydrogen	-3114/-3532/-4043	-1627/-1719/-2277	-139/-265/-455

*********** Lifecycle Emissions ***********
(Values in 1000 metric tonnes of CO2 or equivalent)

Scenario	Fossil Fuel Power Source		
	Coal	Gas CCGT	Gas CCGT + CCS
No Storage	4535/5025/5577	2513/3003/3984	576/1042/2084
Full Storage	788/874/969	437/522/692	100/181/362
Total Reduction	-4894/-5531/-6301	-2582/-2790/-3695	-284/-530/-963
Pumped Storage	-308/-341/-379	-171/-204/-270	-39/-71/-141
Hydrogen	-3439/-3811/-4229	-1905/-2277/-3021	-437/-790/-1580